

1

CHAPTER

Life Processes



Previous Year Questions

Multiple Type Questions

1. A stomata closes when: [CBSE 2024]

- (i) it needs carbon dioxide for photosynthesis.
- (ii) it does not need carbon dioxide for photosynthesis.
- (iii) water flows out of the guard cells.
- (iv) water flows into the guard cells.

The correct reason(s) in this process is/are:

- (a) (i) only
- (b) (i) and (iii)
- (c) (ii) and (iii)
- (d) (ii) and (iv)

2. Which of the following statement(s) is (are) true about human heart? [CBSE 2024]

- (A) Right atrium receives oxygenated blood from lungs through pulmonary artery.
- (B) Left atrium transfers oxygenated blood to left ventricle which sends it to various parts of the body.
- (C) Right atrium receives Deoxygenated blood through vena cava from upper and lower body.
- (D) Left atrium transfers oxygenated blood to aorta which sends it to different parts of the body.

- (a) (A)
- (b) (A)
- (c) (B) and (C)
- (d) (B) and (D)

3. An organism which breaks down the food material outside the body and then absorbs it is [CBSE 2024]

- (A) a plant parasite, *Cuscuta*
- (B) an animal parasite, Tapeworm
- (C) a bacteria, *Rhizobium*
- (D) a fungi, *Rhizopus*

- (a) (B) only

- (b) (A) and (D)

- (c) (B) and (C)

- (d) (B) and (D)

4. In human beings, when the process of digestion is completed, the (i) proteins, (ii) carbohydrates, and (iii) fats are respectively finally converted into:

[CBSE 2024]

- (a) (i) Amino acids, (ii) glucose and (iii) fatty acids
- (b) (i) Amino acids, (ii) glucose, (iii) fatty acids and glycerol
- (c) (i) Glucose, (ii) fatty acids and glycerol (iii) amino acids
- (d) (i) Sugars (ii) amino acids. (iii) fatty acids and glycerol

5. The process in which transport of soluble products of photosynthesis takes place in plants is known as:

[CBSE 2024]

- (a) Transpiration
- (b) Evaporation
- (c) Conduction
- (d) Tans-location

6. In human respiratory system, when a person breathes in, the position of ribs and diaphragm will be: [CBSE 2024]

- (a) lifted ribs and curve/dome shaped diaphragm.
- (b) lifted ribs and flattened diaphragm.
- (c) relaxed ribs and flattened diaphragm.
- (d) relaxed ribs and curve/dome shaped diaphragm.

7. **Assertion (A):** The rate of breathing in aquatic

organisms is much faster than in terrestrial organisms.

Reason: The amount of oxygen dissolved in water is very high as compared to the amount of oxygen in air. **[CBSE 2024]**

- (a) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
- (b) Both Assertion (A) and Reason (R) are true and Reason is not the correct explanation of Assertion (A)
- (c) Assertion (A) is true but Reason (R) is false.
- (d) Assertion (A) is false but Reason (R) is true.

8. **Assertion (A):** In the human heart ventricles have thicker muscular walls than atria.

Reason (R): Ventricles have to pump the blood into various organs. **[CBSE 2024]**

- (a) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
- (b) Both Assertion (A) and Reason (R) are true and Reason is not the correct explanation of Assertion (A)
- (c) Assertion (A) is true but Reason (R) is false.
- (d) Assertion (A) is false but Reason (R) is true.

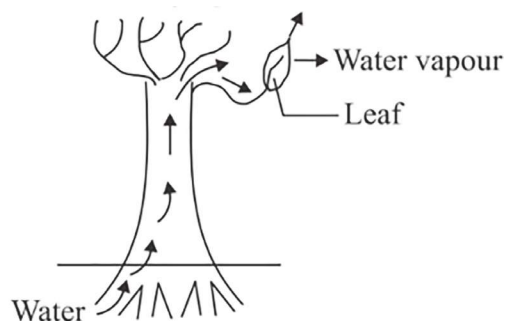
9. Water in the root enters due to **[CBSE 2023]**

- (a) the function of the root to absorb water
- (b) difference in the concentration of ions between the root and the soil
- (c) excess water present in the soil
- (d) diffusion of water in the roots

10. As compared to terrestrial organisms, the rate of breathing in aquatic organisms is **[CBSE 2023]**

- (a) faster because they need more oxygen for their survival.
- (b) faster because the amount of dissolved oxygen in water is fairly low.
- (c) slower because the amount of dissolved oxygen in water is fairly low.
- (d) slower because the capacity of water of dissolving atmospheric air is limited.

11. Observe the following diagram and identify the process and its significance from the following options: **[CBSE 2023]**



- (a) Evaporation : maintains water contents in leaf cells.
- (b) Transpiration: creates a suction force which pulls water inside the plant.
- (c) Excretion: helps in excreting out waste water from the plant.
- (d) Translocation : helps in transporting materials from one cell to another.

12. The process in which loss of water in the form of vapours from the aerial parts of plants takes place is X, which helps in Y. Here, X and Y respectively are **[CBSE 2023]**

- (a) Transpiration and photosynthesis
- (b) Transpiration and temperature regulation
- (c) Translocation and movement of soluble products of photosynthesis in phloem
- (d) Translocation and absorption of water and minerals from soil by roots.

13. **Assertion (A):** The inner walls of the small intestine have finger like projections called villi which are rich in blood.

Reason (R): These villi have a large surface area to help the small intestine in completing the digestion of food.

[CBSE 2023]

- (a) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
- (b) Both Assertion (A) and Reason (R) are true and Reason is not the correct explanation of Assertion (A)
- (c) Assertion (A) is true but Reason (R) is false.
- (d) Assertion (A) is false but Reason (R) is true.

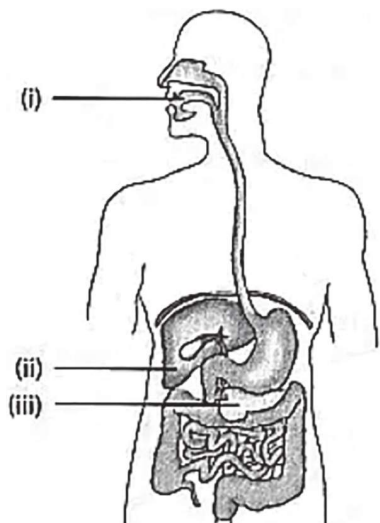
14. **Assertion (A):** The walls of atria are thicker than those of the ventricles.

Reason (R): Ventricles have to pump blood into

various organs at high pressure. [CBSE 2023]

- (a) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
 (b) Both Assertion (A) and Reason (R) are true and Reason is not the correct explanation of Assertion (A)
 (c) Assertion (A) is true but Reason (R) is false.
 (d) Assertion (A) is false but Reason (R) is true.

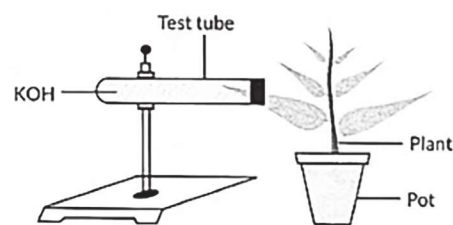
15. In human alimentary canal, the specific enzyme/ juice secreted in locations (i), (ii) and (iii) are [CBSE 2022]



- (a) (i) Amylase (ii) Pepsin (iii) Bile
 (b) (i) Amylase (ii) Bile (iii) Trypsin
 (c) (i) Lipase (ii) Amplase (iii) Pepsin
 (d) (i) Trypsin (ii) Bile (iii) Amylase

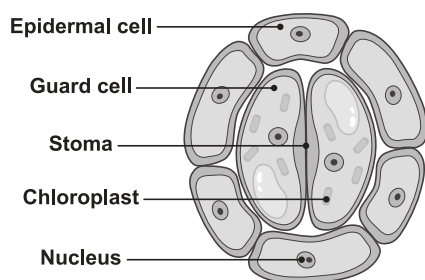
16. Read the following and answer the question from (i) to (iv), [CBSE 2022]

Take a healthy potted plant with elongated leaves. Select a leaf and insert about one half of this leaf in a test tube containing KOH and make it air tight. Place the set-up in sun for two hours. Take out the leaf from the test tube and dip it in boiling water for a few minutes. Put this leaf in a beaker with alcohol and boil it in a water bath. Wash the leaf with water and then dip the leaf in iodine solution for a few minutes. The portion of the leaf dipped in KOH solution will not show any change when dipped in iodine solution



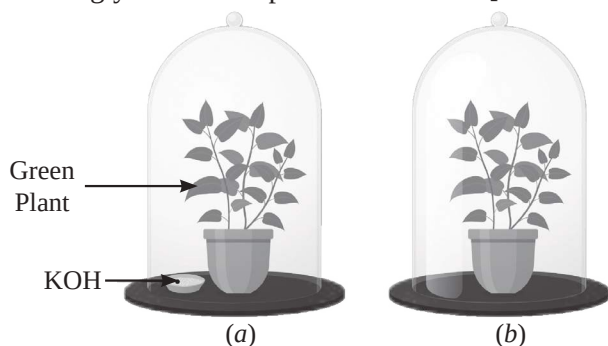
- (i) The function of KOH taken in the test tube is to absorb:
 (a) released water vapours
 (b) released CO_2
 (c) released O_2
 (d) chlorophyll II.
- (ii) On the basis of this activity, we may conclude that the factor for photosynthesis is:
 (a) carbon dioxide (b) oxygen
 (c) chlorophyll (d) water vapour
- (iii) The event that does not occur in photosynthesis is:
 (a) absorption of light energy by chlorophyll
 (b) reduction of carbon dioxide to carbohydrates
 (c) oxidation of carbon to carbon dioxide
 (d) conversion of light energy to chemical energy.
- (iv) Iodine solution gives blue-black colour with:
 (a) starch (b) proteins
 (c) glucose (d) fats
17. The correct statements with reference to single celled organisms are: [CBSE 2022]
 (i) complex substances are not broken down into simpler substances.
 (ii) simple diffusion is sufficient to meet the requirement of exchange of gases.
 (iii) specialised tissues perform different functions in the organism.
 (iv) entire surface of the organism is in contact with the environment for taking in food.
 (a) (i) and (iii) (b) (ii) and (iii)
 (c) (ii) and (iv) (d) (i) and (iv)
18. Which one of the following conditions is true for the state of stomata of a green leaf shown in the given diagram? [CBSE 2022]





- (a) Large amount of water flows into the guard cells.
- (b) Gaseous exchange is occurring in large amount.
- (c) Large amount of water flows out from the guard cells.
- (d) Large amount of sugar collects in the guard cells.

19. A student was asked to write a stepwise procedure to demonstrate that carbon dioxide is necessary for photosynthesis. He wrote the following steps. The wrongly worded step is: [CBSE 2022]



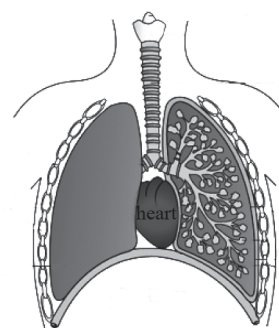
- (a) both potted plants are kept in dark room for at least three days
- (b) bottom of the bell jars is sealed to make them air tight
- (c) both potted plants are kept in sunlight after the starch test
- (d) a leaf from both the plants is taken to test the presence of starch.

20. The length of small intestine in a deer is more as compared to the length of small intestine of a tiger. The reason for this is: [CBSE 2022]
- (a) mode of intake of food
 - (b) type of food consumed
 - (c) presence or absence of villi in intestines
 - (d) presence or absence of digestive enzymes.
21. The sequence of anaerobic respiration in our muscle cells during heavy exercise is [CBSE 2022]

- (a) Glucose $\xrightarrow{\text{Cytoplasm}}$ Pyruvate $\xrightarrow{\text{Muscle cells}}$ Lactic acid + Energy
- (b) Glucose $\xrightarrow{\text{Mitochondria}}$ Pyruvate $\xrightarrow{\text{Muscle cells}}$ Carbon dioxide + Water
- (c) Glucose $\xrightarrow{\text{Cytoplasm}}$ Pyruvate $\xrightarrow{\text{Muscle cells}}$ Ethanol + Carbon dioxide
- (d) Glucose $\xrightarrow{\text{Mitochondria}}$ Pyruvate $\xrightarrow{\text{Muscle cells}}$ Ethanol + Lactic acid

22. The energy released during cellular respiration is used to synthesise: [CBSE 2022]
- (a) ribosomes
 - (b) RBC
 - (c) ATP
 - (d) mitochondria

23. Which of the following statements are correct in reference to the role of A (shown in the given diagram) during a breathing cycle in human beings? [CBSE 2022]

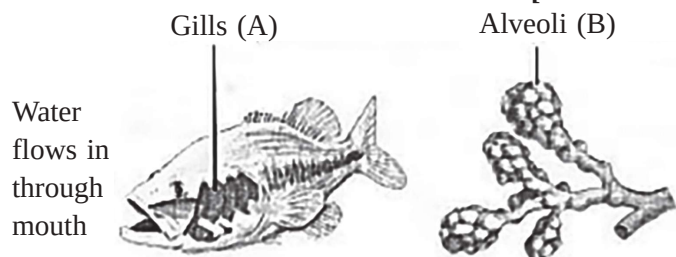


- (i) It helps to decrease the residual volume of air in lungs.
 - (ii) It flattens as we inhale.
 - (iii) It gets raised as we inhale.
 - (iv) It helps the chest cavity to become larger.
 - (a) (ii) and (iv)
 - (b) (iii) and (iv)
 - (c) (i) and (ii)
 - (d) (i), (ii) and (iv)
24. The function of the lining of mucus in the nasal passage of human beings is to: [CBSE 2022]
- (a) increase the temperature of inhaled air
 - (b) move the air in and out
 - (c) filter the air that we breathe in
 - (d) absorb oxygen from the air.
25. In living organisms during respiration which of the following products are not formed if oxygen is not available? [CBSE 2022]
- (a) Carbon dioxide + Water
 - (b) Carbon dioxide + Alcohol
 - (c) Lactic acid + Alcohol

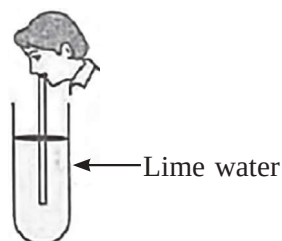
(d) Carbon dioxide + Lactic Acid

26. Respiratory structures of two different animals—a fish and a human being are shown.

Observe (A) and (B) and select one characteristic that hold true for both of them. [CBSE 2022]



- (a) Both are placed internally in the body of animal.
(b) Both have thin and moist surface for gaseous exchange.
(c) Both are poorly supplied with blood vessels to conserve energy.
(d) In both, the blood returns to the heart after being oxygenated.
27. Observe the diagram of an activity given below. What does it help to conclude, when the person exhales into the test-tube? [CBSE 2022]



- (a) Percentage of carbon dioxide is more in inhaled air.
(b) Fermentation occurs in the presence of oxygen.
(c) Percentage of carbon dioxide is more in the exhaled air.
(d) Fermentation occurs in the presence of carbon dioxide.
28. The separation of the right side and the left side of heart is useful to: [CBSE 2022]
- (a) keep a check on oxygenated blood from mixing with deoxygenated blood.
(b) Allow a slow supply of oxygen in the body
(c) Supply energy to animals with low energy needs
(d) Often change their body temperature.
29. In spring, sugar stored in root or stem tissue of plants is transported to the buds for: [CBSE 2022]

- (a) the energy needs of the buds to grow
(b) temperature regulation
(c) balancing the storage in different organs
(d) diffusion process

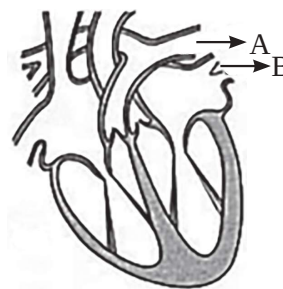
30. Upward movement of water in tall trees is due to [CBSE 2022]

- (a) translocation (b) excretion
(c) photosynthesis (d) transpiration

31. Thin walled blood vessels are called: [CBSE 2022]

- (a) aorta (b) capillaries
(c) arteries (d) vena cava

32. Consider the following statements in connection with the functions of the blood vessels marked A and B in the diagram of a human heart as shown. [CBSE 2022]



- (i) Blood vessel A - It carries carbon dioxide rich blood to the lungs.
(ii) Blood vessel B - It carries oxygen rich blood from the lungs.
(iii) Blood vessel B-Left atrium relaxes as it receives blood from this blood vessel.
(iv) Blood vessel A - Right atrium has thick muscular wall as it has to pump blood to this blood vessel.

The correct statements are

- (a) (i) and (ii) only
(b) (ii) and (iii) only
(c) (ii), (iii) and (iv)
(d) (i), (ii) and (iii)

33. Identify the two components of phloem tissue that help in transportation of food in plants.

[CBSE 2022]

- (a) Phloem parenchyma and sieve tubes
(b) Sieve tubes and companion cells



- (c) Phloem parenchyma and companion cells
- (d) Phloem fibres and sieve tubes

34. Assertion (A): Nitrogen is an essential element for plant growth and is taken up by plants in the form of inorganic nitrates or nitrites. **[CBSE 2022]**

Reason (R): The soil is the nearest and richest source of raw materials like nitrogen, phosphorus and other minerals for the plants.

- (a) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
- (b) Both Assertion (A) and Reason (R) are true and Reason is not the correct explanation of Assertion (A)
- (c) Assertion (A) is true but Reason (R) is false.
- (d) Assertion (A) is false but Reason (R) is true.

35. Assertion (A): The rate of breathing in aquatic organisms is much slower than that seen in terrestrial organisms. **[CBSE 2022]**

Reason (R): The amount of oxygen dissolved in water is very low as compared to the amount of oxygen in air.

- (a) Both (A) and (R) are true and is the correct explanation of .
- (b) Both (A) and (R) are true but is not the correct explanation of (A)
- (c) (A) is true, but (R) is false.
- (d) (A) is false, but is true.

Very Short Answer Type Questions

36. We need to water the soil in plants on a regular basis. But it ultimately reaches the leaves of the plant. Explain how this takes place. **[CBSE 2024]**

OR

Name the type of nutrition exhibited by *Amoeba*. Explain how food is taken in and digested by this organism?

37. State one role of each of the following in human digestive system: **[CNSE 2024]**

- (i) Hydrochloric acid

- (ii) Villi
- (iii) Anal Sphincter
- (iv) Lipase

38. (i) Why is respiratory pigment needed in multicellular organisms with large body size? **[CNSE 2024]**

- (ii) Give reasons for the following:
 - (a) Rings of cartilage are present in the throat.
 - (b) Lungs always contain a residual volume of air.
 - (c) The diaphragm flattens and ribs are lifted up when we breathe in.
 - (d) Walls of alveoli contain an extensive network of blood vessels.

39. Photosynthesis takes place in the leaves and the food prepared by it reaches other parts of the plants. Name the process involved and explain it.

[CBSE 2024]

40. (a) Sometimes while running the athletes suffer from muscle cramps. Why? How is the respiration in this case different from aerobic respiration?

OR

- (b) Write the other name given to lymph. State its two functions. **[CBSE 2024]**

41. Explain in brief two ways by which leaves of a plant help in excretion. **[CBSE 2023]**

42. (i) Name the type of blood (oxygenated/deoxygenated) transported by each of the following mentioning the path (i.e. from one organ (which place) to another (which place)). **[CBSE 2023]**

- (a) Vena cava
- (b) Pulmonary artery

OR

(ii) With the help of a schematic flow chart, show the breakdown of glucose in a cell to provide energy

- (a) in the presence of oxygen
- (b) in lack of oxygen

43. Name the part of the human excretory system where nephrons are found. Write the structure and function of nephrons. **[CBSE 2023]**

44. (a) Why is it important to prevent oxygenated and deoxygenated blood from mixing in birds and

mammals?

- (b) Which animals can tolerate some mixing of the oxygenated and deoxygenated blood streams? On what factor does the body temperature of these animals depend? [CBSE 2023]

45. Write one specific function of each of the following organs in relation to excretion in human beings: [CBSE 2023]

- (i) Renal Artery
- (ii) Urethra
- (iii) Glomerulus
- (iv) Tubular part of nephron

46. Two green plants are kept separately in oxygen-free containers, one in the dark and the other in sunlight. It was observed that plants kept in the dark could not survive longer. Give a reason for this observation. [CBSE 2023]

47. (a) List the events in proper sequence that take place during the process of photosynthesis.

OR

- (b) Explain in brief two ways by which leaves of a plant help in excretion. [CBSE 2023]

48. Give the name of the enzyme present in the fluid in our mouth cavity. State the gland which produces it. What would happen to the digestion process if this gland stops secreting this enzyme? [CBSE 2023]

Short Answer Type Questions

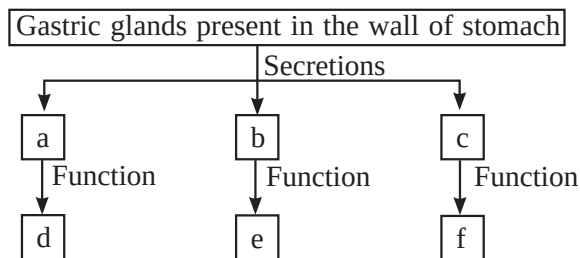
49. (i) How does *Paramecium* obtain its food? [CBSE 2023]
- (ii) List the role of each of the following in our digestive system:
- (a) Hydrochloric acid
 - (b) Trypsin
 - (c) Muscular walls of the stomach
 - (d) Salivary amylase
50. (a) With the help of an activity, explain the action of saliva on the food we eat. [CBSE 2023]
- (b) Why is bile juice important in the process of digestion?

51. What is the other name of “tissue fluid”? Write its two functions. [CBSE 2023]

52. What will happen if: [CBSE 2023]

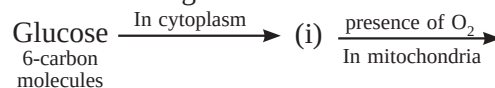
- (a) Xylem tissue in a plant is removed.
- (b) We are injured and start bleeding?

53. Complete the following flow chart as per the given instructions. [CBSE 2021]



54. (a) Describe the structure and function of the basic filtering unit of the kidney.
- (b) List two factors on which the reabsorption of water from urine depends. [CBSE 2021]
55. (a) In the process of respiration, state the function of alveoli. [CBSE 2021]
- (b) The rate of breathing in aquatic organisms is much faster than that in terrestrial organisms. Give reasons.

- (c) Complete the following pathway showing the breakdown of glucose.



(ii) + H₂O + Energy

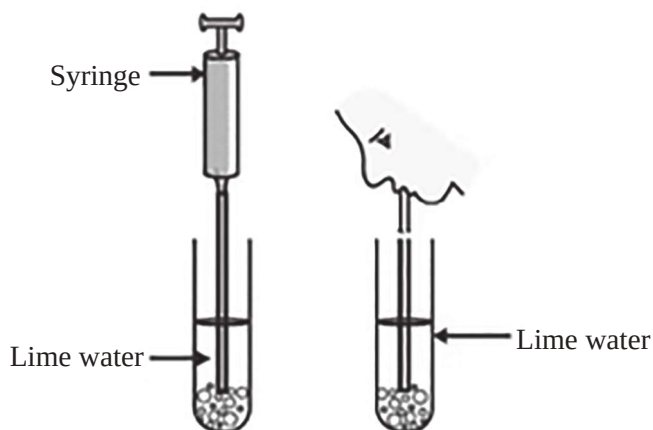
56. (a) Why is there a difference in the rate of breathing between aquatic organisms and terrestrial organism? Explain
- (b) Draw a diagram of the human respiratory system and label it - the pharynx, trachea, lungs, diaphragm and alveolar sac on it. [CBSE 2021]

Long Answer Type Questions

57. (a) Design an experiment to demonstrate that carbon dioxide is essential for photosynthesis. Write the observation and conclusion of the experiment. [CBSE 2024]

OR

- (b) (1)



In the experimental set-up shown above in diagram (1) atmospheric air is being passed into lime water. Time taken by water to turn milky in both the test tubes is different. Give reason.

(ii) Draw the diagram of an open stomatal pore and label (i) Guard cells, and (ii) Chloroplast on it. Mention two functions performed by stomata.

58. (a) With the help of an activity, explain the action of saliva on the food we eat. **[CBSE 2023]**
 (b) Why is bile juice important in the process of digestion?
59. (i) What is double circulation? **[CBSE 2023]**
 (ii) Why is the separation of the right side and the left side of the heart useful? How does it help birds and mammals?
60. (i) Plants absorb water from the soil. Explain how it is taken up and transported from the soil. **[CBSE 2021]**
 (ii) When we are injured and start bleeding, it requires the loss of blood from the system to be minimised. What will happen if the blood loss is not stopped? Is there anything the system could do on its own to prevent the loss?
61. In the human body, the site of absorption of digested food is the small intestine. How is the process of absorption carried out, and why is the absorption of digested food necessary? **[CBSE 2021]**
62. (a) State the role played by the following in the process of digestion: **[CBSE 2021]**
 (i) Enzyme trypsin

- (ii) Enzyme lipase
 (b) List two functions of finger-like projections present in the small intestine.

63. Give reasons: **[CBSE 2021]**
 (a) Ventricles have thicker muscular walls than atria.
 (b) The transport system in plants is slow.
 (c) Circulation of blood in aquatic vertebrates differs from that in terrestrial vertebrates.
 (d) During the daytime, water and minerals travel faster through the xylem as compared to the night.
 (e) Veins have valves, whereas arteries do not.
64. (a) "Blood circulation in fishes is different from the blood circulation in human beings": Justify the statement.
 (b) Describe "blood circulation" in human beings. **[CBSE 2021]**
65. (a) Name the organs that form the excretory system in human beings.
 (b) Describe in brief how urine is produced in the human body. **[CBSE 2021]**
66. (a) Why is nutrition necessary for the human body?
 (b) What causes the movement of food inside the alimentary canal?
 (c) Why is the small intestine in herbivores longer than in carnivores?
 (d) What will happen if mucus is not secreted by the gastric glands? **[CBSE 2021]**
67. Human digestive system is a tube running from mouth to anus. Its main function is to breakdown complex molecules present in the food which cannot be absorbed as such into smaller molecules. These molecules are absorbed across the walls of the tube and the absorbed food reaches each and every cell of the body where it is utilised for obtaining energy.
 (a) Name the glands present in the buccal cavity and write the components of food on which the secretion of these glands act upon.
 (b) Two organs have a sphincter muscle at their exit. Name them.
 (c) What will happen if:
 (i) mucus is not secreted by the gastric glands.
 (ii) Villi are absent in the small intestine.

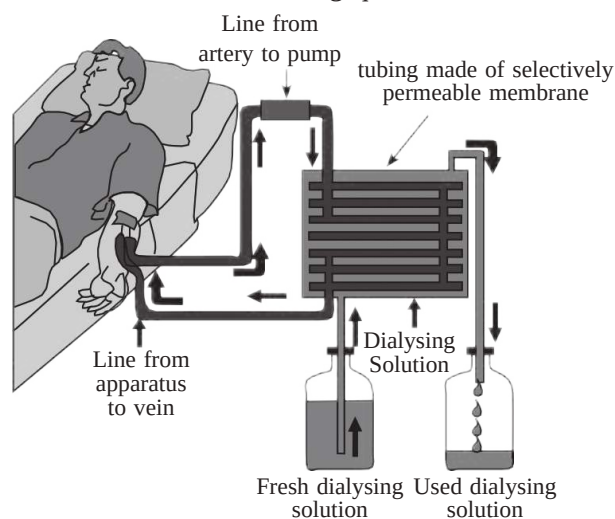
OR

- (c) "Bile juice does not contain any enzyme, yet it has important roles in digestion." Justify the statement.

Case-Based Type Questions

68. Read the following and answer the questions from (i) to (iv). [CBSE 2022]

The figure shown below represents a common type of dialysis called as haemodialysis-It removes waste products from the blood, such as excess salts, and urea which are insufficiently removed by the kidney in patients with kidney failure. During the procedure, the patient's blood is cleaned by filtration through a series of semi-permeable membranes before being returned to the blood of the patient. On the basis of this answer the following questions.



- (i) The haemodialyser has semi-permeable lining of tubes which help:
- (a) to maintain osmotic pressure of blood
 - (b) to filter nitrogenous wastes from the dialysis solution
 - (c) in passing the waste products in the dialysing solution
 - (d) to pump purified blood back into the body of the patient.
- (ii) Which one of the following is not a function of artificial kidney?
- (a) To remove nitrogenous wastes from the blood
 - (b) To remove excess fluids from the blood.
 - (c) To reabsorb essential nutrients from the blood.
 - (d) To filter and purify the blood.
- (iii) The 'used dialysing' solution is rich in:
- (a) urea and excess salts
 - (b) blood cells
 - (c) lymph
 - (d) proteins.
- (iv) Which part of the nephron in human kidney, serves the function of reabsorption of certain substances:
- (a) glomerulus
 - (b) bowman's Capsule
 - (c) tubules
 - (d) collecting Duct



Explanations

Previous Year Questions

Multiple Type Questions

1. (c) Stomata are small openings on the surface of leaves that allow gases to enter and exit the plant. They close when the plant doesn't need carbon dioxide for photosynthesis (which is when it's not producing food and when water flows out of the guard cells (which helps the plant to conserve water). Therefore, the correct answers are (ii) and (iii).
2. (c) The correct statements about the human heart are (B) and (C). The left atrium does transfer oxygenated blood to the left ventricle, which then pumps it to the body. The right atrium receives deoxygenated blood from the body through the vena cava, while (A) is incorrect because the right atrium actually receives deoxygenated blood, not oxygenated, and (D) is misleading as the left atrium transfers blood to the left ventricle, not directly to the aorta.
3. (d) *Rhizopus* is a saprophytic fungus as it obtains nutrition from dead and decaying matter. It consumes soil's dead and decaying organic matter which is broken down into simpler compounds.
4. (b) During digestion in humans, proteins are broken down into amino acids, carbohydrates into glucose, and fats into fatty acids and glycerol. Therefore, the correct answer is (b) because it accurately describes the final products of these macro-nutrients after digestion.
5. (d) This process refers to the movements of the soluble products of photosynthesis, like sugars, from the leaves where they are made to other parts of the plant where they are used or stored. Translocation is essential for distributing energy throughout the plant.
6. (b) The correct answer is (b) lifted ribs and flattened diaphragm. When a person breathes in, the diaphragm contracts and flattens, while the ribs are lifted upward and outward. This movement creates more space in the chest cavity, allowing air to fill the lungs.
7. (c) The correct answer is (c) Assertion (A) is true., but Reason (R) is false. Aquatic organisms do

breathe faster than terrestrial organisms because they need to extract enough oxygen from water, which has less oxygen available than air. Therefore, while the assertion is correct, the reason given is incorrect because it states that water has a high amount of dissolved oxygen, which is not true compared to air.

8. (a) Ventricles have thicker walls than atria as they have to pump blood to the organs with high pressure.
9. (b) Water needed by plants is absorbed by roots. The root hair absorbs water from soil by osmosis. Difference in concentration of ions is created between roots and soil which enables water to enter into roots to compensate difference in concentration.
10. (b) In aquatic organisms, the rate of breathing is higher.

These organisms utilize oxygen dissolved in water.

The amount of dissolved oxygen in water is relatively low compared to the oxygen available in the air. Therefore, aquatic organisms need to breathe faster to obtain sufficient oxygen for their survival.

11. (b)
 - Transpiration pull aids in the upward movement of water from the roots to the leaves.
 - A significant amount of water evaporates through the stomata on the leaf surface.
 - This evaporation process generates a suction force, pulling water up through the plant from the roots.

12. (b)

• Process (X): Transpiration

Transpiration is the process where water is lost in the form of vapours from the aerial parts of plants primarily through the stomata in the leaves.

Significance (Y): Temperature Regulation

- Transpiration helps in temperature regulation by cooling the plant. As water evaporates from the leaf surfaces, heat is dissipated, thus maintaining optimal temperature conditions for various physiological processes, including photosynthesis.

13. (c) Reason is false because villi increases the surface area for greater absorption of digested food.

14. (d) **Assertion (A):** The walls of atria are thicker than those of the ventricles.

This statement is false. In reality, the walls of the ventricles are thicker than those of the atria.

Ventricles have thicker muscular walls because they pump blood out of the heart to the lungs and the rest of the body, which requires more force (pressure).

Reason (R): Ventricles have to pump blood into various organs at high pressure.

This statement is true. Ventricles indeed pump blood into the arteries and throughout the body at high pressure to ensure blood reaches all organs and tissues.

Therefore, while Reason (R) correctly explains the physiological need for thicker ventricular walls due to high-pressure pumping Assertion (A) is incorrect because the ventricular walls are thicker than atrial walls, not the other way around.

15. (b) Saliva (i) is secreted in the mouth. Saliva contains an enzyme called salivary amylase for digestion. Liver (ii) secretes bile to emulsify fat.

The pancreas (iii) secretes pancreatic juice which contains enzyme trypsin for digestion of proteins.

16. (i) (b) In the given experiment, KOH (Potassium hydroxide) in the test tube absorbs carbon dioxide, thus, due to the absence of CO_2 , the leaf fails to produce starch which proves that carbon dioxide is necessary for photosynthesis.

- (ii) (a) The given experiment demonstrates the requirement of carbon dioxide for photosynthesis. When KOH absorbs the available CO_2 from the portion of leaf dipped in it, formation of sugar and starch is inhibited and the portion of leaf dipped in KOH did not show any change when dipped in iodine solution. Thus, it is proved that CO_2 is necessary for photosynthesis.

- (iv) (c) Following are the three events that occur during the process of photosynthesis

- Absorption of light energy by chlorophyll.
 - Conversion of light energy to chemical energy and splitting of water molecules into hydrogen and oxygen.
 - Reduction of carbon dioxide to carbohydrates.
- (iv) (a) The brown coloured iodine will turn blue-black when it reacts with starch.

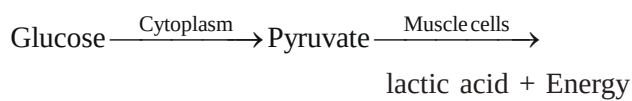
17. (c) Unicellular organisms are one-celled and perform all the life processes that are essential for maintaining the life of cell or organisms like nutrition, respiration, reproduction excretion, etc. In single celled organisms such as *Amoeba*, complex substances are broken into simpler substances.

18. (c) The opening and closing of the stomatal pore depends on the turgidity of the guard cells. The given figure shows a closed stomatal pore which occurs when guard cells lose water and become flaccid.

19. (c) Both potted plants should be kept in sunlight for about two hours before the starch test.

20. (b) Herbivores such as deer eating grass need a longer small intestine to allow the cellulose to be digested. Meat is easier to digest, hence carnivores like tigers have a shorter small intestine.

21. (a) When there is a lack of oxygen in our muscle cells, pyruvate is converted into lactic acid and energy. Therefore, the correct pathway of glucose metabolism in our muscle cells during heavy exercise is:

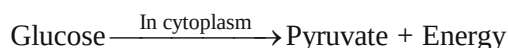


22. (c) The energy released during cellular respiration is immediately used to synthesise a molecule called ATP which is energy currency of the cell and is used as fuel for cellular activities.

23. (a) During inhalation, we lift our ribs and flatten our diaphragm (A), and the chest cavity becomes larger as a result. Because of this, air is sucked into the lungs and fills the expanded alveoli. When our ribs and diaphragm return to their normal position and the chest cavity becomes smaller, exhalation occurs. During breathing, the lungs always contain a residual volume of air, so that there is sufficient time for oxygen to be absorbed and for the carbon dioxide to be released. The residual volume remains unchanged in normal conditions.

24. (c) The mucus lining in the nasal passage helps to moisten and warm the inhaled air and also trap dust particles (i.e., helps in filtering the inhaled air).

25. (a) In aerobic respiration, glucose is completely broken down to CO_2 and H_2O with the production of a large amount of energy (ATP)





Thus, CO₂ and water are formed as a result of aerobic respiration and will not be formed in the absence of oxygen (anaerobic respiration).

26. (b) Both have thin and moist surfaces for gaseous exchange. This characteristic is essential for both gills in fish and alveoli in humans as it allows oxygen to efficiently diffuse into the blood and carbon dioxide to diffuse out. These surfaces are specially adapted to maximize the area available for gas exchange, enhancing the efficiency of respiration in both organisms.
27. (c) When a person blows air from mouth into the test tube containing lime water, the lime water turns into a white precipitate which appears milky. This occurs because the air we exhale has more carbon dioxide.
28. (a) The right and left sides of the heart are separated by a muscular wall that prevents mixing of oxygenated with deoxygenated blood. Such separation allows a highly efficient supply of oxygen to the body.
29. (a) In spring, new buds are formed. These buds need more energy to grow than other parts of the plant. This energy comes from the sugar stored in root or stem tissue and is transported to the buds by phloem.
30. (d) In tall trees, the driving force to move water upward is transpiration. During transpiration, as the water evaporates through the stomata in the leaves, it creates a negative pressure in the leaves and xylem tissues.
31. (b) Capillaries are blood vessels which have thin walls and are one-cell thick. Exchange of materials between blood and surrounding cells takes place across these thin-walled capillaries. These also take waste products away from tissues.
32. (d) Blood vessel A represents pulmonary artery and blood vessel B represents pulmonary vein. Oxygen-rich blood from the lungs comes to the left atrium via pulmonary vein. The left atrium relaxes when it receives blood.
33. (b) The two components of phloem tissue that help in transportation of food in plants are sieve tubes and companion cells.
34. (b) The correct answer is (b) Both (A) and (R) are true but (R) is not the correct explanation of

(A). Nitrogen is indeed essential for plant growth, and plants absorb it mainly as inorganic nitrates or nitrites. However, while the soil is a rich source of nutrients for plants, the reason provided does not directly explain why nitrogen is taken up in that specific form, so it's not a correct explanation of the assertion.

35. (d) The amount of dissolved oxygen in water is fairly low as compared to the amount of oxygen in the air but the rate of breathing in aquatic organisms is much faster than that seen in terrestrial organisms because the aquatic animals have to breathe rapidly to take in sufficient oxygen.

Very Short Answer type Questions

36. When water is lost through stomata in the leaves by transpiration, it creates a suction force/transpiration pull. Due to which water is pulled up through xylem of the roots to the leaves.

OR

The mode of nutrition in *Amoeba* is known as holozoic nutrition. It involves the ingestion, digestion and egestion of food material. *Amoeba* takes in food using temporary finger-like extensions of the cell surface which fuse over the food particle forming a food vacuole. Inside the food vacuole, complex substances are broken down into simpler ones which then diffuse into the cytoplasm. The remaining undigested material is moved to the surface of the cell and thrown out.

37. Role of:
 - (i) Hydrochloric acid: Creates an acidic medium for facilitating the action of enzyme/ kills microorganisms.
 - (ii) Villi: Increases the surface area for absorption of digested food.
 - (iii) Anal Sphincter: Exit of waste material from anus is regulated.
 - (iv) Lipase: Breakdown/digestion of emulsified fats or lipids
38. (i) Diffusion/Diffusion pressure alone cannot take care of oxygen delivery to all parts of the body.
 - (ii) Reasons:
 - (a) To ensure that the air-passage does not collapse.
 - (b) There is sufficient time for oxygen to be absorbed and for the carbon dioxide to be released.

(c) Chest cavity becomes larger.

(d) Because exchange of gases takes place in the alveoli.

39. (i) Translocation

(ii) Transport of soluble products or food prepared by photosynthesis, through phloem in the sieve tubes with the help of companion cells, both in upward and downward directions by utilising energy.

40. (a) (i) Formation of lactic acid in muscles causes cramps.

(ii) Aerobic respiration takes place in the presence of oxygen whereas the respiration taking place above is due to lack of oxygen. End products of aerobic respiration are energy whereas in the above case, lactic acid + Energy is formed.

OR

(b) Tissue fluid/ Extracellular fluid Functions:

i. Carries digested and absorbed fats from the intestine.

ii. Drains excess fluid from extracellular space back into the blood.

iii. Fight against infections.

41. Most of the carbon dioxide is released through stomata in leaves. Plants can get rid of excess water by the process of transpiration, i.e., evaporative loss of water. Many plant waste products are stored in leaves, which are excreted when shed off.

42. (i) Vena cava - deoxygenated blood from body to heart.

Pulmonary artery - deoxygenated blood from heart to lungs.

OR

(ii) Glucose $\xrightarrow{\text{In cytoplasm}}$ Pyruvate

$\xrightarrow[\text{(in mitochondria)}]{\text{Presence of oxygen}}$ $\text{CO}_2 + \text{Water} + \text{Energy}$

Glucose $\xrightarrow{\text{In cytoplasm}}$ Pyruvate

$\xrightarrow[\text{In human muscles cells}]{\text{lack of oxygen}}$ Lactic acid + Energy

43. Nephrons are found in kidneys.

Structure: A cluster of thin-walled capillaries (glomerulus) associated with cup-shaped end of a tube called Bowman's capsule. This further extends into a tubular part which ends in collective ducts.

Function: Filtration of nitrogenous waste from blood to form urine. Reabsorption of useful materials from

the filtrate.

44. (a) Birds and mammals have high energy needs to maintain their body temperature. The separation of oxygenated and deoxygenated blood allows highly efficient supply of oxygen to maintain their body temperature.

(b) Amphibians/reptiles.

The body temperature of these animals depends upon the temperature of the environment.

45. (i) Renal artery carries blood from the aorta to the kidneys for filtration.

(ii) The urethra receives urine stored in the bladder and releases it outside.

(iii) Glomerulus filters the blood passing through it and initiates urine formation.

(iv) The tubular part of the nephron allows selective reabsorption of useful substances like glucose, amino acids, salts, and water into the blood capillaries.

46. The plant kept in the dark could not survive longer because it will not be able to produce the oxygen required for its respiration by the process of photosynthesis. Oxygen is evolved as the byproduct of photosynthesis during the photolysis of water and helps plants to survive longer than is kept in sunlight.

47. (a) Events taking place in the process of photosynthesis are as follows:

(i) Absorption of light energy by chlorophyll.

(ii) Conversion of light energy to chemical energy and splitting of water molecules into hydrogen and oxygen.

(iii) Reduction of carbon dioxide to carbohydrates.

OR

(b) Two ways by which leaves of a plant help in excretion are:

(i) Excess water is excreted by transpiration.

(ii) Oxygen as waste product of photosynthesis is excreted through stomata.

48. Salivary-amylase is present in the fluid in mouth cavity. Salivary gland secretes the salivary amylase. Salivary amylase acts on the carbohydrates and digests them into simple sugars. So, if the salivary glands are unable to produce salivary amylase, digestion of carbohydrates will be hampered in the body.



Short Answer type Questions

49. (i) In *Paramecium*, a unicellular protozoan, the hair-like outgrowth cilia are present on the entire surface and help collect the food. They sweep the food inside the body through the oral groove.

(ii) (a) Role of hydrochloric acid: It helps to maintain the acidic pH in the stomach required for activation of the enzyme pepsin that digests proteins in the stomach.

(b) Role of trypsin in our digestion system: Trypsin breaks down protein into smaller peptides in the duodenum of the small intestine. It helps in digesting dietary protein by breaking the chain of amino acids. It also activates some proenzymes present in pancreatic juice.

(c) Role of muscular walls in the stomach in our digestive system. The muscular walls in the stomach contract periodically and thereby help in the churning and mixing of the food with the digestive enzymes and HCL. It helps in chemical digestion.

(d) Role of salivary amylase in our digestive system: Salivary amylase found in saliva breaks down the starch and converts it into the simplest sugar.

50. (a) The action of saliva on the food we eat is explained with the help of activity as follows: Take two test tubes, A and B. In test tube A, put one teaspoon full of boiled rice, and in test tube B, keep one teaspoon full of boiled rice after chewing it for 3 to 5 minutes. Add 3.4 mL of water in both test tubes and add a few drops of iodine solution to each test tube. In test tube A, the colour of the rice changes because of the absence of any enzyme, while in test tube B, no colour change is observed because when rice is chewed, the amylase enzyme present in saliva breaks down the starch of rice to simple sugars.

(b) Bile juice breaks down fats into fatty acids that can be taken into the body by the digestive tract. In addition, it provides alkaline fluid in the intestine to neutralize the acidic pH of food that comes from the stomach.

51. Tissue fluid is also called lymph. The functions of tissue fluid are:

(i) It carries digested and absorbed fat from the intestine.

(ii) It drains excess fluid from extracellular space back into the blood.

52. (a) If the xylem is removed, transport of water and minerals from the soil would not occur, which leads to the wilting of leaves and, ultimately, the death of the plant.

(b) In case of any injury, when bleeding occurs, platelets circulate around the body and form a mesh-like network or clot at the site of injury.

53. a - Hydrochloric acid (HCl)

b-Pepsin

c-Mucus

d- HCl makes medium acidic for the activation of an enzyme pepsin.

e-Pepsin acts in an acidic medium, which breaks down proteins into peptones.

f - The mucus protects the inner lining of the stomach from the corroding action of HCl.

54. (a) Nephron is the structural and functional unit of the kidney. Its main role is to regulate the concentration of water and soluble substances like sodium and other salts by filtering the blood, reabsorbing the excess water that is needed, and excreting the rest as urine.

(b) Two factors on which the reabsorption of water from urine depends are:

- ❖ The amount of excess water which is present in the body.
- ❖ The amount of dissolved waste that needs to be excreted out of the body.

55. (a) Functions of alveoli are:

- (i) They increase the surface area for the exchange of gases.
- (ii) The thin walls of alveoli facilitate the rapid exchange of oxygen and carbon dioxide between alveolar air and blood.

(b) Aquatic animals like fishes obtain dissolved oxygen from water present through their gills. The amount of dissolved oxygen is quite small compared to the amount of oxygen in the air. Therefore, to obtain the required amount of oxygen from water, aquatic animals have to breathe much faster than terrestrial organisms.

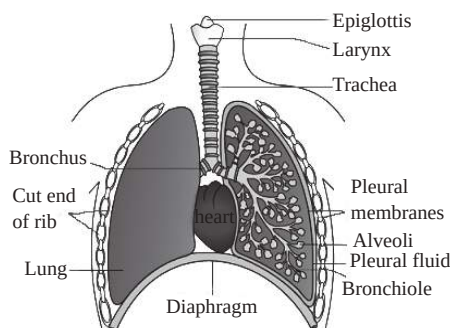
(c)
$$\text{Glucose} \xrightarrow{\text{In cytoplasm}} \text{(i) Pyruvate} \xrightarrow[\text{In mitochondria}]{\text{Presence of } O_2}$$

 6-carbon molecule 3-carbon molecule + energy
 (ii) $CO_2 + H_2O + \text{Energy}$

56. (a) Terrestrial organisms inspire atmospheric oxygen, while aquatic organisms thrive on the dissolved

oxygen present in water. Oxygen diffuses through water at a much slower rate than in air. A terrestrial organism has the advantage of utilizing greater amounts of oxygen at a faster rate with lesser effort, whereas aquatic organisms have to put more effort into obtaining the same amount of oxygen; therefore, breathing in aquatic organisms is much faster than in terrestrial organisms.

(b) The labelled diagram of the human respiratory system is as follows:



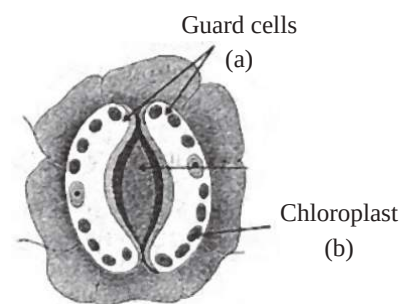
Long Answer type Questions

57. (a) Take two healthy potted plants, A and B, of nearly the same size.

- ❖ Keep them in darkness for three days. (Destarch the plant)
- ❖ Place a watch glass containing potassium hydroxide by the side of potted plant A but not in potted plant B.
- ❖ Cover both the plants with separate bell jars and seal the bottom of the jars with Vaseline.
- ❖ Keep both the plants in sunlight for two hours.
- ❖ Pluck one leaf each from both the plants and test for the presence of starch with iodine solution.
- ❖ Observation: The leaf of the potted plant A with KOH did not turn blue-black. The leaf of the potted plant B turns blue-black.
- ❖ **Conclusion:** KOH absorbs CO_2 so photosynthesis did not occur in potted plant A.

OR

(b) (i) In set up (i) lime water turns milky in more time as compared to set up (ii) because the air we exhaled contains high percentage of CO_2 as compared to atmospheric air.



Open Stomatal Pore

Two labellings: (i) Guard Cells

(ii) Chloroplast Two functions performed by stomata:

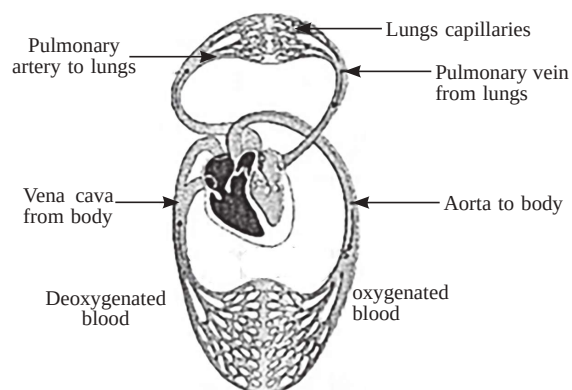
Gaseous exchange

Transpiration

58. (a) The action of saliva on the food we eat is explained with the help of activity as follows: Take two test tubes, A and B. In test tube A, put one teaspoon full of boiled rice, and in test tube B, keep one teaspoon full of boiled rice after chewing it for 3 to 5 minutes. Add 3.4 mL of water in both test tubes and add a few drops of iodine solution to each test tube. In test tube A, the colour of the rice changes because of the absence of any enzyme, while in test tube B, no colour change is observed because when rice is chewed, the amylase enzyme present in saliva breaks down the starch of rice to simple sugars.

(b) Bile helps in emulsification of fat; bile juice breaks down fats into fatty acids that can be taken into the body by the digestive tract. In addition, it provides alkaline fluid in the intestine to neutralize the acidic pH of food that comes from the stomach.

59. (i) Double circulation is a mechanism in which blood circulates twice through the heart in one complete cycle. Systemic circulation and pulmonary circulation are two pathways through which blood flows in double circulation.



- (ii) The separation of the right and left sides of the heart helps in the separation of oxygenated and deoxygenated blood and allows the highly efficient supply of oxygen to the body. Birds and mammals are warm-blooded animals, and their metabolism is quite complex. They require a constant supply of oxygen. Without proper separation, oxygenated blood can mix with deoxygenated blood, which can cause problems in birds and mammals. Separation of the right side and left side of the heart ensures the circulation of oxygenated blood to different body parts.
60. (i) From minerals and water needed by the plants are absorbed by roots. The root hairs absorb water from the soil by the process of osmosis and take in minerals by the process of diffusion. Thus, a difference in concentration of ions is created between the roots and the soil which enables the water to enter into the roots to compensate for the difference in concentration. This water, along with dissolved minerals from the root hair, passes into xylem vessels through cells of the cortex, endodermis, and pericycle, and then ascent of sap takes place from the xylem of roots into the xylem of stem.
- (ii) Bleeding leads to a loss of pressure, which reduces the efficiency of the pumping system. When an injury is caused, the blood platelets release certain chemicals which are called the platelet factors e.g. thromboplastin. These platelet factors help in the clotting of blood.
61. The small intestine is the main region for the absorption of digested food. It consists of the following parts: The inner surface of the small intestine has millions of tiny finger-like projections called villi. These villi increase the surface area for efficient food absorption. Within these villi, many blood vessels are present that absorb the digested food and carry it to the bloodstream. The small intestine in human beings is the site of the complete digestion of food like carbohydrates, proteins, and fats. The absorption of digested food is necessary because an organism takes the complex organic materials by the process of absorption and then utilizes absorbed nutrients for various metabolic processes. With the help of these metabolic processes, the body uses these food nutrients for energy, growth, and cell repair.
62. (a) (i) **Enzyme trypsin:** This enzyme is produced by the pancreas in an inactive form called trypsinogen. Trypsin converts the remaining proteins into peptones, and the peptones into peptides and amino acids.
- (ii) **Enzyme lipase:** It is secreted by the pancreas and small intestine. Lipase converts fats into fatty acids and glycerol.
- (b) Internally, the wall of the small intestine is provided with long finger-like projections called villi. Two functions of villi are:
- (i) The villi greatly increase the absorptive surface area of the inner lining of the small intestine.
- (ii) Villi are richly supplied with blood vessels that carry absorbed food to all cells of the body, where it is utilized for obtaining energy.
63. (a) Since ventricles have to pump blood into various organs with high pressure, they have thicker walls than atria.
- (b) The transport system in plants is less elaborate than in animals, as plants are less active, so their cells do not need to be supplied with materials quickly.
- (c) The aquatic vertebrates, like fish, have gills to oxygenate blood. The flow of blood in a fish is single circulation because the blood passes through the heart only once in one complete cycle of the body. Terrestrial vertebrates, like birds and humans, have double circulation as the blood travels to the heart twice in one complete cycle of blood, and they have lungs for oxygenation of blood.
- (d) During the day time, water and minerals travel faster through the xylem because the rate of transpiration is higher during the day time.
- (e) The lumen of veins has valves, which allow the blood to flow in only one direction. Thus preventing the backflow of blood.
64. (a) Fishes have only two chambers in their heart. The blood is pumped to the gills for oxygenation and from there it passes directly to the rest of the body. Thus, the blood goes only once through the heart during one cycle of passage through the body. This type of circulation is termed as single circulation. In human beings, during circulation, blood travels twice through the heart in one complete cycle of the body, which is called double circulation. The

pathway of blood from the heart to the lung and back to the heart is called pulmonary circulation, and the pathway of blood from the heart to the rest of the body and back to the heart is called systemic circulation.

(b) In human being, deoxygenated blood from the body tissues is poured into the right atrium.

Contraction of the heart forces it into the right ventricle. From the right ventricle, deoxygenated blood goes to the lungs through the pulmonary artery. Oxygenated blood from the lungs is returned to the left atrium and then into the left ventricle via pulmonary vein. The left ventricle forces the oxygenated blood to the whole body. Thus, to make one complete round or circulation circuit around all body parts, the blood passes through the heart twice. This is known as the double circulation of blood.

65. (a) The excretory system (urinary system) in human beings consists of a pair of kidneys, a pair of ureters, a urinary bladder and a urethra.

(b) In the kidney, the wastes are converted to urine by three processes:

(i) **Ultrafiltration:** A large amount of water, along with certain harmful substances like urea, uric acid, K^+ , ammonium salts, creatinine, etc., and certain useful substances like glucose, amino acids, Na^+ , etc. pass through glomerular capillaries and glomerular membrane into the cavity of Bowman's capsule of nephrons under pressure. The filtrate so formed is called nephric filtrate, which is moved towards the ureter.

(ii) **Selective reabsorption:** A large amount of water and sodium, a whole of glucose and amino acids, and a small amount of urea are passed back from the nephric filtrate into the blood capillaries. It occurs either by back diffusion (i.e., water and urea) or active transport (i.e., Na^+ , glucose, and amino acids). It generally occurs in the PCT (Proximal convoluted tubule) of nephrons.

(iii) **Tubular secretion:** Certain harmful chemicals like uric acid, creatinine, K^+ , etc., are passed from blood capillaries surrounding the nephron into nephric filtrate by active transport. It generally occurs in the DCT (Distal convoluted tubule) of nephrons. Now, the fluid is termed as urine and is excreted out of the excretory organs.

66. (a) Nutrition is necessary for the human body because the human body continuously requires energy for life activities like respiration, circulation, excretion, etc. Energy is required even when we are sleeping because a number of biological processes keep on occurring. All these processes require energy, and this energy is obtained from nutrition. It is also needed for the growth and repair of the human body.

(b) The wall of the alimentary tract contains muscles that can contract and expand alternatively. The contraction and expansion movements of the wall of the food pipe is called peristaltic movement. The peristaltic movement moves the partially digested food in all the digestive organs throughout the alimentary canal.

(c) The small intestine in herbivores is longer than in carnivores because herbivores eat plants, rich in cellulose, which takes longer time for complete digestion by the enzymes present in symbiotic bacteria. Therefore, they have longer small intestines. Carnivores feed on flesh, which is easier to digest and does not contain cellulose. Therefore, they have shorter intestines for the digestion of food eaten by them.

(d) Gastric glands secrete HCl, mucus, rennin, and pepsin enzymes. The mucus protects the inner lining of the stomach from the action of acidic HCl and enzymes. In the absence of mucus, there would be erosion of the inner lining of the stomach leading to acidity and ulcers.

67. (a) Salivary glands; Starch/Carbohydrate

(b) Stomach, Anus

(c) (i) The inner lining of the stomach will not be protected from the action of acid.

(ii) Digested food will not be absorbed. / Absorption area will be reduced

OR

Emulsification of fats.

Acidic medium has to be made alkaline for the pancreatic enzymes to act.

Case-Based type Questions

68. (i) (c) The semi-permeable lining of the tubes of haemodialyser allow the passage of waste products urea, creatinine, etc, from the blood to the dialysing fluid.

(ii) (c) Artificial kidney is not involved in the reabsorption of essential nutrients from the blood.



- (iii) (a) The used dialysing fluid/solution is rich in waste products like urea and excess salts because during the passage of patient's blood through tubes with a semi-permeable lining, the waste products from the blood pass into the dialysing fluids by diffusion.
- (iv) (c) The function of tubules of nephron is to allow the selective reabsorption of useful substances like glucose, amino acids, salts and water into the blood capillaries.

2

CHAPTER

Control and Coordination



Previous Year Questions

Multiple Choice Type Questions

- In a nerve cell, the site where the electrical impulse is converted into a chemical signal is known as: **[CBSE 2024]**
 - Axon
 - Dendrites
 - Neuromuscular junction
 - Cell body
- Select out of the following a gland which does NOT occur as a pair in the human body : **[CBSE 2024]**
 - Pituitary
 - Ovary
 - Testis
 - Adrenal
- A plant growth inhibitor hormone which causes wilting of leaves is called : **[CBSE 2024]**
 - Auxin
 - Cytokinin
 - Absciscic acid
 - Gibberellin
- The correct sequence of events when someone's hand touches a hot object unconsciously : **[CBSE 2024]**
 - Receptors in skin → Motor neuron → Relay neuron → Sensory neuron → Effector muscle in arm
 - Receptors in skin → Relay neuron → Sensory neuron → Motor neuron → Effector muscle in arm
 - Receptors in skin → Sensory neuron → Relay neuron → Motor neuron → Effector muscle in arm
 - Receptors in skin → Sensory neuron → Effector muscle in arm → Motor neuron → Relay neuron
- Sense organ in which olfactory receptors are present is : **[CBSE 2024]**
 - Nose
 - Skin
 - Tongue
 - Inner ear
- Select from the following the correct statement about tropic movement in plants : **[CBSE 2023]**
 - It is due to stimulus of touch and temperature.
 - It does not depend upon the direction of stimulus received.
 - It is observed only in roots and not in stems.
 - It is a growth related movement.

- In plants the role of cytokinin is : **[CBSE 2023]**
 - Promote cell division.
 - Wilting of leaves.
 - Promote the opening of stomatal pore.
 - Help in the growth of stem.
- The part in which gustatory receptors are present in our body is : **[CBSE 2023]**
 - inner ear
 - skin
 - tongue
 - inner lining of nose
- Walking in a straight line and riding a bicycle are the activities which are possible due to a part of the brain. Choose the correct location and name of this part from the given table : **[CBSE 2023]**

	Part of the Brain	Name
(a)	Fore brain	Cerebrum
(b)	Mid brain	Hypothalamus
(c)	Hind brain	Cerebellum
(d)	Hind brain	Medulla

- Sensory nerve of a reflex arc carries information from the receptor cells to the : **[CBSE 2023]**
 - spinal cord
 - brain
 - muscles of the effector organ
 - bones of the receptor organ

Very Short Type Questions

11. (a) How is the movement of leaves of a sensitive plant different from the downward movement of the roots ?

OR

- (b) There is a hormone which regulates carbohydrate, protein and fat metabolism in our body. Name the hormone and the gland which secretes it. Why is it important for us to have iodised salt in our diet ? **[CBSE 2024]**
12. Where are auxins synthesized ? How do they promote phototropism ? **[CBSE 2024]**
13. Name the part of brain which is responsible for the following actions: **[CBSE 2023]**
- (i) Maintaining posture and balance
 - (ii) Beating of heart
 - (iii) Thinking
 - (iv) Blood pressure

Short Answer Type Questions

14. Define reflex action. With the help of a flow chart show the path of a reflex action such as sneezing. **[CBSE 2024]**
15. (a) How is the brain and spinal cord protected in human beings ?
- (b) State one main function of
- (i) Medulla and
 - (ii) Cerebellum. **[CBSE 2023]**
16. Where are auxins synthesised in a plant? Which organ of the plant shows:
- (i) Positive phototropism
 - (ii) Negative geotropism
 - (iii) Positive hydrotropism **[CBSE 2023]**
17. Name a plant hormone responsible for bending of a shoot of a plant when it is exposed to unidirectional light. How does it promote phototropism ? **[CBSE 2023]**
18. Define geotropism. Draw a labelled diagram of a plant showing geotropic movements of its parts. **[CBSE 2020]**

Long Answer Type Questions

19. (a) (i) Define a reflex arc. Why have reflex arcs

evolved in animal? Trace the sequence of events which occur, when you suddenly touch a hot object.

(ii) Name the part of nervous system which helps in communication between the central nervous system and other parts of the body. What are the two components of this system?

OR

(b) (i) Leave of 'chhui-mui' plant begin to fold up and droop in response to a stimulus. Name the stimulus and write the cause for such a rapid movement. Is there any growth involved in the movement?

(ii) Define geotropism in plants. What is meant by positive and negative geotropism? Give one example of each type. **[CBSE 2024]**

Case Study

Thyroid Stimulating Hormone (TSH) stimulates thyroid gland to produce thyroxine.

Table: TSH levels during pregnancy

Stage of pregnancy	Normal (mU/L)	Low (mU/L)	High (mU/L)
First trimester	0.2- 2.5	< 0.2	2.5 - 10
Second trimester	0.3 – 3.0	< 0.3	3.01 – 4.5
Third trimester	0.8 – 5.2	< 0.8	> 5.3

It is important to monitor TSH levels during pregnancy. High TSH levels and hypothyroidism can especially affect chances of miscarriage. Therefore, proper medication and consultation with a doctor is required to regulate/control the proper functioning of the thyroid gland.

- (i) Give the full form of TSH.
 - (ii) State the main function of TSH.
 - (iii) Why do TSH levels in pregnant women need to be monitored?
 - (iv) A pregnant woman has TSH level of 8.95 mU/L. What care is needed for her? **[2020]**
2. Answer question numbers (a) to (d) on the basis of your understanding of the following information and related studied concepts.

Thyroid gland is a bilobed structure situated in our neck region. It secretes a hormone called thyroxine. Iodine is necessary for the thyroid gland to make thyroxine. Thyroxine regulates carbohydrate, protein and fat metabolism in the body. It promotes growth of body tissues also. When there is an excess of thyroxine in the body, a person suffers from

hyperthyroidism and if this gland is underactive it results in hypothyroidism. Hyperthyroidism is diagnosed by blood tests that measure the levels of thyroxine and Thyroid Stimulating Hormone (TSH). Hypothyroidism is caused due to the deficiency of iodine in our diet resulting in a disease called goitre. Iodised salt can be included in our diet to control it.

- (a) Where is thyroid gland situated in our body ?
- (b) State the function of thyroxine in human body.
- (c) What is hyperthyroidism ?
- (d) How can we control hypothyroidism ? **[2020]**

3. Questions numbers (a) to (d) are based on table given below.

Study the table in which the levels of Thyroid Stimulating Hormone (TSH) in women are given and answer the questions that follow on the basis of understanding of the following paragraph and the related studied concepts.

Age Range	Normal (mU/L)	Low (mU/L)
18-29 years	0.4-2.34 mU/L	< 0.4 mU/L

30-49 years	0.4-4.0 mU/L	< 0.4 mU/L
50-79 years	0.46-4.68 mU/L	< 0.46 mU/L

Women are at greater risk for developing abnormal TSH levels during menstruation, while giving birth and after going through menopause. Around 5% of women in the United States have some kind of thyroid problem compared to 3% of men. Despite claims that high TSH increases your risk for heart disease, a 2013 study found no link between high TSH and heart diseases. But a 2017 study showed that older women are especially at risk for developing thyroid cancer if they have high TSH levels along with thyroid nodules.

- (a) A 35 year old woman has TSH level 6.03 mU/L. What change should she bring in her diet to control this level ?
- (b) When do women face a greater risk of abnormal TSH level ?
- (c) State the consequence of low TSH level.
- (d) Name the mineral that is responsible for synthesis of hormone secreted by thyroid gland. **[CBSE 2020]**



Explanations

Multiple Choice Type Questions

1. (c) This is the place where the electrical impulse traveling along a nerve cell is converted into a chemical signal to communicate with a muscle cell. At this junction, neurotransmitters are released, allowing the nerve cell to signal the muscle to contract.
2. (a) The pituitary gland is a single gland located at the base of the brain and is often referred to as the “master gland” because it controls many other hormone glands in the body. In contrast, the ovaries, testis, and adrenal glands occur in pairs (one on each side of the body).
3. (c) Absciscic acid is a plant hormone that acts as a growth inhibitor and helps plants respond to stress, such as drought. It causes the stomata to close, reducing water loss and leading to wilting of leaves when the plant is under stress.
4. (c) Receptors in skin → Sensory neuron → Relay neuron → Motor neuron → Effector muscle in arm.

When you touch something, hot, the receptors in your skin detect the heat and send a signal through sensory neurons to the relay neurons in the spinal cord. Then, relay neurons send the signal to motor neurons, which activate the effector muscles in your arm to pull away quickly, helping you avoid injury.
5. (a) Olfactory receptors, which are responsible for the sense of smell, are located in the nasal cavity of the nose. When you inhale, these receptors detect different scents and send signals to the brain, allowing you to identify various smells.
6. (d) Tropic movements depend upon the direction of stimulus received.
7. (a) Cytokinins primarily promotes cell division.
8. (c) Gustatory receptors are present on tongue.
9. (c) Cerebellum controls posture and balance of the body.
10. (a) Reflex arc is pathway taken by nerve impulses and responses in a reflex action, i.e., from receptor organs like skin to spinal cord and from spinal cord to effector organs like muscles.

Very Short Type Questions

11. (a)

	Movement of leaves of sensitive plant	Downward movement of roots
(i)	Stimulus is touch.	Stimulus is gravity.
(ii)	No growth is involved in the movement.	Growth is involved in the movement
(iii)	Non directional	Directional

OR

(b) Thyroxine

Thyroid gland

Iodine is necessary for thyroid gland to make thyroxine hormone. Deficiency of iodine in our diet causes goitre.

12. (i) Synthesized at shoot tip/root tip

(ii) When light falls on one side of the plant, auxin diffuses towards the shady side of shoot. The concentration of auxin stimulates the cells to grow longer on the side of shoot which is away from light. Thus plant appears bent towards light/phototropism.

13. (i) Cerebellum

(ii) Medulla oblongata

(iii) Cerebellum

(iv) Medulla oblongata

Short Answer Type Questions

14. Reflex action is a sudden/spontaneous/immediate action in response to the environment/stimulus e.g. sneezing.

Stimulus → Receptor (Nose) → Sensory neuron → Spinal cord (Relay neuron) → Motor neuron → Effector (Muscles) → Response

15. (a) Brain and spinal cord are protected by skeleton-brain by the cranium and spinal cord by vertebral column.

(b) (i) Medulla- It controls involuntary actions like heart beat, respiration, etc.



(ii) Cerebellum- It maintains posture and balance of body.

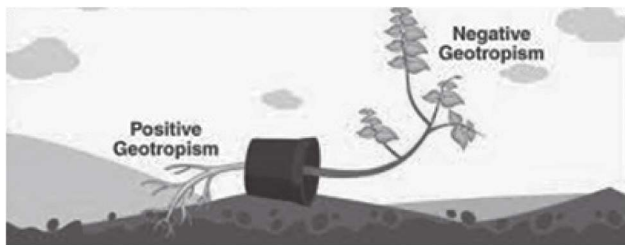
16. Auxin is synthesised at tip of the stem apices or shoots.

(i) Shoots

(ii) Shoots

(iii) Roots (bend towards water)

17. Auxin is responsible for bending of a shoot of a plant when it is exposed to unidirectional light. In the presence of sunlight, the auxin present in the stem starts accumulating in the region away from sunlight, due to which the concentration of auxin in the region of the stem increases, resulting in bending of a shoot of a plant when it is exposed to unidirectional light. Thus, auxin promote phototropism that is the regulation of physiology or development in response to day length.
18. The movement of the growth of the roots downwards and the shoots upwards under the stimuli of gravity is called geotropism. The movement of the part of the plant towards or away from the stimulus gravity.



Long Short Type Questions

19. (a) (i) The pathway in which impulses travel during the reflex action is called a reflex arc.

Because the thinking part of the brain is not fast enough/for quick response to avoid injury.

Reflex arc:

Hot Plate (stimulus) → Receptor (skin) → Sensory neuron → Spinal cord (Relay neuron) → Motor neuron → Effector (Muscles) → Response

(ii) Peripheral Nervous System

Components: Cranial Nerves; Spinal Nerves

OR

(b) (i) Touch

The shape of the leaves changes by changing the amount of water in them.

No, there is no growth involved in this type of movement.

Case Study

- (i) Thyroid stimulating hormone (TSH).
(ii) TSH from pituitary gland stimulates thyroid gland to release thyroxine.
(iii) Thyroxine regulates basal metabolism which is very crucial at the time of pregnancy as maternal body feeds the child. Thyroxine also inhibits TSH through negative feedback. Hence, TSH levels in blood are in turn regulated by Thyroxine which is dependent on iodine. If the level of thyroxine in blood is low and TSH is high, the chances of miscarriage increases.
So, to protect the developing foetus, TSH level must be maintained.
(iv) If TSH levels are high, one of the possible reason is less production of thyroxine which might be due to poor levels of iodine in her diet. She should look at her salt consumption as salt is a good source of iodine.
- (a) In the neck region
(b) Thyroxine regulates carbohydrate, proteins and fat metabolism in the body./ It promotes growth of body tissue.
(c) Excess of secretion of thyroxine in the body / overactivity of the thyroid gland
(d) Can be controlled by including iodised salt in our diet.
- (a) She should monitor iodine intake in her diet.
(b) During menstruation / during pregnancy and after going through menopause. (any two)
(c) Low TSH level leads to swelling of neck region / disease called goitre.
(d) Iodine

3

CHAPTER

How Do Organisms Reproduce?



Previous Year Questions

Multiple Choice Type Questions

1. Identify the mode of asexual reproduction in the following organism: [2024]



- (a) Fragmentation
(b) Multiple fission
(c) Budding
(d) Binary fission
2. Part(s) of a flower which attracts insects for pollination is (are) [2024]
(a) petals and sepals.
(b) anther and stigma.
(c) petals only.
(d) sepals only.
3. The plants that can be raised by the method of vegetative propagation are: [2024]
(a) Sugarcane, roses, grapes
(b) Sugarcane, mustard, potato
(c) Banana, orange, mustard
(d) Papaya, mustard, potato
4. A zygote is formed by the fusion of a male gamete and a female gamete. The number of chromosomes in the zygote of a human is: [2024]
(a) 23 (b) 44

(c) 46 (d) 92

5. The incorrect statement about placenta is: [2024]
(a) It is a disc embedded in the uterine wall.
(b) It contains villi on the embryo's side of the tissue.
(c) It has a very small surface area for glucose and oxygen to pass from mother to the embryo.
(d) The embryo gets nutrition from the mother's blood through it.
6. Chromosomes:
(i) carry hereditary information from parents to the next generation.
(ii) are thread like structures located inside the nucleus of an animal cell.
(iii) always exist in pairs in human reproductive cells.
(iv) are involved in the process of cell division. [2024]

The correct statements are :

- (a) (i) and (ii)
(b) (iii) and (iv)
(c) (i), (ii) and (iv)
(d) (i) and (iv)
7. Assertion (A) and Reason (R), answer these questions selecting the appropriate option given below: [2024]
Assertion (A): Offsprings produced by asexual reproduction are genetically similar to the parents.
Reason (R): Asexual reproduction involves a single parent.
(a) Both are true and is the correct explanation of .
(b) Both (A) and (R) are true and is not correct explanation of .

- (c) (A) is true, but is false.
(d) (A) is false, but (R) is true.

Very Short Type Questions

8. Define fragmentation. [2021 C]
9. After examining a prepared slide under the high power of a compound microscope, a student concludes that the given slide shows the various stages of binary fission in a unicellular organism. Write two observations on the basis of which such a conclusion may be drawn. [2019]
10. Define multiple fission. Give its one example. [2019, Foreign 2014]
11. Newly formed DNA copies may not be identical at times. Give one reason. [NCERT, AI 2017]
12. When a cell reproduces, what happens to its DNA? [AI 2017]
13. Name the method by which *spirogyra* reproduces under favourable conditions. Is this method sexual or asexual? [Delhi 2017]
14. How does *plasmodium* reproduce? Is this method sexual or asexual? [Delhi 2017]

Short Answer Type Questions

15. What are sexually transmitted diseases? List two examples each of diseases caused due to (i) bacterial infection and (ii) viral infection. Which device or devices may be used to prevent the spread of such diseases? [2019]
16. Draw labelled diagram to show the following parts in an embryo of a pea seed: Cotyledon, plumule, radicle. [2019]
17. Reproduction is one of the most important characteristic of living beings. Give three reasons in support of the statement. [AI 2017]
18. State the basic requirement for sexual reproduction. Write the importance of such reproductions in nature. [Delhi 2017]
19. List any two steps involved in sexual reproduction and write its two advantages. [Delhi 2017]

20. State the changes that take place in the uterus when:
 - (a) Implantation of embryo has occurred.
 - (b) Female gamete/egg is not fertilised.

[Delhi 2017]

21. What is vegetative propagation? State two advantages and two disadvantages of this method.

[AI 2017]

22. Describe reproduction by spores in *Rhizopus*.

[AI 2017]

Long Answer Type Questions

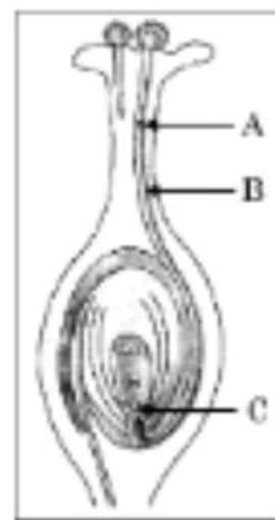
23. (a) (i) What are spores? On which structures are they formed? How do they overcome unfavourable conditions? Name the organism which multiplies with the help of these structures.

[2024]

- (ii) Give two reasons why some plants are grown by the method of vegetative propagation. List two methods used to grow plants vegetatively.

OR

- (b) (i) Study the diagram given below and name the parts marked as A, B and C. What happens when B reaches C in the ovary? Mention its significance.



- (ii) Write the post fertilisation changes that occur in a flower.
24. (i) What happens when:
 - (1) Leaves of *Bryophyllum* fall on the soil?
 - (2) *Planaria* is cut into many pieces?
 - (3) Sporangia of *Rhizopus* on maturation liberate spores?

Mention the modes of reproduction in each of the above three cases.

- (ii) Write the changes that occur in a flower once the fertilization has taken place. **[2023]**

25. Name the part/organ of the human female reproductive system **[2022]**

- where contraceptive devices such as loop or copper-T are placed to prevent pregnancy.
- which is blocked to prevent the transfer of eggs.
- where formation of germ cells as ova takes place.
- from where the embryo gets nutrition from the mother's blood.

26. Mention the changes that occur in the following after fertilisation in a flower:

- Petals
- Zygote
- Ovary
- Ovule **[Term-11,2021 C]**

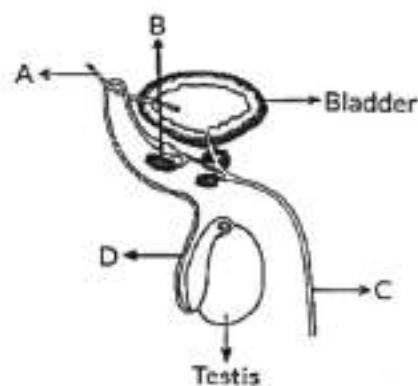
27. (a) List three different categories of contraceptive methods.

- Why has Government of India prohibited prenatal sex determination by law? State its benefits in the long run.
- Unsafe sexual act can lead to various infections. Name two bacterial and two viral infections caused due to unsafe sex. **[2020]**

28. The growing size of the human population is a cause of concern for all people. The rate of birth and death in a given population will determine its size. Reproduction is the process by which organisms increase their population. The process of sexual maturation for reproduction is gradual and takes place while general body growth is still going on. Some degree of sexual maturation does not necessarily mean that the mind or body is ready for sexual acts or for having and bringing up children. Various contraceptive devices are being used by human beings to control the size of population.

- List two common signs of sexual maturation in boys and girls.
- What is the result of reckless female feticide?
- Which contraceptive method changes the hormonal balance of the body?
- Write two factors that determine the size of a population. **[2020]**

29. Based on the given diagram answer the questions given below:



- Label the parts A, B, C and D.
- Name the hormone secreted by testis and mention its role.
- State the functions of and in the process of reproduction. **[2020]**

30. (a) What is puberty?

- Describe in brief the functions of the following parts in the human male reproductive system.

- Testes
- Seminal vesicle
- Vas deferens
- Urethra
- Why are testes located outside the abdominal cavity?
- State how sperms move towards the female germ cell. **[2020]**

31. (a) Identify the modes of asexual reproduction in each of the following organisms:

- Hydra*
- Planaria*
- Amoeba*
- Spirogyra*
- Rhizopus*

- List three advantages of vegetative propagation.
- Why cannot fertilisation take place in flowers if pollination does not occur? **[2020]**

32. Draw a neat diagram showing fertilisation in a flower and label

- pollen tube
- Male germ cell and
- Female germ cell on it.

Explain the process of fertilisation in a flower.

What happens to the (i) ovary and (ii) ovule after fertilisation? **[2020]**

33. (a) Name the mode of reproduction of the following organisms and state the important feature of each mode:

(i) *Planaria*

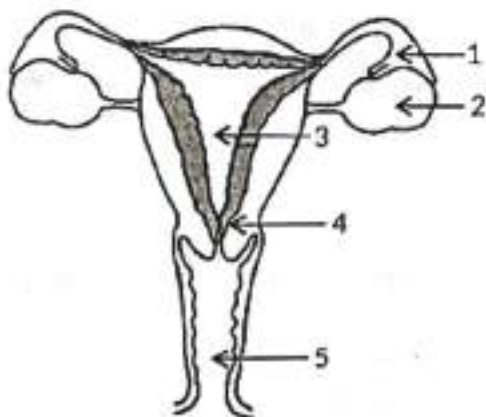
(ii) *Hydra*

(iii) *Rhizopus*

(b) We can develop new plants from the leaves of *Bryophyllum*. Comment.

(c) List two advantages of vegetative propagation over other modes of reproduction. [2020]

34. (a) Identify the given diagram. Name the parts 1 to 5



(b) What is contraception? List three advantages of adopting contraceptive measures.

[NCERT Exemplar, Delhi 2019]

35. (a) Draw a diagram of human female reproductive system and label the parts:

(i) which produce an egg

(ii) where fertilisation takes place

(b) List two bacterial diseases which are transmitted sexually.

(c) What are contraceptive devices? Give two reasons for adopting contraceptive devices in humans.

[AI 2019]

36. Define pollination. Explain the different types of pollination. List two agents of pollination. How does suitable pollination lead to fertilisation?

[Delhi 2019]

37. (a) Name the organ that produces sperms as well as secretes a hormone in human males. Name the hormone it secretes and write its functions.

(b) Name the parts of the human female reproductive system where fertilisation occurs.

(c) Explain how the developing embryo gets nourishment inside the mother's body.

[Delhi 2017]

38. (a) What is variation? How is variation created in a population? How does the creation of variation in a species promote survival?

(b) Explain how, offspring and parents of organisms reproducing sexually have the same number of chromosomes. [CBSE 2017-18 C]

39. List three techniques that have been developed to prevent pregnancy. Which one of these techniques is not meant for males? How does the use of these techniques have a direct impact on the health and prosperity of a family?

[NCERT Exemplar, AI 2017]



Explanations

Previous Year Questions

Multiple Choice Questions

1. (c) In the organism shown, you can observe a small new organism growing from the body of the larger one.

This is typical example of budding, where a new organism develops from an outgrowth or bud due to cell division at one particular site. This method is common in yeast and some invertebrates like *hydras*.

2. (c) Petals are often brightly colored and fragrant, which helps attract insects for pollination. They serve as visual cues that guide insects to the flower, encouraging them to visit and transfer pollen from one flower to another, aiding in plant reproduction.
3. (a) Vegetative propagation is a method of asexual reproduction where new plants are grown from parts of existing plants, such as stems, roots, or leaves. Sugarcane, roses, and grapes can be easily propagated using this method, allowing for the rapid production of new plants with desirable traits.
4. (c) In humans, a zygote is formed when a male gamete (sperm) and a female gamete (egg) fuse together. Each gamete contains 23 chromosomes, so when they combine, the zygote has a total of 46 chromosomes, which is the normal number for human cells.
5. (c) This statement is incorrect because the placenta actually has a large surface area, which allows for efficient transfer of nutrients like glucose and oxygen from the mother's blood to the embryo. The placenta is designed to maximize the exchange of these essential substances to support the developing fetus.
6. (d)

Statement (i) is true because chromosomes carry genetic information from parents to their offspring, determining traits. Statement (iv) is also true as chromosomes play a crucial role in cell division,

ensuring that genetic material is accurately copied and distributed. However, statement (iii) is incorrect because human reproductive cells (sperm and egg) have only one set of chromosomes, not pairs.

7. (a)

Offsprings produced by asexual reproduction are genetically identical to their parent because asexual reproduction involves only one parent, resulting in no mixing of genetic material. This is why the assertion about the genetic similarity of offspring and the reason about single-parent involvement are both true and directly related.

Very Short Type Questions

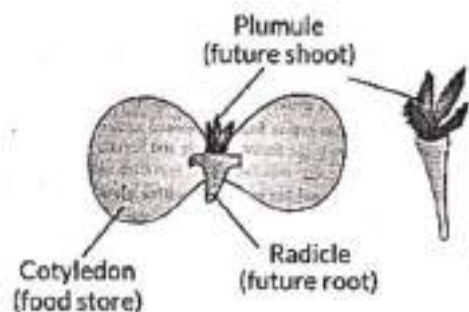
8. Fragmentation is the mode of reproduction in which parent body breaks into two or more fragments and each fragment develops into a new individual. E.g., *Spirogyra*.
9. The two observations that was taken by the student that concludes that the given slide shows the various stages of binary fission in a unicellular organism are as follows:
Presence of unicellular and uninucleate organism showing irregular outline.
Division of nucleus of parent cell into two equal halves.
10. Multiple fission is an asexual mode of reproduction in which the parent organism splits to form many new organisms at the same time. Multiple fission occurs in *Plasmodium*.
11. When a cell reproduces, DNA replication occurs which results in formation of two similar copies of DNA. The process of copying the DNA leads to some variations each time. As a result, the DNA copies produced are similar to each other but sometimes may not be identical.
12. When a cell reproduces, DNA replication occurs which forms two similar copies of DNA.
13. The method by which *Spirogyra* reproduces under favorable conditions is fragmentation. This is an

asexual mode of reproduction.

14. *Plasmodium* reproduces through multiple fission method. In this method, the parent organism splits to form many new organisms at the same time. This is an asexual method of reproduction.

Short Answer Type Questions

15. The diseases that are spread by sexual contact with an infected person are called sexually transmitted disease (STDs).
- (i) Bacterial infection causes gonorrhoea, syphilis.
 - (ii) Viral infection causes AIDS, genital herpes. STDs can be prevented by using male and female condoms.
16. Embryo of pea seed is shown as follows:



17. Reproduction is one of the most important characteristics of living beings because:
- (i) It is essential for existence and continuity of a species.
 - (ii) It helps to pass genetic information to next generation.
 - (iii) It brings variations in next generation which is the basis for evolution.
18. The basic requirement for sexual reproduction is involvement of both sexes, i.e., male and female, to produce an offspring. It takes place by the combination of gametes which come from two different parents.
- The importance of sexual reproduction in nature are:
- (i) Fusion of male and female gametes coming from two different and sexually distinct individuals, exhibit diversity of characters in offspring.
 - (ii) Meiosis during gametogenesis provides opportunities for new combination of genes, which leads to variation required for evolution and plays a prominent role in the origin of new

species. Variations lead to the appearance of such characters, which fit to the changing environment, resulting in the survival of the species.

19. The two main steps involved in sexual reproduction are:
- (i) Formation of male and female gametes.
 - (ii) Fusion of a male gamete with a female gamete to form a new cell called zygote by the process of fertilisation.
- The two important advantages of sexual reproduction are:
- (i) It promotes diversity of characters in the offspring through genetic variations.
 - (ii) It plays an important role in continuous evolution of better organisms that may lead to the origin of new species.

20. (a) Implantation is the close attachment of the blastocyst (young multicellular embryo) to the uterine wall. It is followed by a number of developmental changes in the thickened wall of uterus. An intimate connection between the fetal membrane and the uterine wall called placenta is formed. This is a disc which is embedded in the uterine wall. The placenta serves as the nutritive, respiratory and excretory organ of the fetus.
- (b) When the female gamete/egg is not fertilised, this lining is not needed any longer. So, the lining slowly breaks and comes out through vagina as blood and mucus. This cycle takes place every month and is known as menstrual cycle.
21. Vegetative propagation is a type of asexual reproduction in which the plant parts other than seeds are used as a propagule.

Advantages of vegetative propagation:

- (i) Desirable character of the plant can be preserved through generation.
- (ii) Seedless plants can be grown via this method.

Disadvantages of vegetative propagation:

- (i) Plants produced by this method possess less vigour and are more prone to diseases.
- (ii) Plants produced by this method show no genetic variation.

22. Fungus *Rhizopus* reproduces by spore formation. During the growth of *Rhizopus*, small rounded, bulb-like structures develop at the top of the erect hyphae. Such structures are called sporangia. Inside each sporangium, : nucleus divides several times.

Each nucleus gets surrounded by a little amount of cytoplasm to become spore. Large number of spores are formed inside each sporangium. After sometime sporangium bursts and spores are released in the air. When these spores land on food or soil, under favourable conditions, they germinate into new individuals.

Long Answer Type Questions

23. (a) (i)

- Spores are reproductive structures that detach from the parent and give rise to a new individual.
- Sporangium / Sporangia
- Covered by thick walls to protect them from unfavourable conditions.
- *Rhizopus*

(ii)

- Plants which have lost the capacity to produce seeds.
- Plants bear flowers and fruits earlier so as to reduce time.
- To get genetically similar plants.

Methods: Layering and Grafting

OR

(b) (i)

- A - Male Germ Cell/Male Gamete; B - Pollen tube; C - Female Germ Cell / Female Gamete.
- B carries male germ cell and this germ cell fuses with C (female germ cell) to form a zygote.
- **Significance:** Zygote is capable of growing into a new plant.

(ii) Fertilization is the process of the fusion of the male and the female gamete to form a diploid zygote. After fertilization, the zygote divides several times to form an embryo. The fertilized ovule forms a seed. The seed contains an embryo, enclosed in a protective covering, called the seed coat. As the seed grows further, other floral parts wither and fall off. This leads to the growth of the ovary, which enlarges and ripens to become a fruit with a thick wall called the pericarp.

24. (i) (1) When the *bryophyllum* leaf falls on the wet soil, the buds present in the notches along

the leaf margin develop into new plants. This is one of the example of vegetative propagation by leaves.

- (2) When *planaria* accidentally cut into many pieces then its each piece grows into a complete organism. This is one of the example of regeneration.
- (3) The sporangia of *Rhizopus* contain cells or spores that can eventually develop into new *Rhizopus* individuals when it bursts on maturation. *Rhizopus* reproduce asexually by the formation of the spores (sporulation).
- (ii) After fertilization, the fertilized egg (or zygote) divides several times to form an embryo within the ovule. The ovule develops a tough coat around it and is gradually converted into a seed. The ovary of flower develops and becomes a fruit (with seeds inside it). The other parts of flower like sepals, petals, stamens, stigma and style dry up and fall off. Only the ovary is left behind. So, at the place on plant where we had a flower originally, we now have a fruit (which is the ovary of the flower containing seeds). A fruit protects its seeds.

25. (a) The part/organ of the human female reproductive system where contraceptive devices such as loop or copper-T are placed to prevent pregnancy is the uterus or womb. These devices are inserted into the uterus to inhibit fertilization and implantation of a fertilized egg.
- (b) The part/organ of the human female reproductive system that is blocked to prevent the transfer of eggs is the fallopian tubes. These tubes connect the ovaries to the uterus, and they are blocked or sealed off in certain contraceptive methods or sterilization procedures to prevent the eggs from reaching the uterus for fertilization.
- (c) The part/organ of the human female reproductive system where the formation of germ cells as ova takes place is the ovaries. Ovaries are responsible for producing and releasing mature eggs, or ova, during the menstrual cycle.
- (d) The part/organ of the human female reproductive system from where the embryo gets nutrition from the mother's blood is the placenta. The placenta is a temporary organ that develops during pregnancy and attaches to the uterine wall. It facilitates the exchange of nutrients, oxygen, and waste products



between the mother's blood and the developing fetus.

26. (a) Petals: After fertilization, the petals of the flower may start to wither and fall off. This is because their primary function of attracting pollinators has been fulfilled, and they are no longer needed for reproduction.
- (b) Zygote: After fertilization, the zygote is formed. The zygote undergoes mitotic divisions and develops into an embryo. It is the beginning of a new plant's life cycle.
- (c) Ovary: After fertilization, the ovary develops into a fruit. The ovary wall thickens and matures, protecting the developing seeds inside. The fruit helps in the dispersal of seeds and ensures the survival of the next generation.
- (d) Ovule: After fertilization, the ovule develops into a seed. The ovule contains the fertilized egg, which develops into an embryo. The ovule undergoes changes in structure and becomes the seed coat, which protects the embryo and provides nutrients for its growth and development.
27. (a) Three different categories of contraceptive methods are:
- (i) Barrier methods, i.e., use of condoms, etc.
- (ii) Oral contraceptive methods, i.e., use of oral pills etc.
- (iii) Surgical methods, i.e., vasectomy and tubectomy.
- (b) Prenatal sex determination was banned in India in 1994. This was done to prevent sex selective abortion. It is being used to kill the normal female fetus. This killing of the unborn girl child is called female foeticide which is reducing the number of girls drastically in some societies of our country. Due to reckless female foeticide, male-female sex ratio is declining at an alarming rate. Its benefit in the long run is that the female-male ratio could be maintained for a healthy society.
- (c) Bacterial diseases due to unsafe sex are gonorrhoea and syphilis. Viral diseases due to unsafe sex are AIDS and genital herpes.
28. (i) Two common signs of sexual maturation in boys and girls are:
- (a) Growth of pubic hair and extra hair in the armpits.
- (b) Development of oily skin and pimples.
- (ii) Female feticide is reducing the number of girls drastically in our country, due to which male-female sex ratio is also declining.
- (iii) Chemical contraceptive method changes the hormonal balance of the body.
- (iv) The rate of birth and death in a given population

will determine the size of a population.

29. (a) A-Ureter
B - Seminal vesicle
C - Urethra
D - Vas deferens
- (b) Testes produce male sex hormone testosterone. Hormone testosterone brings about the development of secondary sexual characters during puberty in boys like growth of facial hair, deepening of voice, build up of muscle mass and also regulates formation of sperms.
- (c) Seminal vesicles (B) release its contents into the ejaculatory duct during ejaculation. Urethra (C) carries sperms from the vas deferens through the penis.
30. (a) The age at which the sex hormones begin to be produced and the boy and girl becomes sexually mature, i.e., able to reproduce is called puberty.
- (b) (i) **Testes:** The two testes in male are the sites where male gametes, i.e., sperms are formed. Testes also produce the male sex hormone called testosterone.
- (ii) **Seminal vesicles:** Seminal vesicles are one pair of sac-like structures near the base of bladder. Seminal fluid is a watery alkaline fluid that contains nutrients (fructose) which serve as a source of energy for the sperm. Each seminal vesicle releases its contents into the ejaculatory duct during ejaculation.
- (iii) **Vas deferens:** This is a straight tube, about 40 cm long, which carries the sperms to the seminal vesicles, where mucus and a watery alkaline fluid containing fructose, mix with the sperms.
- (iv) **Urethra:** It is a long tube that arises from urinary bladder. Urethra carries urine from the bladder as well as sperms from the vas deferens, through the penis.
- (c) Testes are located outside the abdominal cavity because sperm formation requires a lower temperature than normal body temperature. The temperature of the testes in the scrotum is about 2°C lower than normal body temperature. This temperature is ideal for sperm formation and development.
- (d) The sperms present in the testes of man are introduced into the vagina of the woman through penis during copulation. Millions of sperms are

released into the vagina at one time. The sperms are highly active and mobile. They travel from here upward through the uterus at the top of fallopian tube within five minutes.

31. (a) The modes of asexual reproduction in each of the following organisms are:

- (i) *Hydra* - Budding
- (ii) *Planaria* - Fragmentation and Regeneration
- (iii) *Amoeba* - Binary Fission
- (iv) *Spirogyra* - Fragmentation and Asexual spore formation

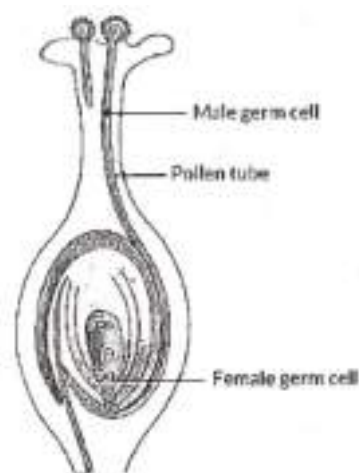
(v) *Rhizopus* - Spore formation and Fragmentation

(b) Three advantages of vegetative propagation are:

- **Genetic uniformity:** Vegetative propagation produces offspring that are genetically identical to the parent plant. This ensures that desirable traits, such as high yield or disease resistance, are preserved in the offspring.
- **Faster propagation:** Vegetative propagation allows for rapid multiplication of plants compared to sexual reproduction, which requires the development of seeds and germination.
- **Preservation of unique characteristics:** Some plants have unique and desirable characteristics that cannot be reliably passed on through seeds. Vegetative propagation allows for the preservation and multiplication of these characteristics.

(c) Fertilisation cannot take place in flowers if pollination does not occur because pollination is the transfer of pollen grains from the anther to the stigma of a flower. The pollen grains contain the male gametes (sperm cells), while the stigma contains the female gametes (egg cells). Fertilisation can only occur when the sperm cells reach the egg cells. Without pollination, there is no transfer of pollen to the stigma, and thus, the sperm and egg cells cannot meet for fertilisation to take place.

32. Diagram showing fertilisation in a flower is as follows:



Fertilisation, in plants, occurs when the male gamete present in pollen grain fuses with the female gamete (or egg) present in ovule. When a pollen grain falls on the stigma of the carpel, it bursts open and grows a pollen tube downwards through the style towards the female gamete in the ovary. Male gametes move down the pollen tube. The pollen tube enters the ovule in the ovary. The tip of pollen tube bursts and male gametes come out of pollen tube. In ovary, the male gamete of pollen combines with the female gamete or egg present in ovule to form a fertilised egg. After fertilisation, (i) ovule develops into seed (ii) ovary develops into fruit.

33. (a) (i) *Planaria* - Regeneration

- Regeneration of organism from its cut body parts occurs by the process of growth and development.
- Regeneration is an asexual mode of reproduction common in lower plants and animals.

(ii) *Hydra* - Budding

- In budding, a small part of the body of the parent organism grows out as a bud which on detaching from a new organism.

(iii) *Rhizopus* - Spores

Spores are usually produced in sporangia. Spore formation is a common method of an asexual reproduction in bacteria and most of the fungi.

(b) The leaves of a *Bryophyllum* have special type of buds in their margins. These buds may get detached from the leaves fall to ground and then grow to produce new *Bryophyllum* plants. The buds can also drop to the ground together with the leaf and then grow to produce new plants.

(c) **Advantages of vegetative propagation are:**

- The new plants produced by artificial vegetative propagation are exactly like the parent plants.

- Many plants can be grown from one plant by vegetative propagation.

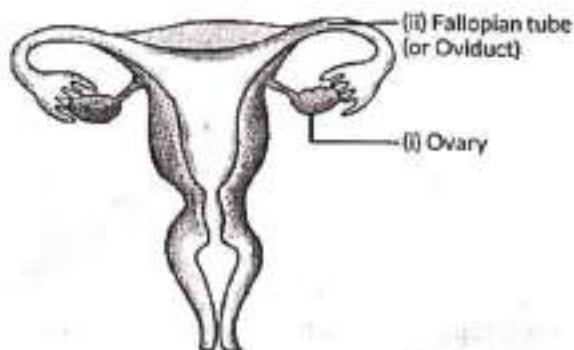
34. (a) The given diagram is the sectional view of human female reproductive system. The labelled parts are:

1. Funnel of fallopian tube or oviduct
2. Ovary
3. Uterus or womb
4. Cervix
5. Vagina

(b) Contraception is the avoidance of pregnancy. Three advantages of adopting contraceptive methods are:

- (i) They prevent frequent or unwanted pregnancies.
- (ii) They prevent the transfer of sexually transmitted infections.
- (iii) They help to regulate the population growth.

35. (a) The sectional view of human female reproductive system is as follows:



- (b) Gonorrhoea and syphilis are two bacterial diseases which are transmitted sexually.
- (c) Contraceptive devices are those devices which are used to prevent pregnancy. It includes diaphragm, condom and intrauterine devices. Contraceptive methods are adopted:
 - (i) to avoid unwanted birth.
 - (ii) to keep the population of a country under control.

36. The process of transfer of pollen grains from anther of a flower to the stigma of the same flower or another flower of the same species is known as pollination. Pollination may be of two major types- (i) self pollination and (ii) cross pollination.

- **Self pollination** is the transfer of pollen grains from the anther to the stigma of the same flower, or to the stigma of another flower of the same plant. This pollination generally takes place in bisexual flowers because they have both male and female gametes in them.
- **Cross pollination** is the transfer of pollen grains

from the anther of a flower of one plant to the stigma of a flower of another plant of the same species. This occurs in unisexual as well as bisexual flowers. Two agents of pollination are wind and water. Pollination results in the deposition of related pollen grains over the receptive stigma of the carpel. Pollen grains after landing on stigma, absorb water, swell and then germinate to produce pollen tubes.

Many pollen tubes grow into the stigma, but only one passes through the style and then moves towards the ovary. Two non- motile male gametes are formed inside the tube during its growth through the style. After reaching the ovary, pollen tube enters the ovule through the micropyle. The tip of the tube finally pierces the micropylar end of the embryo sac. After penetration, the tip of pollen tube ruptures releasing two male gametes into the embryo sac. The mature embryo sac consists of an egg apparatus (one haploid egg and two synergids), two polar nuclei and three antipodal cells. During the act of fertilisation, one male gamete fuses with the egg to form the diploid zygote.

37. (a) The male organ is testis. It secretes the hormone testosterone and regulates the formation of sperms. - It brings about changes in the appearance of boys at the time of puberty.
 - (b) Fertilisation occurs in the oviduct.
 - (c) The developing embryo gets nourishment from the mother's blood with the help of a special tissue, called placenta. The placenta provides a large surface area for the passage of glucose and oxygen from the mother's blood to the embryo.
38. (a) Variation is occurrence of differences between organisms. Variations are created in a population in different ways :
 - (i) There may be minor changes/errors during DNA copying mechanism which happens before any cell division. These variations may go on accumulating from previous generations, ultimately leading to visible changes.
 - (ii) In sexually reproducing organisms, the traits of two individuals combine and give rise to new combination in the progeny. This also leads to variation.
 - (iii) Thus combining variations from two or more individuals would thus create new combinations of variations. Organisms with suitable variations will have better chances of survival. Depending on the nature of variation, different individuals would have different kind of advantages.

- (b) In sexually reproducing organisms each cell has two copies of each chromosome, one each from the male and female parent. During gamete formation, one chromosome from each pair goes to a gamete.

Hence, the gametes have half the number of chromosomes, but one chromosome of each pair. When two gametes combine, they restore the normal number of chromosome in the progeny.

39. Methods developed to prevent pregnancy are:

- (i) Barrier method, i.e., use of condoms, diaphragm, etc.

- (ii) Oral contraceptive method, i.e., use of oral pills.
(iii) Surgical method, i.e., vasectomy and tubectomy. Out of these methods, chemical method is not meant for males.

Use of these techniques help to keep control over number of children in a family, which directly affects prosperity of a family. One of the most common reason for deterioration of women's health is frequent conception. Controlled childbirth will directly affect women health and this will indirectly affect the prosperity of family and nation.



4

CHAPTER

Heredity and Evolution



Previous Year Questions

Multiple Choice Questions

1. Consider the following statements:

[CBSE 2024]

- (i) The sex of a child is determined by what it inherits from the mother.
- (ii) The sex of a child is determined by what it inherits from the father.
- (iii) The probability of having a male child is more than that of a female child.
- (iv) The sex of a child is determined at the time of fertilisation when male and female gametes fuse to form a zygote.

The correct statements are:

- (a) (i) and (iii)
 - (b) (ii) and (iv)
 - (c) (iii) and (iv)
 - (d) (i), (iii) and (iv)
2. In an experiment to study independent inheritance of two separate traits: shape and colour of seeds, the ratio of the different combinations in F_2 progeny would be: [CBSE 2024]
- (a) 1:3
 - (b) 1:2:1
 - (c) 9:3:3:1
 - (d) 9:1:1:3
3. A cross made between two pea plants produces 50% tall and 50% short pea plants. The gene combination of the parental pea plants must be [CBSE 2024]
- (a) Tt and Tt
 - (b) TT and Tt
 - (c) Tt and tt
 - (d) TT and tt
4. A cross between two tall pea plants resulted in offsprings having a few dwarf plants. The gene-

combination of the parental plants must be

[CBSE 2024]

- (a) Tt and Tt
 - (b) Tt and tt
 - (c) TT and tt
 - (d) TT and Tt
5. **Assertion (A):** Human female has a perfect pair of sex chromosome.
- Reason (R):** Sex chromosome contributed by the human male in the zygote decides the sex of a child.
- [CBSE 2024]
- (a) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
 - (b) Both Assertion (A) and Reason (R) are true and Reason (R) is not the correct explanation of Assertion (A).
 - (c) Assertion (A) is true but Reason (R) is false.
 - (d) Assertion (A) is false but Reason (R) is true.
6. A cross between pea plant with white flowers (w) and pea plant with violet flowers (VV) resulted in F_2 progeny in which ratio of violet (VV) and white (ww) flowers will be: [CBSE 2023]
- (a) 1 : 1
 - (b) 2 : 1
 - (c) 3 : 1
 - (d) 1 : 3
7. Consider the following two statements:
- (i) The trait that expresses itself in F_1 generation.
 - (ii) The trait that keeps on passing from one generation to another.
- The appropriate terms for the statements (i) and (ii) respectively are [CBSE 2023]
- (a) Recessive trait; Dominant trait
 - (b) Dominant trait; Recessive trait

- (c) Dominant trait; Inherited trait
- (d) Recessive trait; Inherited trait

8. Assertion (A): In humans, if gene (B) is responsible for black eyes and gene (b) is responsible for brown eyes, then the colour of eyes of the progeny having gene combination Bb, bb or BB will be black only.

Reason (R): The black colour of the eyes is a dominant trait. **[CBSE 2023]**

- (a) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
 - (b) Both Assertion (A) and Reason (R) are true and Reason (R) is not the correct explanation of Assertion (A).
 - (c) Assertion (A) is true but Reason (R) is false.
 - (d) Assertion (A) is false but Reason (R) is true.
- 9. Assertion (A):** Human population show a great deal of variations in traits.

Reason (R): All variations in a species have equal chances of surviving in the environment in which they live. **[CBSE 2023]**

- (a) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
- (b) Both Assertion (A) and Reason (R) are true and Reason (R) is not the correct explanation of Assertion (A).
- (c) Assertion (A) is true but Reason (R) is false.
- (d) Assertion (A) is false but Reason (R) is true.

Very Short Answer Type Questions

- 10. (a)** List any two pairs of visible contrasting characters of garden pea plants used by Mendel for his experiments stating the dominant and recessive characters in each pair.
- OR
- (b)** In human beings, the probability of getting a male or a female child is 50%. Explain with the help of a flow diagram only. **[CBSE 2024]**
- 11.** The survival of a species is promoted through creation of variations. Illustrate with an example. **[CBSE 2024]**
- 12.** List two differences between dominant traits and recessive traits. What percentage of pea plants in the F_2 generation were with yellow seeds in Mendel's cross between the pea plants having yellow (YY) and green coloured (yy) seeds? **[CBSE 2024]**

13. Mendel worked out the rules of heredity by working on garden pea using a number of visible contrasting characters. He conducted several experiments by making a cross with one or two pairs of contrasting characters of pea plant. On the basis of his observations he gave some interpretations which helped to study the mechanism of inheritance. **[CBSE 2024]**

- (a) When Mendel crossed pea plants with pure tall and pure short characteristics to produce F_1 progeny, which two observations were made by him in F_1 plants?
- (b) Write one difference between dominant and recessive trait.
- (iii) (A) In a cross with two pairs of contrasting characters

RRYY	×	rryy
(Round Yellow)		(Wrinkled Green)

Mendel observed 4 types of combinations in F_2 generation. By which method did he obtain F_2 generation? Write the ratio of the parental combinations obtained and what conclusions were drawn from this experiment.

OR

- (iii) (B) Justify the statement:
"It is possible that a trait is inherited but may not be expressed."
- 14.** What is variation? List two main reasons that may lead to variation in a population. **[CBSE 2022]**
- 15.** What is heredity? **[CBSE 2021]**
- 16.** How many pairs of chromosomes are present in human beings? **[CBSE 2020]**

Short Answer Type Questions

- 17.** The most obvious outcome of the reproductive process is the generation of individuals of similar design, but in sexual reproduction they may not be exactly alike. The resemblances as well as differences are marked. The rules of heredity determine the process by which traits and characteristics are reliably inherited. Many experiments have been done to study the rules of inheritance. **[CBSE 2024]**
- (i) Why an offspring of human being is not a true copy of his parents in sexual reproduction?
 - (ii) While performing experiments on inheritance in plants, what is the difference between F_1 and F_2 generation?
 - (iii) (a) Why do we say that variations are useful for the survival of a species over time?

18. In some families, either rural or urban, females are tortured for giving birth to a female child. They do not seem to understand the scientific reason behind the birth of a boy or a girl. Infact the mother is not responsible for the sex of the child and it has been genetically proved that the sex of a newborn is determined by what the child inherits from the father.

- (a) State the basis on which the sex of a newborn baby is determined in humans.
- (b) Why is the pair of sex chromosomes called a mismatched pair in males?
- (c) How is the original number of chromosomes present in the parents restored in the progeny?

OR

- (c) Explain by giving two examples of the organisms in which the sex is not genetically determined.

[CBSE 2023]

19. "Sex chromosomes in human males and females are XY and X X respectively. Statistical probability of getting either a male or a female child is 50%. Justify this statement giving reason. [CBSE 2022]

20. (i) In a cross between violet flowered plants and white flowered plants, state the characteristics of the plants obtained in the F_1 progeny.
- (ii) If the plants of F_1 progeny are self-pollinated, then what would be observed in the plants of F_1 progeny?
- (iii) If 100 plants are produced in F_1 progeny, then how many plants will show the recessive trait in F_2 generation? [CBSE 2022]

21. A cross was made between green-stemmed tomato plants denoted by (GG) and purple-stemmed tomato plants denoted as (gg) to obtain F_1 progeny.

- (a) What colour of the stem would you expect in their F_1 progeny and why?
- (b) Give the percentage of purple-stemmed plants if F_1 plants are allowed to self-pollinate to produce F_2 progeny.
- (c) Write the ratio between GG and gg plants in the F_2 progeny [CBSE 2022]

22 Sex of an individual is determined by different factors in various species. Some animals rely entirely on the environmental cues, while in some other animals the individuals can change their sex during their life time indicating that sex of some species is not genetically determined. However, in human beings, the sex of an individual is largely determined genetically.

- (a) In what way are the sex chromosomes 'X' and 'Y

different in size? Name the mismatched pair of sex chromosome in humans.

- (b) Write the number of pair/pairs of sex chromosomes present in human beings. In which one of the parent (male/female) perfect pair/pairs of sex chromosomes are present?
- (c) Citing two examples, justify the statement "Sex of an individual is not always determined genetically". [CBSE 2022]

23. The mechanism by which the sex of an individual is determined is called sex-determination. In human beings, sex of a newborn is genetically determined, whereas in some others it is not. There are 46 (23 pairs) chromosomes in human beings. Out of these, 44 (22 pairs) control the body characters and 2 (one pair) are known as sex chromosomes. The sex chromosomes are of two types - X chromosome and Y chromosome. At the time of fertilisation, depending upon which type of male gamete fuses with the female gamete, the sex of the newborn child is decided.

- (a) Why is a pair of sex chromosomes in human beings called a mismatched pair in terms of type and size?
- (b) If the gametes always have half the number of chromosomes, then how is the original number of chromosomes restored in the organism?
- (c) Name two animals whose sex is not genetically determined. Explain the process of their sex determination.

OR

- (c) With the help of a flowchart only, show how sex is genetically determined in human beings.

[CBSE 2022]

24. Mendel blended his knowledge of Science and Mathematics to keep the count of the individual exhibiting a particular trait in each generation. He observed a number of contrasting visible characters controlled in pea plants in a field. He conducted many experiments to arrive at the laws of inheritance.

- (a) What do the F_1 progeny of tall plants with round seeds and short plants with wrinkled seeds look like?
- (b) Name the recessive traits in above case.
- (c) Mention the type of the new combinations of plants obtained in F_2 progeny along with the ir ratio, if F_1 progeny w as allowed to self pollinate.

OR

- (c) If 1600 plants were obtained in F_2 progeny, write the number of plants having traits:
 - (i) Tall with round seeds

(ii) Short with wrinkled seeds

Write the conclusion of the above experiment.

[CBSE 2022]

Long Answer Type Questions

25. Justify the statement "Sex of the children will be determined by what they inherit from their father".

[CBSE 2022]

26. (a) Name the two types of gametes produced by men

(b) Does a male child inherit X chromosome from his father? Justify

(c) How many types of gametes are produced by a human female?

[CBSE 2022]

27. (a) Why did Mendel carry out an experiment to study

inheritance of two traits in garden pea?

(b) What were his findings with respect to inheritance of traits in F_1 and F_2 generation?

(c) State the ratio obtained in the F_2 generation in the above mentioned experiment. [CBSE 2020]

28. A green stemmed rose plant denoted by GG and a brown stemmed rose plant denoted by gg are allowed to undergo a cross with each other.

(a) List your observations regarding:

(i) colour of stem in their F_1 progeny

(ii) percentage of brown stemmed plants in F_2 progeny if plants are self pollinated.

(iii) ratio of GG and Gg in the F_2 progeny.

(b) Based on the findings of this cross, what conclusion can be drawn? [CBSE 2020]



Explanations

Previous Years Questions

Multiple Choice Questions

- (b) The sex of a child is determined by the father, as the sperm can carry either an X or Y chromosome, while the mother's egg always carries an X chromosome (statement ii).

The sex of a child is determined at the time of fertilization when the male and female gametes (sperm and egg) fuse to form a zygote (statement iv).

- (c) The 9 : 3 : 3 : 1 ratio occurs when two traits are inherited independently, as in Mendel's dihybrid cross experiment. It represents the combinations of dominant and recessive traits for both seed shape and color in the F_2 generation.
- (c) In the cross Tt (tall) and tt (short), 50% of the offspring inherit Tt (tall) and 50% inherit tt (short). This results in a 1:1 ratio of tall to short plants, explaining why option (c) is the correct answer.
- (a) The parental plants must be Tt (heterozygous tall) and Tt (also heterozygous tall) because this combination can produce offspring with both tall (TT and Tt) and dwarf (tt) plants. In this case, the expected ratio would be 3 tall to 1 dwarf, which explains the presence of dwarf plants among the offspring.
- (b) The answer is (b) because both the assertion and reason are true, but they explain different things. The human female has two X chromosomes (a perfect pair), and while it's true that the male's sperm decides the sex of the child, this fact is not directly explaining the assertion about the female's sex chromosomes.
- (a) In this cross, the parent with violet flowers (VV) is homozygous dominant, while the white flower parent is homozygous recessive (ww). When they are crossed, all offspring in the F_1 generation will have violet flowers (Vw). If F_1 plants are crossed with each other, the F_2 generation will have a 1 : 1 ratio of violet (Vw) to white (ww) flowers, making option (a) the correct answer.
- (c) In the given statements, the trait that expresses

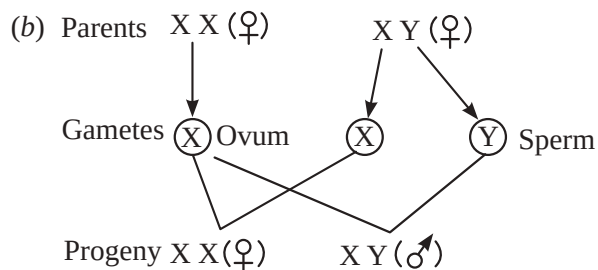
itself in the F_1 generation is called a "dominant trait," while the trait that can be passed down through generations, even if it doesn't show up in every generation, is referred to as an "inherited trait." Therefore, the correct answer is (c) "Dominant trait; Inherited trait."

- (d) The assertion (A) is false because the progeny with the gene combination bb (brown eyes) would not have black eyes; only BB and Bb would. However, the reason (R) is true because the black color is indeed a dominant trait. Therefore, the correct answer is (d), which states that (A) is false but (R) is true.
- (c) Variations get accumulated or discarded as combined effect of environmental factors and reproduction process.

Very Short Answer Type Questions

- (a) 2 visible characters of garden pea plants are:
 - ❖ Tallness (dominant), Dwarfness (recessive)
 - ❖ Yellow seeds (dominant), green seeds (recessive)

OR



- A population of bacteria living in temperate waters that can withstand heat due to the rise in temperature due to global warming and will survive better in a heat wave than the non-variant bacteria having no capacity to tolerate heat wave. Thus, suitable variations promote survival.
-

Dominant Trait

Recessive trait

(i)	When both dominant and recessive traits are inherited, the dominant trait gets expressed	(i)	When both dominant and recessive trait are inherited, the recessive trait does not get expressed
(ii)	A single copy of dominant trait is enough to get it expressed.	(ii)	Both the copies of a trait should be recessive to get it expressed.

75% yellow seeds.

13. (a) All plants Tall

Gene combination: Tt

(b) It is a recessive trait/it cannot be expressed in present in dominant trait.

(c) Tall: Short 3 : 1

Conclusion: Tall trait is dominant and short trait is recessive

14. Variation is the degree of differences in the progeny (off springs) and between the progeny and parents. Two main reasons of variations are mutations and genetic recombination during sexual reproduction.

15. The inheritance of characters (or traits) from the parents to their off springs is called heredity.

16. In human beings. 23 pairs of chromosomes are present in each cell. Out of 23 pairs, 22 pairs of chromosomes carry genes which control somatic traits, these chromosomes are called autosomes. The 23rd pair is called sex chromosomes.

Short Answer Type Questions

17. (i) In sexual reproduction, offspring of human being is not a true copy of his parents because it inherits half of its genetic material from each parent. During the formation of gametes (sperm and egg cells), the genetic material undergoes recombination that leads to shuffling or mixing of genetic material from both parents, resulting in offspring with unique combination of genes.

(ii) F_1 generation is the first filial generation of the offspring from the parents whereas F_2 generation is the second filial generation of the offspring, generated through inbreeding of F_1 generation. F_1 generation can be distinctly different from the parental type whereas F_2 generation always exhibit some parental genotypes.

(iii) Variations allow genetic diversity that is important for the development of new traits and increasing

adaptability towards changes in the environment. It makes the organisms better adapted for survival under changing conditions.

18. (a) Sex of child is determined by what it inherits from the father. A child who inherits X chromosome from her father will be a girl and one who inherits a Y chromosome from father will be a boy.

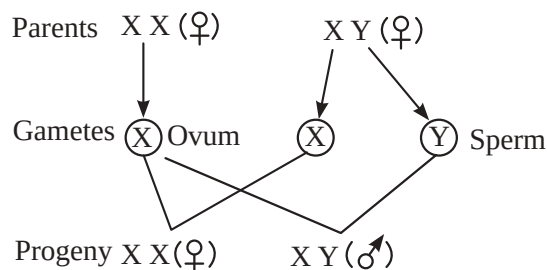
(b) Human beings have 22 pairs of autosomes and one pair of sex chromosomes. The females possess two homomorphic sex chromosomes, named XX while males contain two heteromorphic sex chromosomes, i.e., XY. The Y chromosome is shorter than X chromosome. Therefore a pair of sex chromosomes in is called mismatched pair in terms of type and size males.

(c) Gametes contain half the number of chromosomes of parent. But, when the two gametes (male and female) fuse to form the zygote, the normal diploid condition is restored. Hence, the formation of gametes by meiosis and fusion of male and female gamete help to maintain the number of chromosomes in organism.

OR

(c) Two animals in which sex is not determined genetically are turtle and crocodile. The type of sex in them is determined by environmental factors. In turtles, the temperature of egg incubation has a significant effect on the sex of developing embryos. Males are predominant below 28°C , females above 33°C and equal number of the two sexes between $28-33^{\circ}\text{C}$. In crocodiles, high temperature induces maleness and low temperature femaleness.

19. Human female (XX) produces all gametes (ova) with X-chromosomes, while human male (XY) produces 50% gametes (sperms) with X-chromosome while 50% gametes with Y-chromosome. If sperm having X chromosome fertilises the ovum with X chromosome then a female child will be produced, otherwise a male child will be produced.



Sex of the child (offspring) is determined by the type of sperm that fuses with ovum at the time of fertilisation. Therefore, there is 50% chance of a male



child being born and a 50% chance of a female child being born

20. (i) In the F_1 progeny, the plants obtained will have violet flowers.
 (ii) If the plants of F_1 progeny are self-pollinated, the plants obtained in the F_2 progeny will show a phenotypic ratio of 3:1, with 75% of the plants having violet flowers and 25% having white flowers in F_2 generation.
 (iii) If 100 plants are produced in the F_1 progeny, approximately 25 plants will show the recessive trait of white flowers.
21. (a) F_1 progeny will have green stemmed tomato plants as green is dominant over purple stemmed tomato plants.

(b) If F_1 plants are self pollinated, then the percentage of purple stemmed plant in F_2 progeny will be 25%.

F_2 progenies:

$\begin{matrix} \text{♂} \\ \text{♀} \end{matrix}$	G	g
G	GG (Green Stemmed)	Gg (Green Stemmed)
g	Gg (Green Stemmed)	gg (Purple Stemmed)

Phenotypic: 3 : 1
 ratio Green stemmed : Purple stemmed

(c) Ratio of GG and gg plant in F_2 generation will be 1:1.

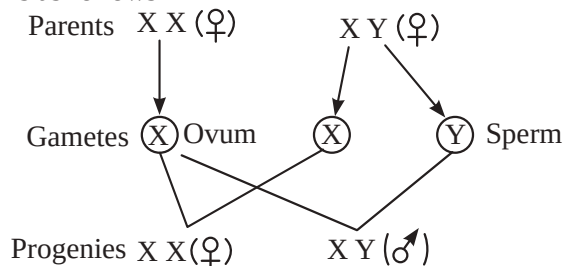
22. (a) X chromosome is morphologically distinct from Y chromosome. Y chromosome is smaller than X chromosome. Hence, they are dissimilar or heteromorphic. Men have mismatched pair of sex chromosome in humans in which one is normal size X while other is a short one called Y.
 (b) Human beings have 22 pair of autosomal chromosomes and one pair of sex chromosome. Women have a perfect pair of sex chromosomes both called X.
 (c) Sex of the individual is not always determined genetically. In some organisms, gender may be determined by environmental factors. For example snails, turtles and lizards sex is determined by the temperature at which fertilised egg are kept.
23. (a) Human beings have 22 pairs of autosomes and one pair of sex chromosomes. The females possess

two homomorphic sex chromosomes, called XX while male have a mismatched pair in which one is a normal sized while the other is a short-one called Y.

(b) Gametes contain half the number of chromosome. But, when the gametes fuse to form the zygote, the normal diploid condition is restored. Hence, the formation of gametes by meiosis helps to maintain the number of chromosomes in organism. (c) Two animals in which sex is not determined genetically are turtle and crocodile. The type of sex in them is determined by environmental factors. In turtles, the temperature of egg incubation has a significant effect on the sex of developing embryos. Males are predominant below 28°C , females above 33°C and equal number of the two sexes between $28-33^\circ\text{C}$. In crocodiles, high temperature induces maleness and low temperature induces femaleness.

OR

- (c) Flow chart of sex determination in human beings is as follows



25. (a) The F_1 progeny will be tall plant with round seeds.
 (b) Recessive traits are short plant and wrinkled seed.
 (c) The new combinations will be tall plant with wrinkled seeds and short plant with round seeds. The ratio will be 9:3:3:1 :: Tall plant with round seed: Tall plant with wrinkled seeds: Short plant with round seed: Short plant with wrinkled seed.

OR

- (c) (i) The F_2 ratio will be 9 : 3 : 3 : 1 :: Tall plant with round seed: Tall plant with wrinkled seeds: Short plant with round seed: Short plant with wrinkled seed.

Tall plant with round seeds

$$= 9/16 \times 1600 = 900$$

- (ii) Short plant with wrinkled seed

$$= 1/16 \times 1600 = 100$$

Long Answer Type Questions

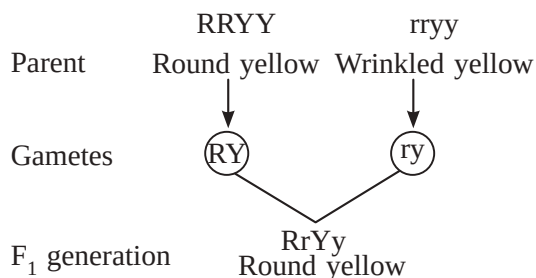
26. The statement "Sex of the children will be determined by what they inherit from their father" is based on the principle of genetic inheritance. The determination of an individual's sex is influenced by the combination

of sex chromosomes inherited from both parents.

In humans, females have two X chromosomes (XX) while males have one X and one Y chromosome (XY). The father contributes either an X or a Y chromosome to the offspring, while the mother always contributes an X chromosome. Therefore, it is the father's genetic contribution that ultimately determines the sex of the child.

During fertilization, if the father's sperm carries an X chromosome, the resulting combination of XX chromosomes will develop into a female child. On the other hand, if the father's sperm carries a Y chromosome, the combination of XY chromosomes will lead to the development of a male child.

27. (a) The two types of gametes produced by men are sperm cells (**spermatozoa**) containing X or Y sex chromosomes. Sperm cells carry either an X or a Y sex chromosome, **determining the sex** of the offspring upon fertilization with a female's egg.
- (b) No, a **male** child does **not inherit an X chromosome** from his father. In humans, sex determination is based on the X and Y chromosomes. **Females have two X chromosomes (XX)**, while **males have one X and one Y chromosome (XY)**. When a child is conceived, they inherit one sex chromosome from each of their parents. The mother always passes on an X chromosome, while the father can pass on either an X or Y chromosome. If the father passes on an X chromosome, the child will be female (XX), and if the father passes on a Y chromosome, the child will be male (XY).
- (c) A human **female** typically **produces one type of gamete**, which is the egg or ovum containing X chromosome.
28. (a) Mendel carried out crosses with two traits to see the interaction and basis of inheritance between them. In a dihybrid cross given by Mendel, it was observed that when two pairs of characters were considered each trait expressed independent of the other.
- (b) For Example, a cross between round yellow and wrinkled green parents.



	RY	rY	Ry	ry
RY	RRYY Round yellow	RrYY Round yellow	RRYy Round yellow	RrYy Round yellow
rY	RrYY Round yellow	rrYY Wrinkled yellow	RrYy Wrinkled yellow	rrYy Wrinkled yellow
Ry	RRYy Round yellow	RrYy Round yellow	RRyy Round yellow	Rryy Round yellow
ry	RrYy Round yellow	rrYy Wrinkled yellow	Rryy Round Green	rryy Wrinkled yellow

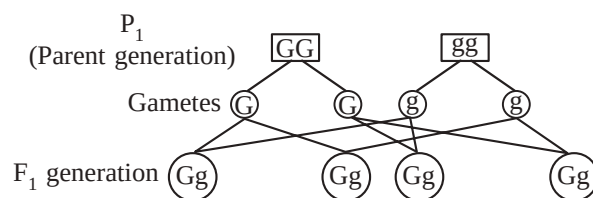
In F₁ generation, all plants are with round yellow seeds. But in F₂ generation, we find all types of plants: Round yellow, Round green, Wrinkled yellow, Wrinkled green.

- (c) F₂ generation ratio: Round-yellow = 9 : Round-green = 3 : Wrinkled-yellow = 3 : Wrinkled-green = 1.

29. (a) (i) Colour of stem in F₁ progeny:

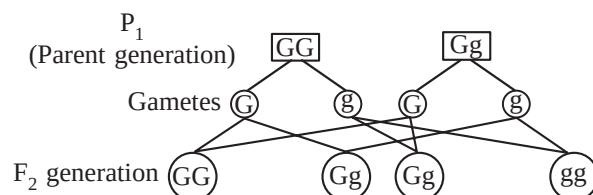
Colour of the stem in F₁ progeny: All green

The colour in the F₁ progeny is green stemmed as green stem colour is dominant.



- (ii) F₁ progeny on self pollination:

Selfing of F₁ generation



F₂ generation Green stemmed: Brown stemmed or 25% of F₂ progeny are brown stemmed rose plant.

- (iii) Ratio of GG and Gg in F₂ progeny:
Genotype of F₂ progeny - GG:Gg

1 : 2

- (b) This is a monohybrid cross. This shows that out of contrasting traits only one dominant trait appears in F₁ generation and the trait which does not express is recessive. On selfing the F₁ plants, both the traits appear in next generation but in a definite proportion.

5

CHAPTER

Our Environment



Previous Year Questions

Multiple Choice Type Questions

1. **Assertion (A):** Accumulation of harmful chemicals is maximum in the organisms at the highest trophic level of a food chain.

Reason (R): Harmful chemicals are sprayed on the crops to protect them from diseases and pests.

- (a) Both Assertion (A) and Reason (R) are true and Reason is the correct explanation of the Assertion (A).
(b) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).
(c) Assertion (A) is true, but Reason (R) is false.
(d) Assertion (A) is false, but Reason (R) is true.

2. Consider the following statements about ozone: [CBSE 2024]

- (A) Ozone is poisonous gas.
(B) Ozone shields the earth's surface from the infrared radiation from the sun.
(C) Ozone is a product of UV radiations acting on oxygen molecule.
(D) At the lower level of the earth's atmosphere, ozone performs most essential function.

The correct statements are

- (a) (A) and (B) (b) (A) and (C)
(c) (B) and (C) (d) (B) and (D)
3. Some wastes are given below: [CBSE 2024]
- (i) Garden waste
(ii) Ball point pen refills
(iii) Empty medicine bottles made of glass
(iv) Peels of fruits and vegetables

- (v) Old cotton shirt

The non-biodegradable wastes among these are:

- (a) (i) and (ii) (b) (ii) and (iii)
(c) (i), (iv) and (v) (d) (i), (iii) and (iv)

4. **Assertion (A):** Oxygen is essential for all aerobic forms of life.

Reason (R): Free oxygen atoms combine with molecular oxygen to form ozone. [CBSE 2024]

- (a) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
(b) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of Assertion (A).
(c) Assertion (A) is true, but Reason (R) is false.
(d) Assertion (A) is false, but Reason (R) is true.

5. Which one of the following is not a natural ecosystem? [CBSE 2024]

- (a) Pond ecosystem (b) Grassland ecosystem
(c) Forest ecosystem (d) Cropland ecosystem

6. **Assertion (A):** The waste we generate daily may be biodegradable or non-biodegradable. **Reason (R):** The waste generated, if not disposed off properly may cause serious environmental problems. [CBSE 2024]

- (a) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
(b) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of Assertion (A).
(c) Assertion (A) is true, but Reason (R) is false.
(d) Assertion (A) is false, but Reason (R) is true.

7. Identify the food chain in which the organisms of the second trophic level are missing: [CBSE 2024]

- (a) Grass, goat, lion

- (b) Zooplankton, Phytoplankton, small fish, large fish
(c) Tiger, grass, snake, frog
(d) Grasshopper, grass, snake, frog, eagle
8. Use of several pesticides which results in excessive accumulation of pesticides in rivers or ponds is a matter of deep concern. Justify this statement. **[CBSE 2023]**
9. (i) Why are crop fields considered as artificial ecosystems?
(ii) Write a common food chain of four steps operating in a terrestrial ecosystem. **[CBSE 2022 C]**
10. What is ozone? How is it formed in the upper layers of the earth's atmosphere? How does ozone affect our ecosystem? **[CBSE 2022]**
11. (a) What is meant by garbage? List two classes into which garbage is classified.
(b) What do we actually mean when we say that "enzymes are specific in their action"? **[CBSE 2022]**
12. What are consumers? Name the four categories under which the consumers are further classified. **[CBSE 2021 C]**
13. How is ozone layer formed? State its importance to all life forms on earth. Why the amount of ozone in the atmosphere dropped sharply in the 1980s? **[CBSE 2020]**
14. (a) Write two harmful effects of using plastic bags on the environment. Suggest alternatives to the usage of plastic bags.
(b) List any two practices that can be followed to dispose off the waste produced in our homes. **[CBSE 2020]**

Very Short Type Questions

15. Differentiate between food chain and food web. In a food chain consisting of deer, grass and tiger, if the population of deer decreases, what will happen to the population of organisms belonging to the first and third trophic levels? **[CBSE 2024]**
16. What are decomposers? List two consequences of their absence in an ecosystem. **[CBSE 2024]**
18. How is ozone formed in the higher levels of the atmosphere? "Damage to the ozone layer is a cause of concern." Justify this statement. **[CBSE 2023]**
19. "Although gardens are created by man but they are considered to be an ecosystem." Justify this statement. **[CBSE 2023]**

20. Name the term used for the materials which cannot be broken down by biological processes. Give two ways by which they harm various components of an ecosystem. **[CBSE 2024]**
21. (i) List two human-made ecosystems.
(ii) "We do not clean a pond in the same manner as we do in an aquarium." Give reason to justify this statement. **[CBSE Term II, 2021-22]**
22. In the following food chain, only 2J of energy was available to the peacocks. How much energy would have been present in Grass? Justify your answer.
Grass Grasshopper Frog Snake Peacock
[CBSE Term II, 2021-22]

Short Answer Type Questions

23. Use of pesticides to protect our crops affects organisms at various trophic levels especially human beings. Name the phenomenon involved and explain how does it happen. **[CBSE 2024]**

Long Answer Type Questions

24. (A) Plants → Deer → Lion. **[CBSE 2024]**
In the given food chain, what will be the impact of removing all the organisms of the second trophic level on the first and third trophic levels? Will the impact be the same for the organisms of the third trophic level in the above food chain if they were present in a food web? Justify.

OR

- (B) A gas 'X' which is a deadly poison is found at the higher levels of the atmosphere and performs an essential function. Name the gas and write the function performed by this gas in the atmosphere. Which chemical is linked to the decrease in the level of this gas? What measures have been taken by an international organization to check the depletion of the layer containing this gas?
25. What is the difference between biodegradable and non-biodegradable substances? List two methods of safe disposal of biodegradable domestic waste. **[CBSE 2023]**
26. Kulhads (disposable cups made of clay) and disposable paper cups both are used as an alternative for disposable plastic cups. Which one of these two can

be considered as a better alternative to plastic cups and why? **[CBSE 2022]**

27. What are human-made ecosystems? Give an example. Can a human-made ecosystem become a self-sustaining ecosystem? Give reason to justify your answer. **[CBSE 2022]**

28. (a) Name the group of organisms which form in the first trophic level of all food chains. Why are they called so?

(b) Why are the human beings most adversely affected by biomagnification?

(c) State one ill-effect of the absence of decomposers from a natural ecosystem. **[CBSE 2022]**

29. (a) We do not clean ponds or lakes, but an aquarium needs to be cleaned regularly. Why?

(b) Why is ozone layer getting depleted at the higher levels of the atmosphere? Mention one harmful effect caused by its depletion. **[2022]**

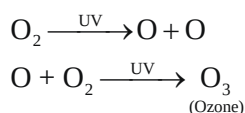


Explanations

Previous Year Questions

Multiple Choice Questions

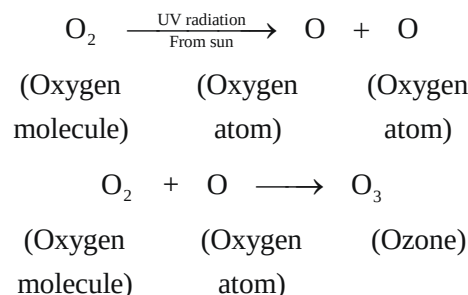
- (b) Harmful chemicals do accumulate in organisms at the highest trophic level due to a process called biomagnification, and these chemicals are often sprayed on crops to protect them. However, the reason does not directly explain why the accumulation is highest at the top of the food chain, so it's not a correct explanation of the assertion.
- (b) Ozone can be harmful at ground level, making it a poisonous gas. Statement (C) is also correct since ozone is formed when ultraviolet (UV) radiation interacts with oxygen molecules. However, statement (B) is incorrect because ozone protects the Earth from UV radiation, not infrared radiation, and statement (D) is misleading because ozone's essential functions are primarily in the upper atmosphere, not at lower levels.
- (b) Non-biodegradable wastes are those that do not break down naturally in the environment. In the given options, ballpoint pen refills (ii) and empty medicine bottles made of glass (iii) are non-biodegradable because they can persist in the environment for a long time. The other items like garden waste, fruit and vegetable peels, and old cotton shirts decompose naturally and are considered biodegradable.
- (b) Assertion (A) correctly states that oxygen is essential for aerobic life forms, while Reason (R) explains a process involving oxygen and ozone formation, which is also true. However, Reason (R) does not explain why oxygen is essential for life, so the correct answer is (b): both are true, but Reason (R) does not explain Assertion (A).
- (d) A natural ecosystem is one that occurs naturally without human interference. Among the options given, a cropland ecosystem (d) is not natural because it is created and managed by humans for farming purposes. In contrast, pond, grassland, and forest ecosystems develop naturally in the environment.
- (b) The answer is (b) because both statements are true: waste can be either biodegradable (which can break down naturally) or non-biodegradable (which cannot), and improper disposal of waste can indeed lead to serious environmental issues. However, the reason does not explain why waste is categorized as biodegradable or non-biodegradable, so it's not a direct explanation of the assertion.
- (c) In a food chain, the second trophic level usually consists of primary consumers that eat producers (like plants). In option (c), tiger, grass, snake, frog, there are no primary consumers (like a herbivore) between the grass (producer) and the snake. Instead, the snake directly feeds on frogs, which means the second trophic level is missing.
- The use of several pesticides results in accumulation of pesticides in rivers and ponds. These chemicals are either washed down into the soil or into the water bodies. From the soil, these are absorbed by the plants along with water and minerals, and from the water bodies these are taken up by aquatic plants and animals and enters the food chain. As these chemicals are not degradable, these get accumulated progressively at each trophic level. As human beings occupy the top level in any food chain, the maximum concentration of these chemicals get accumulated in our bodies i.e., biological magnification. Our food grains such as wheat and rice, vegetables and fruits, and even meat, contain varying amounts of pesticide residues cannot always be removed by washing or other means and causes health hazards.
- (i) Crop fields are the artificial ecosystems because in crop fields, both biotic (living) and abiotic (nonliving) components are manipulated by human beings. Humans can change edaphic factors by adding fertilisers, water, etc. Biotic components may be changed using biocides or adding useful organisms like earthworms etc.
(ii) A food chain consists of various organism at various trophic levels. In terrestrial ecosystem, a common food chain is
Grass → Grasshopper → Frog → Snake
- Ozone (O_3) is a molecule formed by three atoms of oxygen. It is formed in the stratosphere layer of atmosphere when high energy UV rays act on O_2 molecule splitting it into free oxygen (O) atoms. These atoms then combine with molecular oxygen (O_2) to form ozone (O_3).



Ozone shields the surface of the earth from radiations from the sun. The depletion of ozone layer will lead to global warming and some serious health issues such as damage of skin cells that leads to skin cancer, snow blindness or inflammation of cornea, increased fatality of young animals, mutations and reduced immunity.

- 11.(a) Garbage is the waste material especially of domestic refuse. The two classes into which garbage is classified are
 - (i) Biodegradable
 - (ii) Non-biodegradable.
- (b) Enzymes are specific in their action. For example, enzyme maltase acts on sugar maltose but not on lactose or sucrose. Different enzymes may act on the same substrate but give rise to different products. Similarly an enzyme may act on different substrates producing different end products.
12. Consumers are the organisms which are unable to synthesise their own food. Therefore, they utilise materials and energy stored by the producers or eat other organisms. They are known as the heterotrophs. The consumers are of following categories:
 - (i) Primary or first-order consumers: These include the animals which eat plants or plant products. They are called herbivores or primary (first order) consumers. E.g., Cattle, deer, goat, rabbit, hare, rats, mice, grasshoppers etc
 - (ii) Secondary or second order consumers: These include the animals which depend on primary consumers for their food. They are called primary carnivores or secondary (second order) consumers. Secondary consumers can be carnivores or omnivores. E.g., Cats, dogs, foxes, small fish, etc.
 - (iii) Tertiary or third order consumers: These are large carnivores (or top carnivores) which feed on primary and secondary consumers. These are termed as secondary carnivores or tertiary (third order) consumers. Common examples include shark and crocodile, wolves, lion, etc.
 - (iv) Quaternary or fourth order consumers: These are even larger carnivores which feed on secondary carnivores (tertiary consumers). E.g., Tigers, lions and eagles/hawks etc.

13. When high energy ultraviolet radiations react with oxygen present in stratosphere (the higher level of atmosphere) it splits into its constituent atoms. Since these atoms produced are very reactive, they react with molecular oxygen to form ozone.



Ozone shields the surface of the earth from UV radiations from the sun. The depletion of ozone layer will lead to global warming and some serious health issues such as damage of skin cells that leads to skin cancer, snow blindness or inflammation of cornea, increased fatality of young animals, mutations and reduced immunity. In 1980s, the production of CFCs increased which releases active chlorine in the atmosphere. The active chlorine then reacts with ozone molecules present there to convert them to oxygen. This results in thinning of ozone layer. CFCs are used as refrigerants and in fire extinguishers. That is why, amount of ozone in the atmosphere dropped sharply.

14. (a) **Two harmful effects of using plastic bags on the environment:**
 - (i) Plastic bags are non-biodegradable substances which are not acted upon by microbes. So, they cannot be decomposed and therefore persist in the environment for a long time causing harm to the soil fertility and quality.
 - (ii) Plastic bags choke drains which result in waterlogging, that allows breeding of mosquitoes and hence leads to various diseases. Jute bags and cloth bags are the alternatives to the polyethene bags.
- (b) Practices that can be followed to dispose off the waste produced in our homes:
 - (i) Separation of biodegradable and non-biodegradable wastes.
 - (ii) The biodegradable waste can be converted to manure.
 - (iii) Non-biodegradable waste should be disposed off at suitable places from where municipal authorities can pick them up and dispose properly and scientifically.

- (iv) Use discarded bottles and jars to store food items.

Very Short Type Questions

15.

Food chain	Food web
It is a series of organisms feeding on one another at various levels	It is a network of interconnected food chains / series of branching lines which provides a number of feeding connections amongst different organisms.

- Population of grass/ first trophic level will increase.
- Population of tiger/third trophic level will decrease.

16. Decomposers are the microorganisms that break-down the complex organic substances into simple inorganic substances.

Consequences:

- No replenishment of soil
- Foul smell
- Breeding of flies
- Accumulation of dead plants and animals in the environment.
- No recycling of nutrients

18. Ozone (O_3) is a molecule formed by three atoms of oxygen. It is formed in the stratosphere layer of atmosphere when high energy UV rays act on O_2 molecule splitting it into free oxygen (O) atoms. These atoms then combine with molecular oxygen (O_2) to form ozone (O_3).

19. In a garden, various plants like grasses, trees, flower bearing plants such as jasmine, sunflower, rose, and animals like insects, frogs and birds are found. All these living organisms interact with each other and their growth, reproduction and other activities are affected by the abiotic components such as light, water, wind, soil, minerals, etc. of ecosystem. Thus, a garden is considered to be an ecosystem.

20.

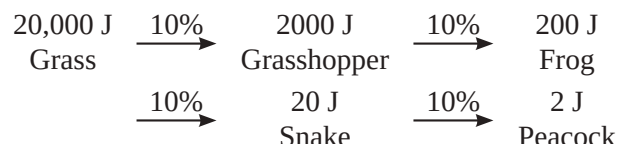
- Non-biodegradable substances
- Two ways:
 - They are inert and persist in the environment for long time and cause pollution.
 - Cause Biological magnification
 - Affect the fertility of soil

21. (i) The two human made ecosystems are aquarium

and garden.

- (ii) We do not clean pond as we do in an aquarium because the waste generated in a pond is acted upon by the decomposers which convert them into simple soluble substances, whereas, in aquarium, the waste gets mixed with water and left untreated due to absence of decomposers.

22. In the given food chain, 20,000 J of energy must have been present in grass. This is because, as per the 10% law of energy transfer, only of energy is transferred to the next trophic level.



Short Answer Type Questions

23. Phenomenon - Biological Magnification/ Biomagnification:

- Pesticides are washed down into the soil and water bodies.
- From the soil pesticides are absorbed by crop plants along with water and minerals and enter the food chain.
- These chemicals are non-biodegradable and get accumulated progressively at each trophic level.
- As human beings occupy the top level in any food chain, the maximum concentration of these chemicals gets accumulated in our bodies.

Long Answer Type Questions

24. (A)

- Number of plants/organisms of first trophic level will increase.
- Number of lions/ organisms of third trophic level will decrease.
- No
- As the organisms of that level will find alternative foods and will not starve to death / food web is more stable where other animals as prey may be available.

OR

- Gas 'X' is Ozone.
- Ozone shields the surface of the earth from ultra-violet (UV) radiations from the sun.
- CFCs (Chlorofluorocarbons)



- Succeeded in forging an agreement to freeze CFC production at 1986 levels / Manufacturing of CFC free refrigerators

25. Differences between biodegradable and non-biodegradable substances are as follows :

Biodegradable Substance	Non-biodegradable Substance
It decomposes naturally in the environment by the action of microorganisms.	It does not decompose naturally.
It is environment friendly.	It is harmful to the environment and causes pollution.
It is made up of natural ingredients.	It is made up of synthetic materials.
It can be converted into manure or recycled.	It can be either reused or recycled.
Examples: Waste paper, wood crumbles, etc.	Examples: Plastic bags, cans, disposable bottles, etc.

Domestic waste can be safely disposed off by composting. In composting, biodegradable domestic wastes, such as left-over food, fruit, vegetable peels, etc., can be buried in pit, dug into ground. They are converted into compost and used as manure.

In landfill a huge pit is made in an open low lying area and wastes are dumped into the pits. Once the pits are full, they are covered with soil and left for decomposition.

26. **Kulhad:**

- Kulhad is a cup that is made using clay or soil.
- This is usually used for serving tea/coffee.
- This was brought in use to replace the plastic cup in trains.
- Plastic is a nonbiodegradable harmful substance that does not decompose in nature and affects the ecosystem or environment negatively.
- Kulhads are made of biodegradable soil, therefore, this was used to replace the plastic and protect the environment and health of humans.

Discontinuation of kulhads:

- Since kulhads are made using clay, it is a practice of harming the environment too.
- The clay is fertile soil and kulhads were discontinued

to avoid its reduction.

- Making of kulhad on a large scale to serve tea passengers in the train was leading to the loss of this top fertile soil.

27. Human-made ecosystems, also known as artificial ecosystems, are environments created or heavily modified by humans to mimic natural ecosystems. These ecosystems are managed and controlled by humans to support specific purposes, such as agriculture, recreation, or habitation.

Examples of Human-Made Ecosystems:

1. **Aquariums:** Tanks or controlled water environments where aquatic life is maintained artificially.
2. **Agricultural Fields:** Farmlands where crops are cultivated for food production.
3. **Zoos:** Areas where animals are kept and cared for outside their natural habitats.
4. **Urban Parks:** Green spaces within cities created for recreational purposes.

Human-made ecosystems are environments created or significantly altered by humans to serve specific purposes

An example of a human-made ecosystem is an agricultural field, where natural landscapes are converted into areas for growing crops

While human-made ecosystems can develop some self-regulating features, they are not as self-sufficient as natural ecosystems like forests or wetlands

They often require continuous management and intervention to maintain their productivity and stability

28. (a) Producers form the first trophic level of all food chains. They are called producers because they are autotrophic organisms which alone are able to manufacture organic food from inorganic raw materials by the process of photosynthesis. They capture sun's energy and convert it into chemical energy. The chemical energy is used in combining raw materials into organic food. This food is used up by themselves and rest enters the food chains as food for consumers.
- (b) Human beings are most adversely affected by biomagnification because they occupy the highest trophic level in any food chain. As in biomagnification, successive concentration of

non-biodegradable substances increases in the trophic level of food chains, so, it leads to most toxicity at highest trophic level. Hence, maximum concentration of chemicals get accumulated most in their body.

- (c) Absence of decomposers will lead to the accumulation of dead remains and waste products of organisms in our natural ecosystem. The decomposers breakdown complex organic substances into simple inorganic substances, so, that it can go into the soil and can be used up by plants.

29. (a) We do not clean pond as we do in an aquarium because the waste generated in a pond is acted

upon by the decomposers which convert them into simple soluble substances, whereas, in aquarium, the waste gets mixed with water and left untreated due to absence of decomposers.

- (b) The ozone layer is getting depleted at the higher levels of the atmosphere due to use of chlorofluorocarbons (CFCs) which are used in refrigerator. Other ozone depleting substances include carbon tetrachloride, hydrofluorocarbons used in fire extinguisher, air i conditioners, etc. Due to the ozone layer depletion, humans will be directly exposed to the ultraviolet radiations of sun. This will result in serious health issues like skin cancer, sunburns, quick ageing, mutations and weak immune system.



1

CHAPTER

Chemical Reactions and Equations



Previous Year Questions

Multiple Choice Questions

1. Select from the following a decomposition reaction in which source in energy for decomposition is light: [CBSE 2024]

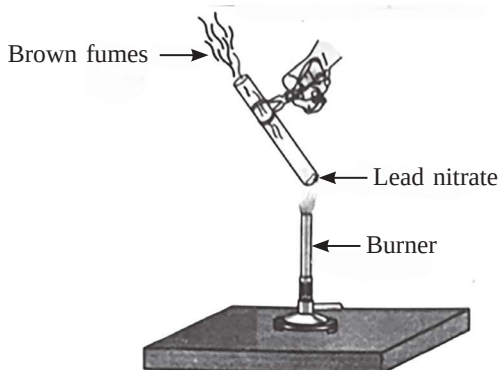
(a) $2\text{FeSO}_4 \rightarrow \text{Fe}_2\text{O}_3 + \text{SO}_2 + \text{SO}_3$
 (b) $2\text{H}_2\text{O} \rightarrow 2\text{H}_2 + \text{O}_2$
 (c) $2\text{AgBr} \rightarrow 2\text{Ag} + \text{Br}_2$
 (d) $\text{CaCO}_3 \rightarrow \text{CaO} + \text{CO}_2$

2. $\text{MnO}_2 + 4\text{HCl} \rightarrow \text{MnCl}_2 + 2\text{H}_2\text{O} + \text{Cl}_2$

The reaction given above is a redox reaction because in this case: [CBSE 2024]

(a) MnO_2 is oxidised and HCl is reduced
 (b) HCl is oxidised
 (c) MnO_2 is reduced
 (d) MnO_2 is reduced and HCl is oxidised

3. The emission of brown fumes in the given experimental set up is due to [CBSE 2023]



(a) Thermal decomposition of lead nitrate which produces brown fumes of nitrogen dioxide.
 (b) Thermal decomposition of lead nitrate which produces brown fumes of lead oxide.

(c) Oxidation of lead nitrate forming lead oxide and nitrogen dioxide.

(d) Oxidation of lead nitrate forming lead oxide and oxygen

4. $\text{MnO}_2 + x\text{HCl} \rightarrow \text{MnCl}_2 + y\text{H}_2\text{O} + z\text{Cl}_2$

In order to balance the above chemical equation, the values of x , y and z respectively are: [CBSE 2023]

(a) 6, 2, 2 (b) 4, 1, 2
 (c) 4, 2, 1 (d) 2, 2, 1

5. Which of the following metals do not corrode in moist air? [CBSE 2023]

(a) Copper (b) Iron
 (c) Gold (d) Silver

6. **Assertion (A):** The colour of aqueous solution of copper sulphate turns colourless when a piece of lead is added to it.

Reason (R): Lead is more reactive than copper, and hence displaces copper from its salt solution.

[CBSE 2023]

(a) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (b) Both (A) and (R) are true but (R) is not the correct explanation of (A).
 (c) (A) is true, but (R) is false.
 (d) (A) is false, but (R) is true.

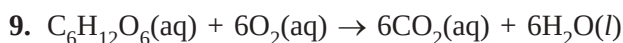
7. A student took sodium sulphate solution in a test tube and added barium chloride solution to it. He observed that an insoluble substance has formed. The colour and molecular formula of the insoluble substance is [CBSE 2021]

(a) Grey, Ba_2SO_4 (c) Yellow, $\text{Ba}(\text{SO}_4)_2$
 (b) White, BaSO_4 (d) Pink, BaSO_4

8. Sodium reacts with water to form sodium hydroxide

and hydrogen gas. The balanced equation which represents the above reaction is [CBSE 2021]

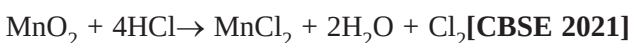
- (a) $\text{Na(s)} + 2\text{H}_2\text{O(l)} \rightarrow 2\text{NaOH(aq)} + 2\text{H}_2\text{(g)}$
- (b) $2\text{Na(s)} + 2\text{H}_2\text{O(l)} \rightarrow 2\text{NaOH(aq)} + \text{H}_2\text{(g)}$
- (c) $2\text{Na(s)} + 2\text{H}_2\text{O(l)} \rightarrow \text{NaOH(aq)} + 2\text{H}_2\text{(g)}$
- (d) $2\text{Na(s)} + \text{H}_2\text{O(l)} \rightarrow 2\text{NaOH(aq)} + 2\text{H}_2\text{(g)}$



The above reaction is a/an [CBSE 2021]

- (a) displacement reaction
- (b) endothermic reaction
- (c) exothermic reaction
- (d) neutralisation reaction

10. Which of the following statements about the reaction given below are correct?



- (i) HCl is oxidized to Cl_2
- (ii) MnO_2 is reduced to MnCl_2
- (iii) MnCl_2 acts as an oxidizing agent
- (iv) HCl acts as an oxidizing agent
- (a) (ii), (iii) and (iv) (b) (i), (ii) and (iii)
- (c) (i) and (ii) only (d) (iii) and (iv) only

11. It is important to balance the chemical equations to satisfy the law of conservation of mass. Which of the following statements of the law is incorrect?

[CBSE 2021]

- (a) The total mass of the elements present in the reactants is equal to the total mass of the elements presents in the products.
- (b) The number of atoms of each element remains the same, before and after a chemical reaction.
- (c) The chemical composition of the reactants is the same before and after the reaction.
- (d) Mass can neither be created nor can it be destroyed in a chemical reaction.

12. Which one of the following reactions is categorised as thermal decomposition reaction? [CBSE 2021]

- (a) $2\text{H}_2\text{O(l)} \rightarrow 2\text{H}_2\text{(g)} + \text{O}_2\text{(g)}$
- (b) $2\text{AgBr(s)} \rightarrow 2\text{Ag(s)} + \text{Br}_2\text{(g)}$
- (c) $2\text{AgCl(s)} \rightarrow 2\text{Ag(s)} + \text{Cl}_2\text{(g)}$
- (d) $\text{CaCO}_3\text{(s)} \rightarrow \text{CaO(s)} + \text{CO}_2\text{(g)}$

13. **Assertion (A):** Burning of natural gas is an

endothermic process.

Reason (R): Methane gas combines with oxygen to produce carbon dioxide and water [CBSE 2021]

- (a) Both (A) and (R) are true and (R) is the correct explanation of (A).
- (b) Both (A) and (R) are true but (R) is not the correct explanation of (A).
- (c) (A) is true, but (R) is false.
- (d) (A) is false, but (R) is true.

14. When aqueous solutions of potassium iodide and lead nitrate are mixed, an insoluble substance separates out. The chemical equation for the reaction involved is [CBSE 2023]

- (a) $\text{KI} + \text{PbNO}_3 \rightarrow \text{PbI} + \text{KNO}_3$
- (b) $2\text{KI} + \text{Pb(NO}_3)_2 \rightarrow \text{PbI}_2 + \text{KNO}_3$
- (c) $\text{KI} + \text{Pb(NO}_3)_2 \rightarrow \text{PbI} + \text{KNO}_3$
- (d) $\text{KI} + \text{PbNO}_3 \rightarrow \text{PbI}_2 + \text{KNO}_3$

15. A metal ribbon 'X' burns in oxygen with a dazzling white flame forming a white ash 'Y'. The correct description of X, Y and the type of reaction is [CBSE 2023]

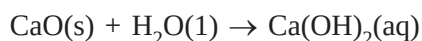
- (a) X = Ca; Y = CaO; Type of reaction = Decomposition
- (b) X = Mg; Y = MgO; Type of reaction = Combination
- (c) X = Al; Y = Al_2O_3 ; Type of reaction = Thermal decomposition
- (d) X = Zn; Y = ZnO; Type of reaction = Endothermic.

16. **Assertion (A):** Reaction of quicklime with water is an exothermic reaction.

Reason (R): Quicklime reaction vigorously with water releasing a large amount of heat. [CBSE 2021]

- (a) Both (A) and (R) are true and (R) is the correct explanation of (A).
- (b) Both (A) and (R) are true but (R) is not the correct explanation of (A).
- (c) (A) is true, but (R) is false.
- (d) (A) is false, but (R) is true.

17. Calcium oxide reacts vigorously with water to produce slaked lime.



This reaction can be classified as: [CBSE 2020]

- A. Combination reaction
- B. Exothermic reaction
- C. Endothermic reaction

D. Oxidation reaction

Which of the following is a correct option?

- (a) (A) and (C) (b) (C) and (D)
(c) (A), (C) and (D) (d) (A) and (B)

OR

When hydrogen sulphide gas is passed through a blue solution of copper sulphate, a black precipitate of copper sulphide is obtained and the sulphuric acid so formed remains in the solution. The reaction is an example of a: [CBSE 2020]

- (a) Combination reaction
(b) Displacement reaction
(c) Decomposition reaction
(d) Double displacement reaction

18. In a double displacement reaction such as the reaction between sodium sulphate solution and barium chloride solution: [CBSE 2020]

- A. exchange of atoms takes place
B. exchange of ions takes place
C. a precipitate is produced
D. an insoluble salt is produced

The correct option is:

- (a) (B) and (D) (c) only (B)
(b) (A) and (C) (d) (B), (C) and (D)

Short Answer Type Questions

19. Translate the following statements into chemical equations and then balance them: [CBSE 2024]

- (i) Solution of barium chloride and aluminium sulphate in water react to give insoluble barium sulphate and the solution of aluminium chloride.
(ii) Aluminium metal reacts with steam to give aluminium oxide and hydrogen gas.

20. Write one chemical equation each for the chemical reaction in which the following have taken place:

- (i) Change in colour
(ii) Change in temperature
(iii) Formation of precipitate

Mention colour change/temperature change (rise/fall)/compound precipitated along with equation.

[CBSE 2024]

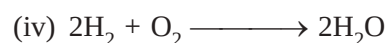
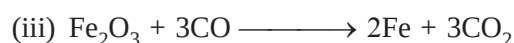
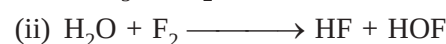
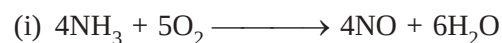
21. (i) Define a decomposition reaction. Write chemical

equation for the reaction that occurs when lead nitrate is heated strongly in a boiling tube.

- (ii) In electrolytic decomposition of water two gases are liberated at the electrodes. Give the mass ratio of the gas liberated at the cathode and at the anode.

[CBSE 2024]

22. (a) Identify the reduction agent in the following reactions: [CBSE 2023]



- (b) Define a redox reaction in term of gain or loss of oxygen.

23. (i) While electrolysis of water before passing the current some drops of an acid are added. Why? Name the gases liberated at cathode and anode. Write the relationship between the volume of gas collected at anode and the volume of gas collected at cathode

- (ii) What is observed when silver chloride is exposed to sunlight? Give the type of reaction involved.

[CBSE 2023]

24. 1 g of copper powder was taken in a China dish and heated. What change takes place on heating? When hydrogen gas is passed over this heated substance, a visible change is seen in it. Give the chemical equations of reactions, the name and the color of the products formed in each case. [CBSE 2020]

25. Identify the type of reactions taking place in each of the following cases and write the balanced chemical equation for the reactions. [CBSE 2019]

- (i) Zinc reacts with silver nitrate to produce zinc nitrate and silver
(ii) Potassium iodide reacts with lead nitrate to produce potassium nitrate and lead iodide

26. 2 g of silver chloride is taken in a china dish and the china dish is placed in sunlight for sometime. What will be your observation in this case? Write the chemical reaction involved in the form of a balanced chemical equation. Identify the type of chemical reaction. [CBSE 2019]



Explanations

Previous Year Questions

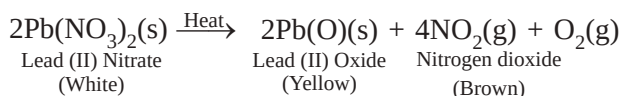
Multiple Choice Questions

- (c) $2\text{AgBr} \xrightarrow{\text{Light}} 2\text{Ag} + \text{Br}_2$
- (d) MnO_2 is reduced and HCl is oxidised.

Explanation: Here HCl is oxidized to Cl_2 and MnO_2 is reduced to MnCl_2 . The reaction in which oxygen is either gained or hydrogen is lost by a substance is called oxidation reaction.

The reaction in which hydrogen is gained or oxygen is lost by a substance is called reduction reaction.

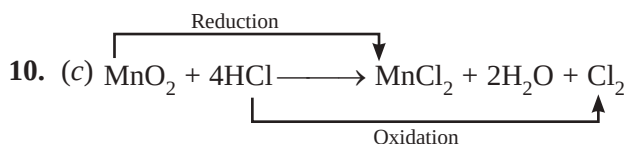
- (a) Thermal decomposition of lead nitrate which produces brown fumes of nitrogen dioxide.



- (c) $\text{MnO}_2 + 4\text{HCl} \longrightarrow \text{MnCl}_2 + 2\text{H}_2\text{O} + \text{Cl}_2$
- (c) Gold metal do not corrode in moist air.
- (a) Both (A) and (R) are true and (R) is the correct explanation of (A).

- (c) A white colour is formed on the surface
 $\text{Na}_2\text{SO}_4(\text{aq}) + \text{BaCl}_2(\text{aq}) \rightarrow \text{BaSO}_4(\text{s}) + 2\text{NaCl}(\text{aq})$
White

- (b) $2\text{Na}(\text{s}) + 2\text{H}_2\text{O}(\text{l}) \rightarrow 2\text{NaOH}(\text{aq}) + \text{H}_2(\text{g})$
- (c) The given reaction represents respiration which is an exothermic reaction.



In the given reaction:

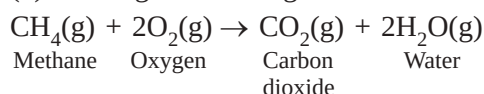
MnO_2 is oxidizing agent and HCl is reducing agent.

- (c) In a chemical reaction, the composition of reactants changes to give new product(s).
- (d)

	Reaction	Type of reaction
(a)	$2\text{H}_2\text{O}(\text{l}) \rightarrow 2\text{H}_2(\text{g}) + \text{O}_2(\text{g})$	Electrolytic decomposition

(b)	$2\text{AgBr}(\text{s}) \rightarrow 2\text{Ag}(\text{s}) + \text{Br}_2(\text{g})$	Photolytic decomposition
(c)	$2\text{AgCl}(\text{s}) \rightarrow 2\text{Ag}(\text{s}) + \text{Cl}_2(\text{g})$	Photolytic decomposition
(d)	$\text{CaCO}_3(\text{s}) \rightarrow \text{CaO}(\text{s}) + \text{CO}_2(\text{g})$	Thermal decomposition

- (d) Burning of natural gas is an exothermic process.



- (b) $2\text{KI}(\text{aq}) + \text{Pb}(\text{NO}_3)_2(\text{aq}) \rightarrow \text{PbI}_2(\text{s}) + 2\text{KNO}_3$

- (b) $2\text{Mg}(\text{s}) + \text{O}_2(\text{g}) \rightarrow 2\text{MgO}(\text{s})$
'X'
'Y'

The type of reaction involved is combination reaction.

- (a) $\text{CaO}(\text{s}) + \text{H}_2\text{O}(\text{l}) \rightarrow \text{Ca}(\text{OH})_2(\text{aq}) + \text{Heat}$
Quicklime
Water
Calcium hydroxide

- (d) (A) and (B)

OR

- (d) Double displacement reaction

- (d) (B), (C) and (D)

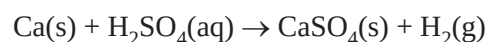
Short Answer Type Questions

- (i) $3\text{BaCl}_2(\text{aq}) + \text{Al}_2(\text{SO}_4)_3(\text{aq}) \rightarrow 2\text{AlCl}_3(\text{aq}) + 3\text{BaSO}_4(\text{s})$

- (ii) $2\text{Al}(\text{s}) + 3\text{H}_2\text{O}(\text{g}) \rightarrow \text{Al}_2\text{O}_3(\text{s}) + 3\text{H}_2(\text{g})$

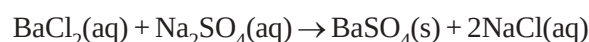
- (i) **Change in colour:** The reaction between lead nitrate solution and potassium iodide solution
 $\text{Pb}(\text{NO}_3)_2(\text{aq}) + 2\text{KI} \rightarrow \text{PbI}_2(\text{s}) + 2\text{KNO}_3(\text{aq})$
White
Yellow

- (ii) **Change in temperature:** The action of dilute sulphuric acid on calcium.



In this reaction, heat evolves.

- (iii) **Formation of precipitate:** The action of barium chloride on sodium sulphate.



Here, $\text{BaSO}_4(\text{s})$ is a white precipitate.

- (i) A type of chemical reaction in which one complex reactant breaks down into two or more simpler

$$2\text{Pb}(\text{NO}_3)_2(\text{s}) \xrightarrow{\text{heat}} 2\text{PbO}(\text{s}) + 4\text{NO}_2(\text{g}) + \text{O}_2(\text{g})$$
$$2\text{H}_2\text{O}(\ell) \longrightarrow 2\text{H}_2(\text{g}) + \text{O}_2(\text{g})$$

At cathode At anode

22. (a) The substance that oxidized is the reducing agent and the substance which gets reduced is the oxidizing agent.

There is the oxidation of ammonia (NH_3) and reduction of oxygen (O_2) taking place so here NH_3 is the reducing agent and O_2 is oxidising agent.

There is the oxidation of H_2O and reduction of F_2 taking place so here H_2O is the reducing agent and F_2 is oxidising agent.

There is the oxidation of CO and reduction of Fe_2O_3 taking place so here CO is the reducing agent and Fe_2O_3 is oxidising agent.

There is the oxidation of H_2 and reduction of O_2 taking place so here H_2 is the reducing agent and O_2 is oxidising agent.

Example: $2\text{Cu} + \text{O}_2 = 2\text{CuO}$

The volume of gas collected at cathode is double in volume of gas collected at anode.

The type of reaction involved is photochemical decomposition reaction.

$$\underset{\text{Brown}}{2\text{Cu}} + \text{O}_2 \xrightarrow{\text{Heat}} \underset{\text{Copper oxide; Black colour}}{2\text{CuO}}$$
$$\underset{\text{Black}}{2\text{CuO}} + \text{H}_2 \xrightarrow{\text{Heat}} \underset{\text{Copper; Brown}}{\text{Cu}} + \text{H}_2\text{O}$$
$$\text{Zn(NO}_3)_2(\text{aq}) + 2\text{Ag(s)}$$

Zinc nitrate
Silver

(ii) $2\text{KI}(\text{aq}) + \text{Pb}(\text{NO}_3)_2(\text{aq}) \longrightarrow$
Potassium Lead nitrate
iodide $\text{PbI}_2(\text{s}) +$

$$\text{PbI}_2(\text{s}) + 2\text{KNO}_3(\text{aq})$$

Lead iodide Potassium
(Yellow ppt.) Nitrate

26. Observation: white silver chloride turns grey in sunlight due to the decomposition of silver chloride into silver and chlorine.

$$\underset{\text{Silver chloride}}{2\text{AgCl(s)}} \xrightarrow{\text{Sunlight}} \underset{\text{Silver}}{2\text{Ag(s)}} + \underset{\text{Chlorine}}{\text{Cl}_2\text{(g)}}$$

Type of chemical reaction – Decomposition reaction

2

CHAPTER

Acids, Bases and Salts



Previous Year Questions

Multiple Choice Questions

1. When 2 mL of sodium hydroxide solution is added to few pieces of granulated zinc in a test tube and then warmed, the reaction that occurs can be written in the form of a balanced chemical equation as: [CBSE 2024]

- (a) $\text{NaOH} + \text{Zn} \rightarrow \text{NaZnO}_2 + \text{H}_2\text{O}$
(b) $2\text{NaOH} + \text{Zn} \rightarrow \text{Na}_2\text{ZnO}_2 + \text{H}_2$
(c) $2\text{NaOH} + \text{Zn} \rightarrow \text{NaZnO}_2 + \text{H}_2$
(d) $2\text{NaOH} + \text{Zn} \rightarrow \text{Na}_2\text{ZnO}_2 + \text{H}_2\text{O}$

2. Consider the following compounds: FeSO_4 ; CuSO_4 ; CaSO_4 ; Na_2CO_3

The compound having maximum number of water of crystallisation in its crystalline form in one molecule is: [CBSE 2024]

- (a) FeSO_4
(b) CuSO_4
(c) CaSO_4
(d) Na_2CO_3

3. Solid Calcium oxide reacts vigorously with water to form Calcium hydroxide accompanied by the liberation of heat. From the information given above it may be concluded that this reaction [CBSE 2024]

- (a) is endothermic and pH of the solution formed is more than 7.
(b) is exothermic and pH of the solution formed is 7.
(c) is endothermic and pH of the solution formed is 7.
(d) is exothermic and pH of the solution formed is more than 7.

4. Tooth decay begins at the pH of [CBSE 2024]

- (a) 5.1 (b) 5.8
(c) 6.5 (d) 8.0

5. Select a pair of natural indicator from the following: [CBSE 2024]

- (a) Litmus and methyl orange
(b) Turmeric and Litmus
(c) Phenolphthalein and methyl orange
(d) Methyl orange and Turmeric

6. The acid produced in our stomach during digestion and the base used to neutralise the excess acid during indigestion respectively are: [CBSE 2024]

- (a) HCl , $\text{Mg}(\text{OH})_2$
(b) HCl , $\text{Ca}(\text{OH})_2$
(c) Amino acids, $\text{Ca}(\text{OH})_2$
(d) Lactic acid, $\text{Mg}(\text{OH})_2$

7. The salt present in tooth enamel is: [CBSE 2024]

- (a) Calcium phosphate (b) Magnesium phosphate
(c) Sodium phosphate (d) Aluminium phosphate

8. An aqueous solution of sodium chloride is prepared in distilled water. The pH of this solution is:

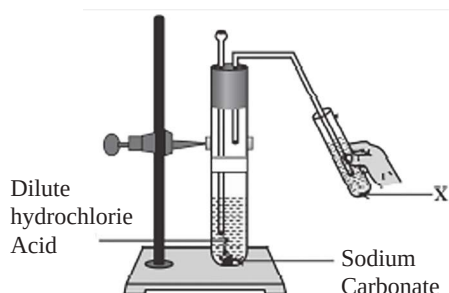
[CBSE 2024]

- (a) 6 (b) 8
(c) 7 (d) 3

9. An aqueous solution 'A' turns phenolphthalein solution pink. When another aqueous solution 'B' is added to the pink solution, the pink colour disappears. Now when a few drops of solution 'A' are added to this reaction, the mixture appears pink again. The respective changes in the nature of the solution are from: [CBSE 2024]

- (a) acidic $\rightarrow$ basic $\rightarrow$ basic
- (b) basic $\rightarrow$ acidic $\rightarrow$ acidic
- (c) acidic $\rightarrow$ basic $\rightarrow$ acidic
- (d) basic $\rightarrow$ acidic $\rightarrow$ basic

10. In the experimental setup given below, it is observed that on passing the gas produced in the reaction in the solution 'X' the solution 'X' first turns milky and then colourless.



The option that justifies the above stated observation is that 'X' is aqueous calcium hydroxide and

[CBSE 2023]

- (a) it turns milky due to carbon dioxide gas liberated in the reaction and after sometime it becomes colourless due to formation of calcium carbonate.
 - (b) it turns milky due to formation of calcium carbonate and on passing excess of carbon dioxide it becomes colourless due to formation of calcium hydrogen carbonate which is soluble in water.
 - (c) it turns milky due to passing of carbon dioxide through it. It turns colourless as on further passing carbon dioxide, sodium hydrogen carbonate is formed which is soluble in water.
 - (d) the carbon dioxide liberated during the reaction turns lime water milky due to formation of calcium hydrogen carbonate and after some time it turns colourless due to formation of calcium carbonate which is soluble in water.
11. When zinc reacts with sodium hydroxide, the product formed is: [CBSE 2023]
- (a) Sodium oxide (b) Sodium zincate
 - (c) Zinc hydroxide (d) Zinc oxide
12. Which of the following gives the correct increasing order of acidic strength? [CBSE 2023]
- (a) Sodium chloride < Acetic acid < Hydrochloric acid
 - (b) Sodium chloride < Hydrochloric acid < Acetic acid
 - (c) Acetic acid < Sodium chloride < Hydrochloric acid

- (d) Hydrochloric acid < Sodium chloride < Acetic acid

13. When Sodium bicarbonate reacts with dilute hydrochloric acid, the gas evolved is: [CBSE 2023]

- (a) Hydrogen; it gives pop sound with burning match stick.
- (b) Hydrogen; it turns lime water milky.
- (c) Carbon dioxide; it turns lime water milky.
- (d) Carbon dioxide; it blows off a burning match stick with a pop sound

14. Sodium hydroxide is termed an alkali while Ferric hydroxide is not because: [CBSE 2023]

- (a) Sodium hydroxide is a strong base, while Ferric hydroxide is a weak base.
- (b) Sodium hydroxide is a base which is soluble in water while Ferric hydroxide is also a base but it is not soluble in water.
- (c) Sodium hydroxide is a strong base while Ferric hydroxide is a strong acid.
- (d) Sodium hydroxide and Ferric hydroxide both are strong base but the solubility of Sodium hydroxide in water is comparatively higher than that of Ferric hydroxide.

15. **Assertion (A):** Reaction of Quicklime with water is an exothermic reaction. [CBSE 2023]

Reason (R): Quicklime reacts vigorously with water releasing a large amount of heat.

- (a) Both (A) and (R) are true and (R) is the correct explanation of (A).
- (b) Both (A) and (R) are true but (R) is not the correct explanation of (A).
- (c) (A) is true but (R) is false.
- (d) (A) is false but (R) is true.

16. Select a pair of olfactory indicators from the following: [CBSE 2023]

- (a) Clove oil and vanilla essence
- (b) Onion and turmeric
- (c) Clove oil and litmus solution
- (d) Vanilla and methyl orange

17. Which of the following statements is true for an amphoteric oxide? [CBSE 2023]

- (a) It reacts only with acid and does not form water.
- (b) It reacts with acid as well as base to form salt and hydrogen gas.
- (c) It reacts with both acid as well as base to form salt and water.
- (d) It reacts only with base and does not form water.

18. Hydronium ions are formed by the reaction between:
[CBSE 2023]

- (a) Sodium hydroxide and water
- (b) Calcium chloride and water
- (c) Hydrogen chloride gas and water
- (d) Ethanol and water

19. Few drops of aqueous solution of ammonium chloride are put on a universal indicator paper. The paper turns pink.

Study the following table and choose the correct option. [CBSE 2023]

	Nature	Ammonium chloride is a salt of ...	Range of pH
(a)	acidic	weak acid and strong base	less than 7
(b)	basic	weak acid and strong base	more than 7
(c)	acidic	strong acid and weak base	less than 7
(d)	basic	strong acid and strong base	7

20. Two salts X and Y are dissolved in water separately. When phenolphthalein is added to these two solutions, the solution 'X' turns pink and the solution Y does not show any change in colour, therefore 'X' and 'Y' are [CBSE 2023]

	(X)	(Y)
(a)	Na_2CO_3	NH_4Cl
(b)	Na_2SO_4	NaHCO_3
(c)	NH_4Cl	Na_2SO_4
(d)	NaNO_3	Na_2SO_4

21. **Assertion (A):** It is advised that while diluting an acid one should add water to acid and not acid to water keeping the solution continuously stirred.

Reason (R): The process of dissolving an acid into water is highly exothermic. [CBSE 2023]

- (a) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of (A).
- (b) Both Assertion (A) and Reason (R) are true but Reason (R) is not the correct explanation of (A).
- (c) Assertion (A) is true, but Reason (R) is false.
- (d) Assertion (A) is false, but Reason (R) is true.

22. A solution turns the colour of turmeric to reddish brown. If the same solution is poured on universal indicator, its colour would change to: [CBSE 2023]

- (a) violet
- (b) blue
- (c) red
- (d) green

23. There are four solutions A, B, C, and D with pH values as follows:

Solution	A	B	C	D
pH	2.0	7.0	8.0	12.0

Which solution(s) would liberate hydrogen gas with zinc? [CBSE 2023]

- (a) A only
- (b) D only
- (c) A and D
- (d) B and C

24. An aqueous solution 'A' turns phenolphthalein solution pink. On addition of an aqueous solution 'B' to 'A' the pink colour disappears. The following statement is true for solution 'A' and 'B': [CBSE 2020]

- (a) A is strongly basic and B is a weak base.
- (b) A is strongly acidic and B is a weak acid.
- (c) A has pH greater than 7 and B has pH less than 7.
- (d) A has pH less than 7 and B has pH greater than 7

Short Answer Type Questions

25. Name the type of chemical reaction in which calcium oxide reacts with water. Justify your answer by giving balanced chemical equation for the chemical reaction. [CBSE 2024]
26. (i) The pH of a sample of tomato juice is 4.6. How is this juice likely to be in taste? Give reason to justify your answer. [CBSE 2024]
- (ii) How do we differentiate between a strong acid and a weak base in terms of ion-formation in aqueous solutions?
- (iii) The acid rain can make the survival of aquatic animals difficult. How?
27. A compound which is prepared from gypsum has the property of hardening when water is mixed in right quantity with it: [CBSE 2024]
- (i) Write common name and the chemical name of this compound.
 - (ii) Give chemical equation for its preparation.
 - (iii) List its two uses.
28. 1 gram of solid sodium chloride was taken in a clean and dry test tube and concentrated sulphuric acid was added to it. [CBSE 2024]
- (i) Name the gas evolved in the reaction.
 - (ii) What will be observed when this gas is tested with (I) dry, and (II) wet blue litmus paper? Write your conclusion about the nature (acidic/basic) of this

gas.

OR

Some metals react with acids to produce salt and hydrogen gas. Illustrate it with an example. How will you test the presence of this gas?

29. (a) (i) A compound 'X' which is prepared from gypsum has the property of hardening when mixed with proper quantity of water.

Identify 'X' and write its chemical formula.

[CBSE 2023]

- (ii) State the difference in chemical composition between baking soda and baking powder.

OR

- (b) Write balanced chemical equation for the reaction that occurs when: [CBSE 2023]

(i) blue coloured copper sulphate crystals are heated.

(ii) Sodium hydrogen carbonate is heated during cooking.

30. Write the chemical composition of tooth enamel. Under what conditions of pH it starts corroding? Explain the reason of tooth decay and suggest one method to prevent it. [CBSE 2023]

31. The industrial process used for the manufacture of caustic soda involves electrolysis of an aqueous solution of compound 'X'. In this process, two gases 'Y' and 'Z' are liberated. 'Y' is liberated at cathode and 'Z', which is liberated at anode, on treatment with dry slaked lime forms a compound 'B'. Name X, Y, Z and B. [CBSE 2023]

32. (i) Why is acidified water considered to be a good conductor of electricity? [CBSE 2023]

(ii) Write a chemical equation showing the ionic products formed on dissolving potassium hydroxide in water.

(iii) Care must be taken while diluting concentrated nitric acid with water. Why?

33. Consider the following salts: [CBSE 2023]

(i) YCl (ii) NH_4X

(iii) Z_2CO_3

(a) What would be the pH of the salt solution if in YCl, Y is sodium? Give reason for your answer.

(b) If in salt NH_4X , X is nitrate, then its solution will give what colour with universal indicator? Why?

(c) What would be the change in colour in blue litmus solution if Z_2CO_3 is added to it and Z is potassium?

34. (a) Sometimes the pH of our mouth gets lower than 5-5. Why? [CBSE 2023]

(b) A basic salt 'X' is obtained by heating baking soda followed by crystallisation. Identify 'X' and state its two industrial uses.

(c) Why do copper sulphate crystals turn white on heating?

35. (a) How is a universal indicator better than litmus solution? [CBSE 2023]

(b) State two features of pH scale.

(c) Why should we not add water to a concentrated acid?

36. (i) Draw a labelled diagram to show the preparation of hydrogen chloride gas in laboratory. [CBSE 2020]

(ii) Test the gas evolved first with dry and then with wet litmus paper. In which of the two cases, does the litmus paper show change in colour?

(iii) State the reason of exhibiting acidic character by dry HCl gas/HCl solution.

37. Give reasons for the following: [CBSE 2020]

(i) Only one half of water molecule is shown in the formula of plaster of Paris.

(ii) Sodium hydrogen carbonate is used as an antacid.

(iii) On strong heating, blue coloured copper sulphate crystals turn white.

38. During electrolysis of brine, a gas 'G' is liberated at anode. When this gas 'G' is passed through slaked lime, a compound 'C' is formed, which is used for disinfecting drinking water. [CBSE 2020]

(i) Write formula of 'G' and 'C'.

(ii) State the chemical equations involved.

(iii) What is common name of compound 'C'? Give its chemical name.

39. Out of HCl and CH_3COOH , which one is a weak acid and why? Explain with the help of an example. [CBSE 2020]

40. A solution 'X' gives orange colour when a drop of it falls on pH paper, while another solution 'Y' gives bluish colour when a drop of it falls on pH paper. What is the nature of both the solutions? Determine the pH of solutions 'X' and 'Y'. [CBSE 2020]

Long Answer Type Questions

41. Salts play a very important role in our daily life. Sodium chloride which is known as common salt is

used almost in every kitchen. Baking soda is also a salt used in faster cooking as well as in baking industry. The family of salts is classified on the basis of cations and anions present in them.

[CBSE 2024]

- (a) Identify the acid and base from which Sodium chloride is formed.
- (b) Find the cation and the anion present in Calcium sulphate.
- (c) "Sodium chloride and washing soda both belong to the same family of salts." Justify this statement.

OR

Define the term pH scale. Name the salt obtained by the reaction of Potassium hydroxide and Sulphuric acid and give the pH value of its aqueous solution.

42. (i) Five solutions A, B, C, D and E when tested with pH paper showed pH as 4, 1, 13, 7 and 10 respectively. Which solution is: [CBSE 2024]

- (1) Strongly acidic
- (2) Strongly alkaline
- (3) Weakly acidic
- (4) Neutral and
- (5) Weakly alkaline?

Arrange the solutions in increasing order of H^+ ion concentration.

- (ii) Write the name and formula of (1) an acidic salt and (2) a basic salt giving the name of the parent acid and parent base used to form the salt in each case.

43. A few crystals of ferrous sulphate were taken in a dry boiling tube and heated. Tiny water droplets were observed in the tube after some time.[CBSE 2024]

- (i) From where did these water droplets appear? Explain.
- (ii) What colour change will be observed during heating?
- (iii) How many molecules of water are attached per molecule of $FeSO_4$ crystal? Write the molecular formula of crystalline forms of (I) Copper sulphate, and (II) Sodium carbonate.
- (iv) State how is Plaster of Paris obtained from gypsum. Write two uses of Plaster of Paris

44. An acid 'X' present in tamarind when mixed with

'Y', produces a mixture 'Z'. 'Z' on addition to a dough when heated makes cakes soft and spongy. 'Y' is prepared from common salt and helps in faster cooking. [CBSE 2024]

- (i) Write the common names of 'X', 'Y' and 'Z', and the chemical formula of 'Y'.
- (ii) How is 'Y' prepared and how does it help in making cakes soft and spongy? Illustrate the reaction with suitable chemical equation.
- (iii) Write the name and chemical formula of a mild base other than 'Y' used as an antacid.

45. The teacher while conducting practicals in the laboratory divided the students into three groups and gave them various solutions to find out their pH and classify them into acidic, basic and neutral solutions.

Group A-Lemon juice, vinegar, colourless aerated drink

Group B-Tomato juice, coffee, ginger juice

Group C-Sodium hydroxide, sodium chloride, lime water [CBSE 2023]

- (a) For the solutions provided, which group is/are likely to have pH value (i) less than 7, and (ii) greater than 7?
- (b) List two ways of determining pH of a solution
- (c) Explain, why the sour substances such as lemon juice are effective in cleaning the tarnished copper vessels.

46. (a) On heating 'X' at 373 K, it loses water molecules and becomes 'Y'. Y is a substance which doctors use for supporting fractured bones in the right position.

[CBSE 2023]

- (i) Identify 'X' and 'Y'.
- (ii) How can 'X' be reobtained from 'Y'?

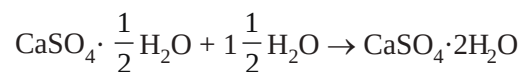
OR

- (b) Two solutions M and N give Red and Blue colour respectively with a universal indicator.

[CBSE 2023]

- (i) In which solution will the hydrogen ion concentration be more? Justify your answer.
- (ii) If both M and N solutions are mixed and the resultant mixture is tested with a universal

indicator, it turns green. What is the nature of the salt formed? Justify your answer.



47. A cloth strip dipped in onion juice is used for testing a liquid 'X'. The liquid 'X' changes its odour. Which type of an indicator is onion juice? 'The liquid 'X' turns blue litmus red. List the observations the liquid 'X' will show on reacting with the following:

[CBSE 2020]

(a) Zinc granules

(b) Solid sodium carbonate

Write the chemical equations for the reactions involved.

48. Define water of crystallization. Give the chemical formula for two compounds as examples. How can it be proved that the water of crystallization makes a difference in the state and colour of the compounds?

[CBSE 2020]





Explanations

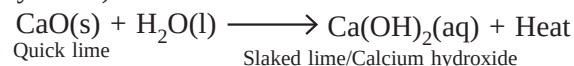
Previous Year Questions

Multiple Choice Questions

- (b) $2\text{NaOH(aq)} + \text{Zn(s)} \rightarrow \text{Na}_2\text{ZnO}_2\text{(aq)} + \text{H}_2\text{(g)}$
- (d) $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$: Green vitriol
 $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$: Blue vitriol
 $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$: Gypsum
 $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$: Washing soda
- (d) It is exothermic reaction and pH of the solution formed is more than 7.
- (a) Tooth decays begins at the pH of 5.1
- (b) Turmeric and Litmus are natural indicators.
- (a) The acid produced in our stomach during digestion and the base used to neutralise the excess acid during indigestion respectively are HCl, Mg(OH)_2
- (a) The salt present in tooth enamel is Calcium phosphate.
- (c) An aqueous solution of sodium chloride is prepared in distilled water. The pH of this solution is 7.
- (d) basic $\rightarrow$ acidic $\rightarrow$ basic
- (b)
- (b)
- (a)
- (c)
- (b)
- (a) Reaction of Quicklime with water is an exothermic reaction because Quicklime reacts vigorously with water releasing a large amount of heat.
- (a) 17. (c) 18. (c)
- (c) 20. (a) 21. (d) 22. (b) 23. (c)
- (c) As the aqueous solution of A turns phenolphthalein solution pink, hence A is basic in nature. On adding an acidic solution, the pink colour will disappear. Hence, B is an acid.

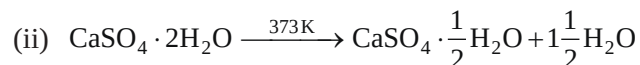
Short Answer Type Questions

25. Combination reaction - Single product is formed (or any other)



26. (i) The taste of tomato juice will be slightly sour. The pH 4.6 indicates that tomato juice is an acid and acids are sour in taste.
- (ii) Acid that give more H^+ ion/ H_3O^+ are strong acids
 Bases that give less OH^- ion are weak bases.
- (iii) Living animals can survive with a pH range of 7.0 to 7.8. So, if the pH of river water becomes low due to acid rain ($\text{pH} < 5.6$), then survival of aquatic animals becomes difficult.

27. (i) Plaster of paris; Calcium sulphate hemihydrate



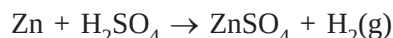
- (iii) Uses:

- Use for making toys
- Materials for decoration
- Making surfaces smooth
- Supporting fractured bones

28. (i) HCl gas was evolved.

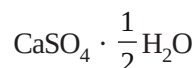
- (ii) (I) No change in colour (II) Wet blue litmus turns red

OR



Hydrogen burns with a pop sound when a burning matchstick is brought near it.

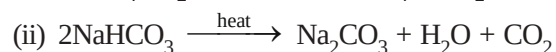
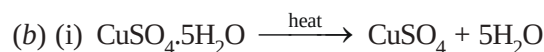
29. (a) (i) X : Plaster of Paris/Calcium sulphate hemihydrate



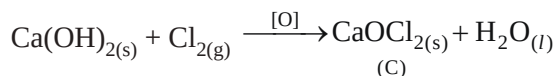
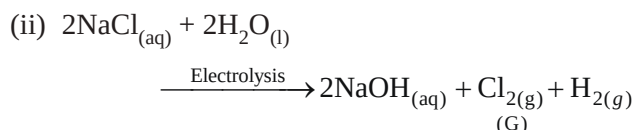
- (i) Baking Soda - NaHCO_3 /Sodium hydrogen carbonate/sodium bicarbonate

Baking powder - A mixture of NaHCO_3 /Baking soda + Tartaric acid/any mild edible acid

OR

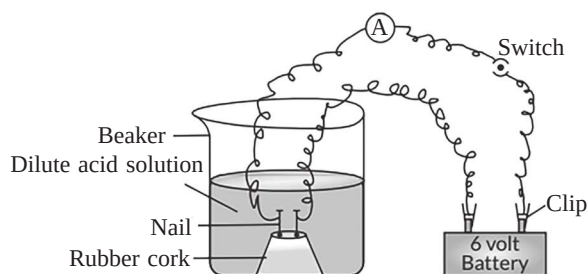


lime, bleaching powder is formed which is used for disinfecting drinking water. Hence, G is Cl_2 and C is CaOCl_2 .



(iii) Common name of C is bleaching powder. Its chemical name is calcium hypochlorite.

39. Out of HCl and CH_3COOH , CH_3COOH is a weak acid because it dissociates partially in the solution. This can be proved with the help of following example. If 1 M HCl and 1 M CH_3COOH are taken in the beaker as shown in the figure, greater deflection is observed in case of HCl which shows that more ions are produced by HCl in solution which produce more current.



40. 'X' is acidic solution having pH range 3-4. 'Y' is basic solution having pH 9.

Long Answer Type Questions

41. (a) Acid-HCl, Base-NaOH
 (b) Cation Ca^{2+} , Anion SO_4^{2-}
 (c) Salts having same cations but different anions belong to the same family of salts. e.g. sodium chloride (NaCl) and Washing Soda/sodium carbonate (Na_2CO_3) both have Na^+ as cation.

OR

A scale for measuring hydrogen ion (H^+) concentration in a solution is called pH scale.

- Potassium Sulphate/ K_2SO_4
- pH = 7

42. (i) (1) Solution B, (2) Solution C, (3) Solution A
 (4) Solution D, (5) Solution E

Increasing order of H^+ ion concentration:



- (ii) 1. Acidic salt: Ammonium chloride/ NH_4Cl
 Parent acid: Hydrochloric acid/HCl
 Parent base: Ammonium hydroxide/ NH_4OH
 2. Basic salt: Sodium carbonate/ Na_2CO_3
 Parent acid: Carbonic acid/ H_2CO_3
 Parent base: Sodium hydroxide/ NaOH

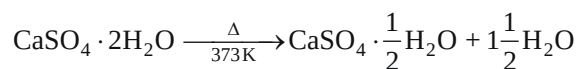
43. (i) The molecules of water of crystallisation in ferrous sulphate crystals get evaporated on heating.

(ii) Green $\rightarrow$ White

(iii) (I) $\text{CuSO}_4 \cdot 7\text{H}_2\text{O}$

(II) $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$

(iv) On heating gypsum ($\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$) at 373 K it loses water molecules



Two uses of plaster of Paris:

- Making toys/material for decoration
- Supporting fractured bones

44. (i) X-Tartaric acid, Y-Baking soda, Z-Baking powder
 $\text{Y} = \text{NaHCO}_3$

(ii) $\text{NaCl} + \text{H}_2\text{O} + \text{CO}_2 + \text{NH}_3 \rightarrow \text{NH}_4\text{Cl} + \text{NaHCO}_3$
 $\text{NaHCO}_3 + \text{H}^+ \rightarrow \text{CO}_2 + \text{H}_2\text{O} + \text{Sodium salt of acid}$
 CO_2 released during heating makes the cake soft and spongy

(iii) Magnesium hydroxide; Mg(OH)_2

45. (a) (i) Group A and B – less than 7

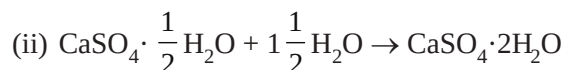
(ii) group C – greater than 7

(b) pH paper and universal indicator

(c) Copper vessel is tarnished due to formation of basic copper oxide. Lemon juice being acidic reacts with copper oxide and the salt formed is washed away.

46. (a) (i) $\text{X} = \text{CaSO}_4 \cdot 2\text{H}_2\text{O}$ /Gypsum/Calcium sulphate dihydrate

$\text{Y} = \text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O}$ /Plaster of Paris/Calcium sulphate hemi-hydrate



OR

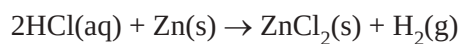
(b) (i) M, it is an acidic solution

(ii) Neutral

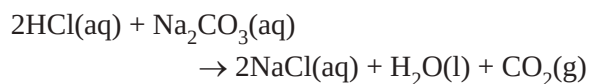
M is an acid and N is a base/Neutralization reaction

47. Onion juice is an olfactory indicator. Since the liquid 'X' turns blue litmus red, therefore, it is an acid.

(a) When liquid 'X' (an acid like HCl) reacts with zinc granules, bubbles of hydrogen will be formed



- (b) When liquid 'X' (an acid like HCl) reacts with sodium carbonate, carbon dioxide gas will be liberated.



- 48.** Water of crystallisation is the fixed number of molecules of water which are loosely bonded to one molecule of salt. For example:



We can prove that by heating the compound containing water of crystallisation, its colour or state change due to the removal of water of crystallisation. For example

- (i) Copper sulphate crystals containing water of crystallisation are blue but on heating they turn white.
- (ii) Similarly plaster of Paris is a white powder and on mixing with water it changes to gypsum which forms a hard solid mass.



3

CHAPTER

Metals and Non-metals



Previous Year Questions

Multiple Choice Questions

1. When 2 mL of sodium hydroxide solution is added to few pieces of granulated zinc in a test tube and then warmed, the reaction that occurs can be written in the form of a balanced chemical equation as:

[CBSE 2024]

- (a) $\text{NaOH} + \text{Zn} \rightarrow \text{NaZnO}_2 + \text{H}_2\text{O}$
(b) $2\text{NaOH} + \text{Zn} \rightarrow \text{Na}_2\text{ZnO}_2 + \text{H}_2$
(c) $2\text{NaOH} + \text{Zn} \rightarrow \text{NaZnO}_2 + \text{H}_2$
(d) $2\text{NaOH} + \text{Zn} \rightarrow \text{Na}_2\text{ZnO}_2 + \text{H}_2\text{O}$
2. A metal and a non-metal that exists in liquid state at the room temperature are respectively: [CBSE 2024]
(a) Bromine and Mercury
(b) Mercury and Iodine
(c) Mercury and Bromine
(d) Iodine and Mercury
3. Oxides of aluminium and zinc are: [CBSE 2024]
(a) acidic (b) basic
(c) amphoteric (d) neutral
4. **Assertion (A):** Hydrogen gas is not evolved when zinc reacts with nitric acid.
Reason (R): Nitric acid oxidises the hydrogen gas produced to water and itself gets reduced. [CBSE 2024]
(a) Both (A) and (R) are true and (R) is the correct explanation of (A).
(b) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(c) (A) is true, but (R) is false.
(d) (A) is false, but (R) is true.
5. A metal 'X' is used in thermite process. When X is

burnt in air it gives an amphoteric oxide 'Y'. 'X' and 'Y' are respectively: [CBSE 2023]

- (a) Fe and Fe_2O_3 (b) Al and Al_2O_3
(c) Fe and Fe_3O_4 (d) Al and Al_3O_4

Short Answer Type Questions

6. (a) Why is potassium kept immersed in kerosene?

OR

- (b) Write the name of an allotrope of carbon.

[CBSE 2021]

7. Given reason why:

[CBSE 2021]

- (a) Gold and silver are used for making jewellery.
(b) A few metals are used for making cooking utensils.

8. What is a displacement reaction? Write balanced chemical equation for a displacement reaction in which iron is a reactant. Name one more element whose behaviour is similar to that of iron in such reactions. Why will this kind of behaviour not be shown by gold? [CBSE 2021]

9. The atomic number of an element is 20. Write its electronic configuration. State whether this element is a metal or a non-metal. What is its valency? Write the name and formula of the compound which this element forms with chlorine. [CBSE 2021]

10. 1 g of copper powder was taken in a China dish and heated. What change takes place on heating? When hydrogen gas is passed over this heated substance, a visible change is seen in it. Give the chemical equations of reactions, the name and the color of the products formed in each case. [CBSE 2020]

11. What would you observe on adding zinc granules to freshly prepared ferrous sulphate solution? Give reason for your answer. [CBSE 2019]

12. An ore on treatment with dilute hydrochloric acid

produces brisk effervescence. Name the type of ore with one example. What steps will be required to obtain metal from the enriched ore? Also write the chemical equations for the reactions involved in the process. [CBSE 2019]

Long Answer Type Questions

13. The metals produced by various reduction processes are not very pure. They contain impurities, which must be removed to obtain pure metals. The most widely used method for refining impure metals is electrolytic refining. [CBSE 2024]

- (i) What is the cathode and anode made of in the refining of copper by this process?
- (ii) Name the solution used in the above process and write its formula.
- (iii) (a) How copper gets refined when electric current is passed in the electrolytic cell?

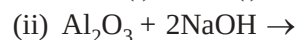
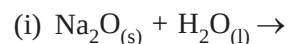
OR

(b) You have two beakers 'A' and 'B' containing copper sulphate solution. What would you observe after about 2 hours if you dip a strip of zinc in beaker 'A' and a strip of silver in beaker 'B'? Give reason for your observations in each case.

14. Almost all metals combine with oxygen to form metal

oxides. Metal oxides are generally basic in nature. But some metal oxides show both basic as well as acidic behaviour. Different metals show different reactivities towards oxygen. Some react vigorously while some do not react at all. [CBSE 2023]

- (a) What happens when copper is heated in air? (Give the equation of the reaction involved).
- (b) Why are some metal oxides categorized as amphoteric? Give one example.
- (c) Complete the following equations:



OR

- (c) On burning Sulphur in oxygen a colourless gas is produced.

(i) Write chemical equation for the reaction.

(ii) Name the gas formed.

(iii) State the nature of the gas.

(iv) What will be the action of this on a dry litmus paper?

15. Carbon cannot reduce the oxides of sodium, magnesium and aluminium to their respective metals. Why? Where are these metals placed in the reactivity series? How are these metals obtained from their ores? Take an example to explain the process of extraction along with chemical equations. [CBSE 2020]



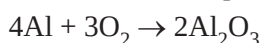
Explanations

Previous Year Questions

Multiple Choice Questions

- (b) $2\text{NaOH} + \text{Zn} \rightarrow \text{Na}_2\text{ZnO}_2 + \text{H}_2$
- (c) Mercury and Bromine
- (c) amphoteric
- (a) Both (A) and (R) are true and (R) is the correct explanation of (A).
- (c) $\text{X} = \text{Al}$ and $\text{Y} = \text{Al}_2\text{O}_3$

Reason: Aluminium is used in thermite process. When aluminium is heated in oxygen it forms aluminium oxide which is amphoteric in nature.



Short Answer Type Questions

- (a) Because potassium is highly reactive and catches fire when kept in open / water

OR

(b) Graphite/Diamond/Fullerene

- (a) Because gold and silver are most malleable / ductile / least reactive.

(b) Because such metals are good conductors of heat / have high melting points.

- A reaction in which more reactive element displaces less reactive element from its compound is known as a displacement reaction.

- $\text{Fe(s)} + \text{CuSO}_4(\text{aq}) \rightarrow \text{FeSO}_4(\text{aq}) + \text{Cu(s)}$
(any other reaction of iron with compound of element less reactive than iron)
- Zinc / lead (or any other element having reactivity comparable to iron)
- As gold is least reactive.

- Ca = 20

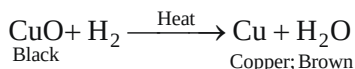
Electronic configuration: 2,8,8,2

- Metal
- Valency: 2
- Name: Calcium Chloride
- Formula: CaCl_2

- A black colour is formed on the surface



Original/brown colour is restored.

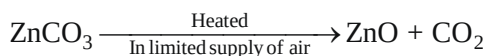


- The solution turns green to colourless due the formation of zinc sulphate and a black coating of iron is formed on zinc surface.

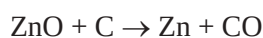
Reason: Zinc is more reactive than iron so it displaces the iron from its salt solution.

- Carbonate ore Zinc carbonate

Calcination



Reduction:



Long Answer Type Questions

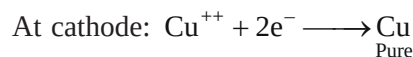
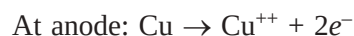
- (i) Cathode – Pure copper

Anode – Impure copper

(ii) Acidified Copper Sulphate; CuSO_4

(iii) (A)

- Pure copper from the anode dissolves into electrolyte and an equivalent amount of pure metal from the electrolyte is deposited on cathode.



- The soluble impurities go into the solution whereas insoluble impurities settle down at the bottom of the anode.

OR

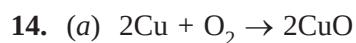
(iii) (B)

In Beaker A :

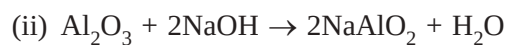
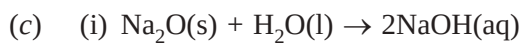
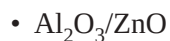
- The blue colour of the solution fades (or becomes colourless)
- Reason – Zn is more reactive than copper

In Beaker B:

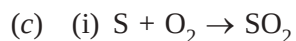
- No change in colour.
- Reason – Silver is less reactive than Copper.



(b) Because amphoteric oxides react with acid and bases to form salt and water.



OR



(ii) Sulphur dioxide

(iii) Acidic

(iv) No change

15. • These metals have more affinity for oxygen than carbon.

- Towards the top of the reactivity series .
- By electrolytic reduction of their molten ores.

Example : Extraction of sodium from molten sodium chloride by electrolysis.

Process:

- Molten NaCl is taken in an electrolytic cell and on passing electricity Na is deposited at cathode and chlorine is liberated at anode

Reactions:

At cathode - $\text{Na}^+ + \text{e}^- \rightarrow \text{Na}$

At anode - $2\text{Cl}^- \rightarrow \text{Cl}_2 + 2\text{e}^-$ (Or any other example)

4 CHAPTER

Carbon and its Compounds



Previous Year Questions

Multiple Choice Questions

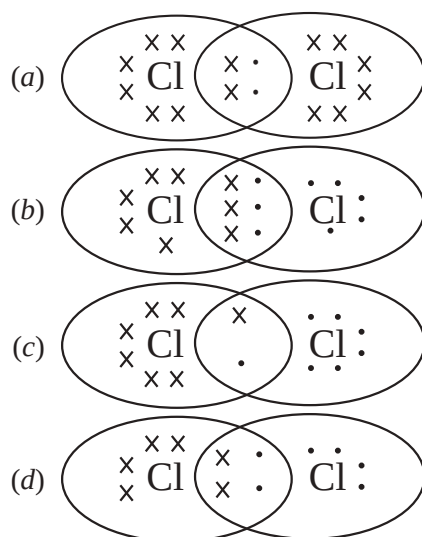
1. Carbon compounds: [CBSE 2024]

- (i) are good conductors of electricity.
- (ii) are bad conductors of electricity.
- (iii) have strong forces of attraction between their molecules.
- (iv) have weak forces of attraction between their molecules.

The correct statements are:

- (a) (i) and (ii) (b) (ii) and (iii)
- (c) (ii) and (iv) (d) (i) and (iii)

2. The election dot structure of chlorine molecule is [CBSE 2023]



3. **Assertion (A)** In a homologous series of alcohols, the formula for the second member is C_2H_5OH and the third member is C_3H_7OH .

Reason (R): The difference between the molecular

masses of the two consecutive members of a homologous series is 144. [CBSE 2020]

- (a) Both A and R are true and R is the correct explanation of the Assertion.
- (b) Both A and R are true but R is not the correct explanation of the Assertion.
- (c) A is true but R is false.
- (d) A is false but R is true.

Short Answer Type Questions

4. "Carbon prefers to share its valence electrons with other atoms of carbon or with atoms of other elements rather than gaining or losing the valence electrons in order to attain noble gas configuration." Give reasons to justify this statement. [CBSE 2022]

5. The atomic number of an element 'X' is 11. [CBSE 2022]

- (i) Write the electronic configurations of X and find its valency.
- (ii) Write the formula and nature of its oxide.

6. (a) Draw the electron dot structure for ethyne.
(b) List two differences between the properties exhibited by covalent compounds and ionic compounds. [CBSE 2022]

7. Name the functional group present in propanone [CBSE 2020]

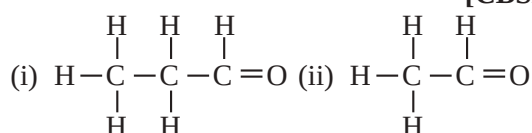
8. In three test tubes A, B and C, three different liquids namely, distilled water, underground water and distilled water in which a pinch of calcium sulphate is dissolved, respectively are taken. Equal amount of soap solution is added to each test tube

and the contents are shaken. In which test tube will the length of the foam (lather) be longest? Justify your answer. **[CBSE 2019]**

9. 3 mL of ethanol is taken in a test tube and warmed gently in a water bath. A 5% solution of alkaline potassium permanganate is added first drop by drop to this solution, then in excess. **[2020]**

- How is 5% solution of KMnO_4 prepared?
- State the role of alkaline potassium permanganate in this reaction. What happens on adding it in excess?
- Write chemical equation of this reaction.

10. Consider the following organic compounds: **[CBSE 2022]**



- Name of functional group present in these compounds.
 - Write the general formula for the compounds of this functional group.
 - State the relationship between these compounds and draw the structure of any other compound having similar functional group.
11. Write the chemical formula and name of the compound which is the active ingredient of all alcoholic drinks. List its two uses. Write chemical equation and name of the product formed when this compound reacts with **[CBSE 2019]**
- Sodium metal
 - Hot concentrated sulphuric acid

Long Answer Type Questions

12. (A) (i) Define a homologous series of carbon compounds.
- Why are the melting and boiling points of C_4H_8 higher than that of C_3H_6 or C_2H_4 ?
 - Why do we NOT see any gradation in chemical properties of a homologous series compounds?
 - Write the name and structures of (i) aldehyde and (ii) ketone with molecular form $\text{C}_3\text{H}_6\text{O}$ **[CBSE 2024]**
- OR
- (B) (i) Write the name and structure of an organic compound 'X' having two carbon atoms in

its molecule and its name is suffixed with -ol'.

- What happens when 'X' is heated with excess concentrated sulphuric acid at 443K? Write a chemical equation for the reaction stating the conditions for the reaction. Also state the role played by concentrated sulphuric acid in the reaction.
- Name and draw the electron dot structure of hydrocarbons produced in the above reaction. **[CBSE 2024]**

13. What is methane? Draw its electron dot structure. Name the type of bonds formed in this compound. Why are such compounds: **[CBSE 2019]**

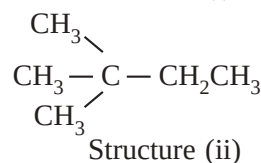
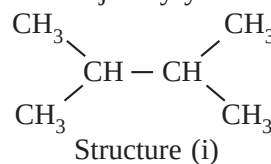
- Poor conductors of electricity? and
- Have low melting and boiling points? What happens when this compound burns in oxygen?

14. (A) (i) Draw the structure of the following compounds: **[CBSE 2023]**

(a) Butanoic acid

(b) Chloropentane

- How are structure (i) and structure (ii) given below related to one another? Give reason to justify your answer.



Draw one more possible structure for above case.

- Differentiate between saturated and unsaturated carbon compounds on the basis of their general formula.

OR

- (B) (i) What happens when a small piece of sodium is dropped in ethanol? Write the equation for this reaction.
- Why is glacial acetic acid called so?
 - What happens when ethanol is heated at 443 K in the presence of conc. H_2SO_4 ? Write the role of conc. H_2SO_4 in this case.
 - Write an equation showing saponification.



Explanations

Previous Year Questions

Multiple Choice Questions

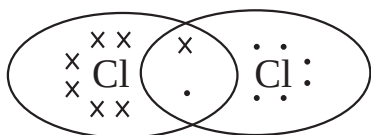
1. (c) (ii) and (iv)

Carbon compounds are bad conductor of electricity and carbon compounds have weak forces of attraction between their molecules.

2. (c) Atomic number of chlorine = 17

Electronic configuration = 2, 8, 7

Electronic dot structure of chlorine molecule:



3. (c) The difference between the molecular masses of the two consecutive members of a homologous series is 14.

Short Answer Type Questions

4. Carbon does not prefer to gain or lose electrons because:

On gaining four electrons it will form C^{4-} ion, which is highly unstable due to the large amount of energy required to overcome inter-electronic forces of repulsion.

On losing four electrons it will form C^{4+} ion, which is highly unstable due to the large amount of energy required to remove four electrons from the carbon atoms.

Therefore, in order to overcome these problems, carbon prefers to share its valence electrons with other atoms of carbon or with atoms of other elements.

5. (i) K L M

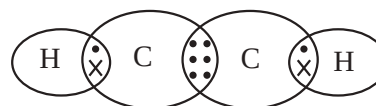
Electronic configuration of 'X' = 2 8 1

Valency of 'X' = 1

- (ii) Formula of oxide of 'X' = X_2O

Nature of Oxide of 'X' = Basic

6. (a) Electron dot structure for ethyne:



Ethyne molecule

- (b) Differences between ionic and covalent compounds:

S.N.	Covalent Compounds	Ionic Compounds
1.	ionic compounds are formed by complete transfer of electrons	Covalent compounds are formed by mutual sharing of electrons
2.	ionic compounds have high melting and boiling points	Covalent compounds have comparatively low melting and boiling points
3.	ionic compounds conduct electricity in aqueous solution or in molten state	Covalent compounds are generally non-conductors of electricity

7. The functional group present in propanone is ketone.

8. In test tube A the length of the foam (lather) will be longest.

Reason: Soap produces good lather (or foam) with soft water (distilled water) only.

Both test tubes B and C contain hard water and soap forms scum in hard water.

9. (i) 5% solution of $KMnO_4$ is prepared by dissolving 5 g of $KMnO_4$ crystals in 95 g of water. [1]

- (ii) Alkaline $KMnO_4$ acts as an oxidising agent. The colour of $KMnO_4$ will not disappear when added in excess.

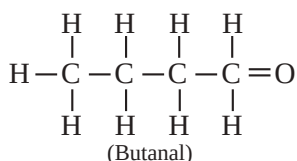


10. (a) The functional group present in these compounds ((i) and (ii)) is aldehyde ($-CHO$).

- (b) The general formula for the compounds having aldehyde functional group is $C_nH_{2n}O$.

- (c) These compounds are the members of the homologous series of aldehydes.

Structure of butanal having aldehyde functional group is



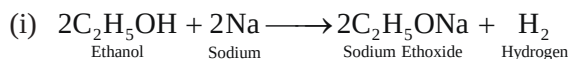
11. Chemical formula of the compound = $\text{C}_2\text{H}_5\text{OH}$

Name of compound = Ethanol

Uses of ethanol

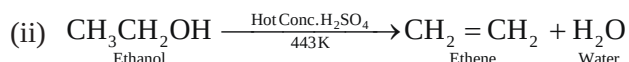
(a) It is used as a solvent in tincture of iodine.

(b) It is used as a solvent in cough syrups.



Name of the products formed = Sodium ethoxide and hydrogen

Name of the products formed Ethene and water.



Name of the products formed = Ethene and water.

Long Answer Type Questions

12. (A) (i) Homologous series may be defined as a series of similarly constituted compounds in which the members possess similar chemical characteristics and the two consecutive members differ in their molecular formula by CH_2 .

(ii) C_4H_8 , C_3H_6 and C_2H_4 belong to the same homologous series because each compound differs by CH_2 units with the preceding one and each compound belongs to the same functional group.

In homologous series, the molecular masses of succeeding members are higher than the preceding members. As the molecular mass increases, the melting and boiling points also increase. So, melting and boiling points of C_4H_8 are higher than the others in the given series.

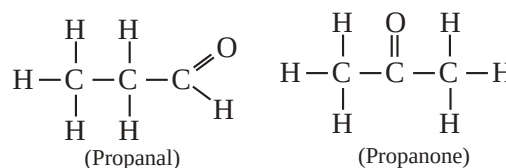
(iii) We do not observe gradation in chemical properties of homologous series compounds because they share a similar functional group and exhibit a regular increase in molecular size and mass with each successive member. This consistent structural pattern results in similar chemical behavior among homologs. Although physical properties such as boiling points and melting points may vary gradually due to increasing molecular size,

chemical properties, including reactivity and functional group behavior, remain relatively constant within a homologous series. This uniformity arises from the consistent presence and arrangement of functional groups, which dictate the compound's chemical behavior.

(iv) The aldehyde and ketone that can be represented by molecular formula $\text{C}_3\text{H}_6\text{O}$ are propanal and propanone respectively.

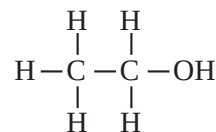
Aldehyde - Propanal - $\text{CH}_3\text{CH}_2\text{CHO}$

Ketone - Propanone - CH_3COCH_3

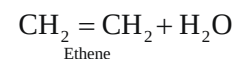
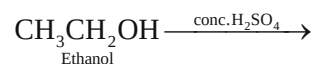


OR

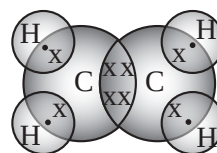
(i) The name and molecular formula of a carbon compound having its name suffixed with “-ol” and having two carbon atoms in a molecule is ethanol and $\text{CH}_3\text{CH}_2\text{OH}$, respectively. Structure of ethanol is:



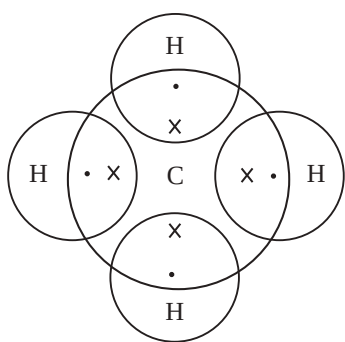
(ii) When ethanol is heated with excess of conc. H_2SO_4 , it leads to dehydration of it giving ethene.



(iii) Name of hydrocarbon produced in the above reaction is ethene.



13. Methane is a compound of carbon with chemical formula CH_4 .



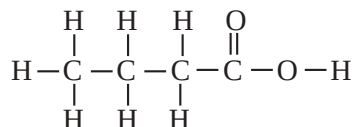
Electron dot structure of methane

Covalent bonds are formed in this compound.

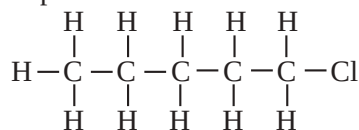
- (i) In covalently bonded molecules, the electrons are shared between atoms and no charged particles are formed. Therefore, such compounds are generally poor conductors of electricity.
- (ii) Covalently bonded molecules are seen to have strong bonds within the molecule, but have weak inter-molecular forces. This gives rise to low melting and boiling points of these compounds.

When methane burns in oxygen, CO_2 , H_2O and a large amount of heat and light is released.

14. (A) (i) (a) Butanoic acid:

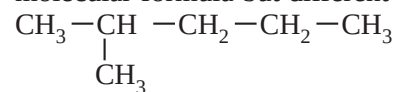


- (b) Chloropentane:



- (ii) Both the structures are structural isomers

of each other (hexane) as both have same molecular formula but different structures.

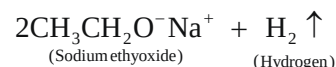
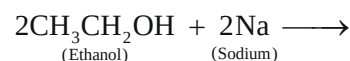


(iii)

Saturated Carbon Compounds		Unsaturated Carbon Compounds
General formula:	formula: (alkanes)	General formula:
$\text{C}_n\text{H}_{2n+2}$	(n = number of C atoms)	C_nH_{2n} (alkenes) or $\text{C}_n\text{H}_{2n-2}$ (alkynes) (n = number of C atoms)

OR

- (B) (i) Ethanol reacts with sodium leading to evolution of H_2 gas and formation of sodium ethoxide.



- (ii) Melting point of pure ethanoic acid is 290 K and hence it often freezes during winters forming white crystals resembling glaciers. This gave rise to its name glacial acetic acid.
- (iii) Heating ethanol at 443 K with concentrated sulphuric acid gives ethene on dehydration. Concentrated H_2SO_4 acts as a dehydrating agent.
- (iv) $\text{CH}_3\text{COOC}_2\text{H}_5 + \text{NaOH} \rightarrow \text{C}_2\text{H}_5\text{OH} + \text{CH}_3\text{COONa}$

1

CHAPTER

Light-Reflection and Refraction



Previous Year Questions

Multiple Choice Questions

- Absolute refractive index of glass and water is $\frac{3}{2}$ and $\frac{4}{3}$ respectively. If the speed of light in glass is 2×10^8 m/s, the speed of light in water is:
[CBSE 2024]
(a) $\frac{9}{4} \times 10^8$ m/s (b) $\frac{5}{2} \times 10^8$ m/s
(c) $\frac{7}{3} \times 10^8$ m/s (d) $\frac{16}{9} \times 10^8$ m/s
- At what distance from a convex lens should an object be placed to get an image of the same size as that of the object on a screen? [CBSE 2024]
(a) Beyond twice the focal length of the lens.
(b) At the principal focus of the lens.
(c) At twice the focal length of the lens.
(d) Between the optical center of the lens and its principal focus.
- In torch lights and head lights of vehicles, the bulb is placed [CBSE 2023]
(a) between the pole and the focus of the reflector.
(b) very near to the focus of the reflector.
(c) between the focus and centre of curvature of the reflector.
(d) at the centre of curvature of the reflector.
- An object is placed in front of a convex mirror at infinity. According to the New Cartesian Sign Convention, the sign of the focal length and the sign of the image distance in this case are respectively:
[CBSE 2023]
(a) +, - (b) -, +

(c) -, - (d) +, +

- A student wants to obtain an erect image of an object using a concave mirror of 10 cm focal length. What will be the distance of the object from mirror?
[CBSE 2023]
(a) Less than 10 cm
(b) 10 cm
(c) between 10 cm and 20 cm
(d) more than 20 cm
- To obtain a magnification of +2 with a concave mirror of radius of curvature 60 cm the object distance must be [CBSE 2023]
(a) -90 cm (b) -45 cm
(c) -30 cm (d) -15 cm
- The laws of reflection hold true for: [CBSE 2020]
(a) plane mirrors only
(b) concave mirrors only
(c) convex mirrors only
(d) all reflecting surfaces
- When an object is kept within the focus of a concave mirror, an enlarged image is formed behind the mirror. This image is: [CBSE 2020]
(a) real (b) inverted
(c) virtual and inverted (d) virtual and erect

Very Short Answer Type Questions

- An object is placed at a distance of 10 cm from a convex mirror of focal length 15 cm. Find the position of the image formed by the mirror.
[CBSE 2024]

10. A person suffering from presbyopia needs bifocal lens. If he needs two lens of power -4.0 dioptre and $+2.0$ dioptre, which one of these two lenses is for the correction of distant vision and what is its focal length? **[CBSE 2024]**

11. Draw a labelled ray diagram to show the path of the reflected ray corresponding to the ray which is incident obliquely to the principal axis, towards the pole of a convex mirror. Mark the angle of incidence and angle of reflection on it. **[CBSE 2019]**

12. A teacher gives a convex lens and a concave mirror of focal length of 20 cm each to his student and asks him to find their focal lengths by obtaining the image of a distant object. The student uses a distant tree as the object and obtains its sharp image, one by one, on a screen. The distances d_1 and d_2 between the lens/mirror and the screen in the two cases and the nature of their respective sharp images are likely to be

Give reason for your answer. **[CBSE 2019]**

- (a) (20 cm, 40 cm) and (erect and erect)
 (b) (20 cm, 40 cm) and (inverted and inverted)
 (c) (20 cm, 20 cm) and (inverted and inverted)
 (d) (20 cm, 40 cm) and (erect and inverted)

Short Answer Type Questions

13. Define the term power of accommodation of human eye. What happens to the image distance in the eye when we increase the distance of an object from the eye? Name and explain the role of the part of human eye responsible for it in this case.

[CBSE 2024]

14. The image of an object formed by lens is of the same size but inverted, if the object distance is 30 cm, calculate

[CBSE 2024]

- (a) (i) The distance between the object and its image

- (ii) Focal length of the lens

- (b) Draw a ray diagram to show the image formed in the above case.

15. Study the data given below showing the focal length of three concave mirrors A, B and C and the respective distances of objects placed in front of the

mirrors:

[CBSE 2024]

Case	Mirror	Focal Length (cm)	Object Distance (cm)
1	A	20	45
2	B	15	30
3	C	30	20

- (i) In which one of the above cases the mirror will form a diminished image of the object? Justify your answer.
 (ii) List two properties of the image formed in case 2.
 (iii) (A) What is the nature and size of the image formed by mirror C? Draw ray diagram to justify your answer.

OR

- (iii) (B) An object is placed at a distance of 18 cm from the pole of a concave mirror of focal length 12 cm. Find the position of the image formed in this case.

16. A concave mirror has a focal length of 20 cm. At what distance from the mirror should a 4 cm tall object be placed so that it forms an image at a distance of 30 cm from the mirror? Also calculate the size of the image formed. **[CBSE 2019]**

17. A real image $2/3$ rd of the size of an object is formed by a convex lens when the object is at a distance of 12 cm from it. Find the focal length of the lens.

[CBSE 2019]

18. (a) A 5 cm tall object is placed perpendicular to the principal axis of a convex lens of focal length 20 cm. The distance of the object from the lens is 30 cm. Find the position, nature and size of the image formed. **[CBSE 2019]**

- (b) Draw a labelled ray diagram showing object distance, image distance and focal length in the above case. **[CBSE 2019]**

Long Answer Type Questions

19. (a) The variation of image distance (v) with object distance (u) for a convex lens is given in the following observation table. Analyse it and answer the questions that follow: **[CBSE 2024]**

S. No.	Object Distance (u) cm	Image Distance (v) cm
1	-150	+30
2	-75	+37.5
3	-50	+50
4	-37.5	+75
5	-30	+150
6.	-15	+37.5

- Without calculation, find the focal length of the convex lens. Justify your answer.
- Which observation is not correct? Why? Draw a ray diagram to find the position of the image formed for this position of the object.
- Find the approximate value of magnification for $u = -30$ cm.

OR

- Define principal axis of a lens. Draw a ray diagram to show what happens when a ray of light parallel to the principal axis of a concave lens passes through it.
 - The focal length of a concave lens is 20 cm. At

what distance from the lens should a 5 cm tall object be placed so that its image is formed at a distance of 15 cm from the lens? Also, calculate the size of the image formed.

- 20.** Draw a ray diagram in each of the following cases to show the formation of image, when the object is placed: **[CBSE 2020]**

- between optical centre and principal focus of a convex lens.
- anywhere in front of a concave lens.
- at 2F of a convex lens.

State the signs and values of magnifications in the above mentioned cases (i) and (ii).

- 21.** An object 4.0 cm in size, is placed 25.0 cm in front of a concave mirror of focal length 15.0 cm. **[CBSE 2020]**

- At what distance from the mirror should a screen be placed in order to obtain a sharp image?
- Find the size of the image.
- Draw a ray diagram to show the formation of image in this case.



Explanations

Previous Year Questions

Multiple Choice Questions

1. (a) $\frac{9}{4} \times 10^8 \text{ m/s}$

Let n_g be the absolute refractive index of glass.

n_w be the absolute refractive index of water.

v_g and v_w be the speed of light in glass and water respectively.

Given, $n_g = \frac{3}{2}$; $n_w = \frac{4}{3}$; $v_g = 2 \times 10^8 \text{ m/s}$

We know that, $n = \frac{c}{v}$

So, $n_g = \frac{c}{v_g} = \frac{3}{2}$ and $n_w = \frac{c}{v_w} = \frac{4}{3}$

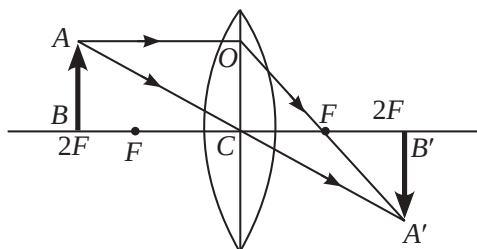
Dividing these two equations,

$$\frac{v_w}{v_g} = \frac{9}{8}$$

$$\Rightarrow v_w = \frac{9}{8} \times 2 \times 10^8 = \frac{9}{4} \times 10^8 \text{ m/s}$$

Hence, speed of light in water is $\frac{9}{4} \times 10^8 \text{ m/s}$.

2. (c) At twice the focal length of the lens.



When the object is placed at twice the focal length, the image formed by the convex lens is real and of the same size as the object

3. (b) Very near to the focus of the reflector

In torches and head lights of vehicles the bulb is placed very near to the focus of the reflector to get a parallel beam of light through which we can see to a longer distance.

4. (b) $-, +$

According to the new sign convention : the focal length of a convex mirror is negative. The image distance for a virtual image formed by a convex mirror is positive.

5. (a) Less than 10 cm

Virtual image is formed when an object is placed between pole and focus of the concave mirror.

6. (d) -15 cm

If radius of curvature is 60 cm then focal length will be 30 cm, $f = -30 \text{ cm}$

If radius of curvature is 60 cm then focal length will be 30 cm, $f = -30 \text{ cm}$.

So the object must be placed between pole and focus. So option (d) is correct.

7. (d) The laws of reflection is valid for all reflecting surfaces.

8. (d) virtual and erect

When an object is placed between pole and focus of concave mirror virtual and erect image is formed.

Very Short Answer Type Questions

9. Given that, $u = -10 \text{ cm}$, $f = +15 \text{ cm}$, $v = ?$

According to the mirror formula,

$$\frac{1}{v} = \frac{1}{f} - \frac{1}{u}$$

where, u = the distance between the object and the pole of the mirror,

v = the distance between the image and the pole of the mirror, and f = focal length

$$\Rightarrow \frac{1}{v} = \frac{1}{15} - \frac{1}{-10} \Rightarrow \frac{1}{v} = \frac{1}{15} + \frac{1}{10}$$

$$\Rightarrow \frac{1}{v} = \frac{2+3}{30} = \frac{5}{30} \Rightarrow v = \frac{30}{5} = 6 \text{ cm}$$

So, when an object is placed at a distance of 10 cm from a convex mirror of focal length 15 cm, the image will be formed at a distance of 6 cm behind

the mirror.

10. A bifocal lens consists of both concave lens and convex lens to help an individual suffering from presbyopia. The upper part of the lens consists of concave lens typically used for distant vision, and the lower part consists of convex lens for near vision.

In this case, the person needs a bifocal lens with a power of -4.0 diopters and $+2.0$ diopters. The positive sign indicates that it is a converging lens, used to correct near vision and the negative sign indicates that it is a diverging lens, used to correct distant vision. Hence, the -4.0 diopter lens is for the correction of distant vision.

We know that,

Focal length (f) of the lens is given by,

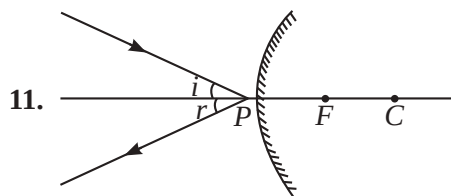
$$f = \frac{1}{P} \text{ where } P \text{ is power.}$$

Given power of lens for distant vision is -4.0 diopters.

$$\text{So, } f = \frac{1}{-4.0} \text{ m}$$

Therefore, the focal length of the -4.0 diopter lens for distant vision is -0.25 m or -25 cm.

Negative sign indicates that it is a concave lens.



12. (c) (20 cm, 20 cm) and (inverted and inverted)

Reason: Only real and inverted image can be obtained on the screen and in both cases the image is formed at the principal focus.

Short Answer Type Questions

13. The ability of the eye lens to adjust its focal length in order to see objects at different distances clearly is known as the power of accommodation of the human eye.

When the object is moved farther away, there is no change in the image distance in the eye. The image

distance remains constant as the distance between the eye lens and the image which is formed on the retina doesn't change.

To maintain clear focus on the retina, the eye needs to change its focal length so that the image is still formed on the retina. The part of the human eye responsible to do this is the ciliary muscle.

14. (a) (i) Magnification of lens

$$= \frac{\text{Image distance}}{\text{Object distance}} = \frac{\text{height of the image}}{\text{height of the object}}$$

Since image of an object formed by lens is of the same size but inverted Magnification of lens = -1

$$M = -1 = \frac{\text{Image distance}}{\text{Object distance}} = \frac{\text{image distance}}{-30}$$

Image distance = -30 cm

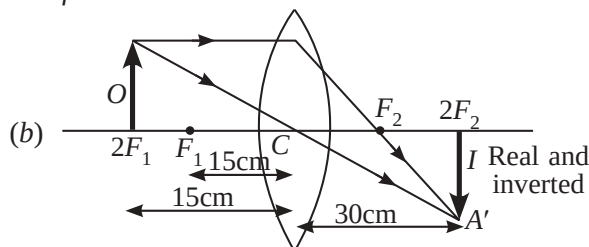
The distance between the object and it's image = $v - u = 30 - (-30) = 60$ cm

(ii) From lens formula,

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

$$\frac{1}{f} = \frac{1}{30} - \frac{1}{-30} = \frac{1}{30} + \frac{1}{30} = \frac{1}{15}$$

$f = 15$ cm



15. (i) In case-1, a diminished image will be formed.

Focal length = 20 cm

$\Rightarrow$ Radius of curvature = 40 cm

Now, object distance = 45 cm

This means that the position of the object is beyond the centre of curvature, hence the nature of the image is real, inverted and diminished.

- (ii) Given: Focal length = 15 cm

$\Rightarrow$ Radius of curvature = 30 cm

Also, object distance = 30 cm

This means the object is placed at the centre of curvature. So, the properties of image formed

will be:

Nature of image = Real and inverted

Size of object = equal to that of the object

(iii) (A) Given: Focal length = 30 cm,

Object distance = 20 cm

This means that the position of the object is between pole and principal focus.

Hence, nature of image = virtual and erect

Size of the image = Enlarged

OR

$$16. \frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{v} = \frac{1}{f} - \frac{1}{u}$$

$$\frac{1}{v} = \frac{1}{20} + \frac{1}{30}$$

$$u = -60 \text{ cm}$$

$$m = \frac{-v}{u}$$

$$m = -\left[\frac{-30}{-60}\right]$$

$$m = -\frac{1}{2}$$

$$\frac{h_2}{h_1} = m$$

$$h_2 = h_1 \times m$$

$$h_2 = 4 \times \left[\frac{-1}{2}\right]$$

$$h_2 = -2 \text{ cm}$$

$$17. m = \frac{v}{u}$$

$$m = -\frac{2}{3}$$

$$-\frac{2}{3} = \frac{v}{-12}$$

$$v = 8 \text{ cm}$$

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}, \frac{1}{f} = \frac{1}{8} - \frac{1}{-12}$$

$$\frac{1}{f} = \frac{1}{8} + \frac{1}{12}$$

$$\frac{1}{f} = \frac{5}{24} \Rightarrow f = \frac{24}{5} = 4.8 \text{ cm}$$

$$18. (a) f = 20 \text{ cm}, u = -30 \text{ cm}$$

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{v} = \frac{1}{f} + \frac{1}{u}$$

$$\frac{1}{v} = \frac{1}{20} + \frac{1}{-30}$$

$$\frac{1}{v} = \frac{1}{60}$$

$$v = 60 \text{ cm}$$

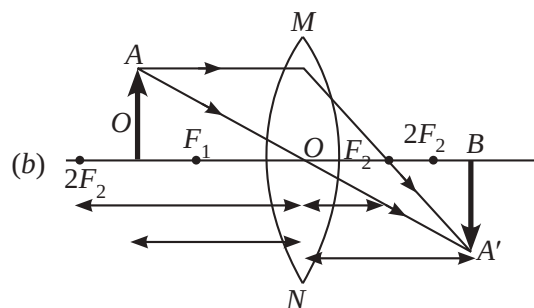
Real, inverted and magnified

$$m = \frac{v}{u}$$

$$m = \frac{60}{-30} = -2$$

$$h' = m \times h$$

$$h' = -2 \times 5 = -10 \text{ cm}$$

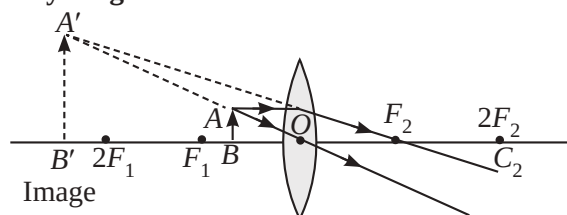


Long Answer Type Questions

19. (a)

- (i) Based on observation 3, where image is formed at the same distance from the optical centre as the object, we can deduce that the lens's radius of curvature is 50 cm. When BYOU'S an object is positioned at $2F$, the image is formed at an equal distance from the lens but on the opposite side. Since we know that the focal length is half of the radius of curvature, the focal length of the lens is +25 cm.
- (ii) Observation 6 is incorrect. In this case, the object is at a distance of -15 cm, i.e., it is placed between F and O . When the object is positioned between F and O , the resulting image forms on the same side as the object, and its nature is virtual. However, it is given that the image distance is +37.5 cm which shows that the image is real and is formed on the other side of the lens which is incorrect.

Ray diagram:



- (iii) Given, $u = -30$ cm, $v = +150$ cm

We know that

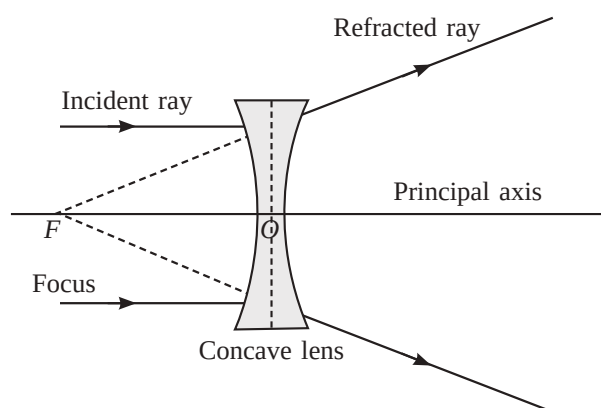
$$\text{Magnification, } m = \frac{v}{u} = -\frac{150}{30} = -5$$

OR

(b)

- (i) An imaginary straight line passing through the two centres of curvature of the spherical surfaces a lens is called principal axis.

Ray diagram for a ray of light parallel to the principal axis in a convex lens:



Hence, when a ray of light is parallel to the principal axis of the concave lens, the refracted ray passes through the focus when extended backwards.

- (ii) Given,

Focal length, $f = -20$ cm (negative for concave lens)

Image distance, $v = -15$ cm

Object height, $h = 5$ cm

Object distance, $u = ?$

Using the lens formula, $\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$

$$\frac{1}{-20} = \frac{1}{-15} - \frac{1}{u} \Rightarrow \frac{1}{u} = \frac{1}{-15} + \frac{1}{20} = \frac{-1}{60}$$

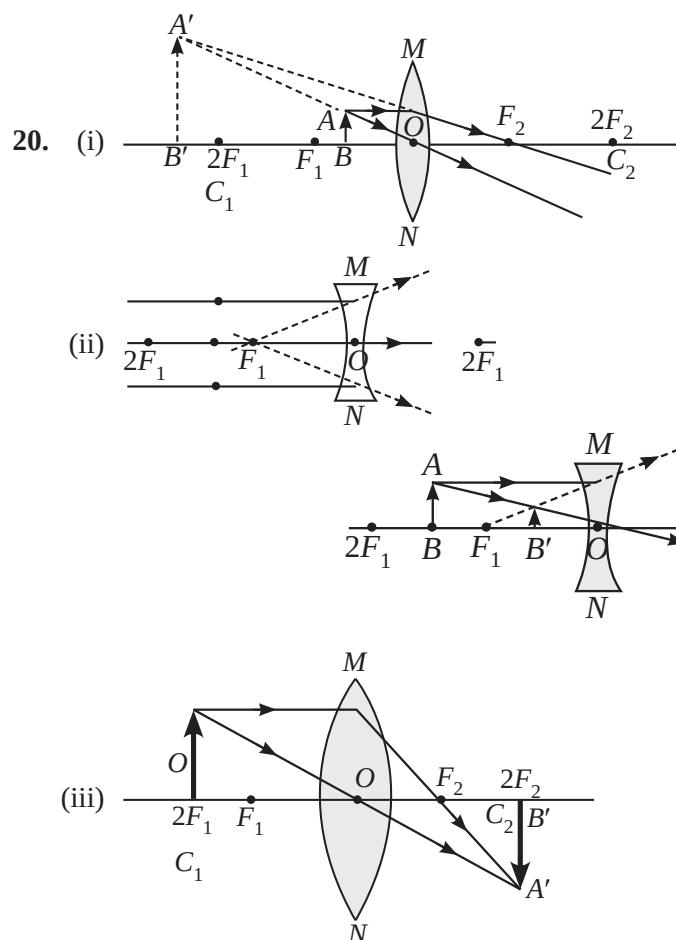
$$\Rightarrow u = -60 \text{ cm}$$

So, the object should be placed at a distance of 60 cm from the concave lens.

Now that we have the object distance u , we can use the magnification formula to find the image height,

$$\text{Magnification (m)} = \frac{h_2}{h_1} = \frac{v}{u}$$

$$h_2 = \frac{h_1 v}{u} = -\frac{5 \times (-15)}{-60} = \frac{5}{4} = 1.25 \text{ cm}$$



In case (i) sign is positive and $m > 1$

(ii) sign is positive and $m < 1$

21. Given $h = +4.0$ cm, $u = -25.0$ cm, $f = -15.0$ cm

- (i) image distance $v = ?$;

$$\text{mirror formula : } \frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\text{or } \frac{1}{v} = \frac{1}{f} - \frac{1}{u} = -\frac{1}{15} - \left(-\frac{1}{25}\right)$$

$$= \frac{-1}{15} + \frac{1}{25} = \frac{-5+3}{75} = \frac{-2}{75}$$

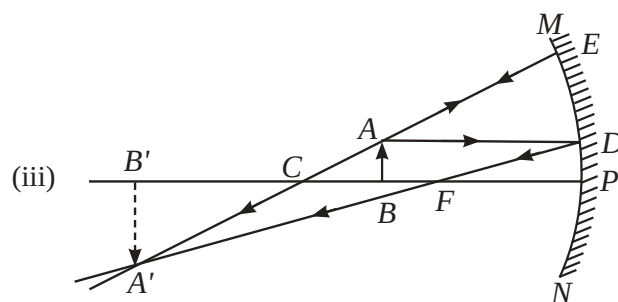
$$v = -37.5 \text{ cm}$$

The screen should be placed 37.5 cm in front of the mirror.

$$(ii) \quad m = \frac{h^1}{h} = -\frac{v}{u}$$

$$h^1 = -\frac{v}{u} \cdot h = -\frac{(-37.5 \times 4)}{-25}$$

$$h^1 = -6.0 \text{ cm (size of the image)}$$



2

CHAPTER

Human Eye and the Colourful World



Previous Year Questions

Multiple Choice Questions

1. The lens system of human eye forms an image on a light sensitive screen, which is called as:[CBSE 2024]
(a) Cornea (b) Ciliary muscles
(c) Optic nerves (d) Retina
2. **Assertion (A):** The rainbow is a natural spectrum of sunlight in the sky.

Reason (R): Rainbow is formed in the sky when the sun is overhead and water droplets are also present in air. [CBSE 2024]

- (a) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).
 - (b) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).
 - (c) Assertion (A) is true, but Reason (R) is false.
 - (d) Assertion (A) is false, but Reason (R) is true.
3. In bifocal lenses used for the correction of presbyopia: [CBSE 2024]
(a) the upper portion is of convex lens for the near vision and lower part is of concave lens for the distant vision.
(b) the upper portion is of convex lens for the distant vision and lower part is of concave lens for the near vision.
(c) the upper portion is of concave lens is for the near vision and lower part is of convex lens for the distant vision.
(d) the upper portion is of concave lens for the distant vision and lower part is of convex lens for the near vision.
 4. Consider the following statements in the context of

human eye:

[CBSE 2024]

- a. The diameter of the eye ball is about 2.3 cm.
- b. Iris is a dark muscular diaphragm that controls the size of the pupil.
- c. Most of the refraction for the light rays entering the eye occurs at the crystalline lens.
- d. While focusing on the objects at different distances the distance between the crystalline lens and the retina is adjusted by ciliary muscles.

The correct statements are

- (a) (A) and (B) (b) (A), (B) and (C)
(c) (B), (C) and (D) (d) (A), (C) and (D)

5. **Assertion (A):** Myopic eye cannot see distant objects distinctly.

Reason (R): For the correction of myopia converging lenses of appropriate power are prescribed by eye-surgeons. [CBSE 2024]

- (a) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).
 - (b) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).
 - (c) Assertion (A) is true, but Reason (R) is false.
 - (d) Assertion (A) is false, but Reason (R) is true.
6. **Assertion (A):** The colour of clear sky appears blue.

Reason (R): Light of blue colour has longer wavelength as compared to the light of red colour so it is scattered more in the upper atmosphere.

[CBSE 2024]

- (a) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).
- (b) Both Assertion (A) and Reason (R) are true, but

Reason (R) is not the correct explanation of the Assertion (A).

(c) Assertion (A) is true, but Reason (R) is false.

(d) Assertion (A) is false, but Reason (R) is true.

7. The phenomena of light involved in the formation of rainbow are: [CBSE 2024]

- (a) Refraction, reflection and dispersion
- (b) Refraction, dispersion and internal reflection
- (c) Reflection, dispersion and internal reflection
- (d) Refraction, dispersion, scattering and total internal reflection

8. The colour of light for which the refractive index of glass is minimum, is: [CBSE 2024]

- (a) Red (b) Yellow
- (c) Green (d) Violet

9. **Assertion (A):** Red light signals are used to stop the vehicles on the road.

Reason (R): Red coloured light is scattered the most so as to be visible from a large distance.

[CBSE 2024]

- (a) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).
- (b) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).

(c) Assertion (A) is true, but Reason (R) is false.

(d) Assertion (A) is false, but Reason (R) is true.

10. When a beam of white light passes through a region having very fine dust particles, the colour of light mainly scattered in that region is: [CBSE 2024]

- (a) Red (b) Orange
- (c) Blue (d) Yellow

Very Short Answer Type Questions

11. A person suffering from an eye defect uses lenses of power -1 D. Name the defect of vision and list its two causes. State the nature (converging/diverging) of the corrective lens. [CBSE 2024]
12. What is presbyopia? Name the type of lenses used for the correction of this defect. State the nature (converging/diverging) of the upper part of such lenses. [CBSE 2024]
13. When do we say that a particular person is suffering from hypermetropia? List two causes of this defect. Name the type of lens used to correct this defect. [CBSE 2024]

14. A person suffering from presbyopia needs bifocal lens. If he needs two lens of power -4.0 dioptre and $+2.0$ dioptre, which one of these two lenses is for the correction of distant vision and what is its focal length? [CBSE 2024]

15. Name the phenomenon of light responsible for Tyndall effect. Write an event where this phenomenon can be observed. [CBSE 2024]

16. What would have been the colour of the sky, if the Earth had no atmosphere? Give reason to justify your answer. [CBSE 2024]

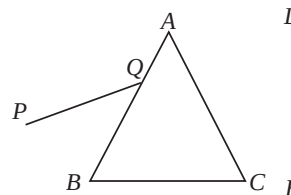
Short Answer Type Questions

17. Study the diagram given below and answer the questions that follow: [CBSE 2024]



- (i) Name the defect of vision represented in the diagram. Give reason for your answer.
- (ii) List two causes of this defect.
- (iii) With the help of a diagram show how this defect of vision is corrected.

18. A narrow beam, PQ of white light is passing through a glass prism ABC as shown in the diagram.



Draw a ray diagram to show the emergent beam as it falls on the screen DE . Also write the phenomenon involved and its cause. Using the second law of refraction state which colour of light must have the highest value of refractive index amongst seven visible colours of light. Justify your answer.

[CBSE 2024]

19. Define the term power of accommodation of human eye. Write the name of the part of eye which plays a major role in the process of accommodation and explain what happens when human eye focuses (i) nearby objects and (ii) distant objects. [CBSE 2024]
20. Draw a ray diagram to show the formation of a rainbow in the sky. On this diagram mark A where dispersion of light occurs, B where internal reflection of light occurs and C where refraction of light occurs. List two necessary conditions to

observe a rainbow.

[CBSE 2024]

21. Name and explain the phenomenon of light due to which the path of a beam of light becomes visible when it enters a smoke filled room through a small hole. Also state the dependence of colour of the light we receive on the size of the particle of the medium through which the beam of light passes.

[CBSE 2024]

22. Define the term power of accommodation of human

eye. What happens to the image distance in the eye when we increase the distance of an object from the eye? Name and explain the role of the part of human eye responsible for it in this case.

[CBSE 2024]

23. When and where does a rainbow appear in the sky? List the phenomena of light in proper sequence which occurs during the formation of a rainbow. Why can a rainbow not be seen from the surface of the moon?

[CBSE 2024]





Explanations

Previous Year Questions

Multiple Choice Questions

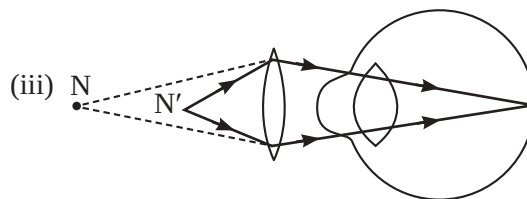
1. (d) 2. (c) 3. (d) 4. (a) 5. (c)
6. (c) 7. (b) 8. (a) 9. (c) 10. (a)

Very Short Answer Type Questions

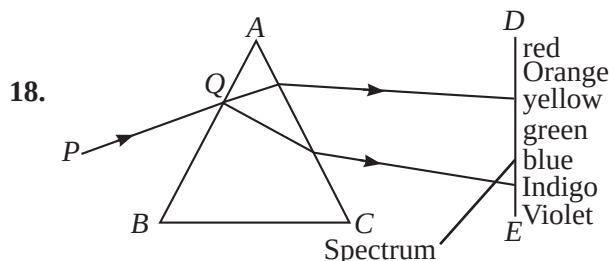
11. (a) The person is suffering from myopia.
The causes for myopia are-
- (1) In certain people, the eyeball becomes elongated.
 - (2) In some cases, the curvatures of the cornea or the lens are such that the eye-lens has a shorter focal length (more power) than usual.
- Corrective lens for myopia should be diverging in nature.
12. Presbyopia is a defect of vision in which person finds difficult to see the nearby object as well as distinct objects clearly.
This defect of vision is corrected by using the suitable lens called bifocal lens.
The upper portion of bifocal lens is a concave lens to focus on the far object.
13. Hypermetropia is also known as far-sightedness. A person with hypermetropia can see distant objects clearly but cannot see nearby objects clearly.
This defect arises either because-
- (i) the focal length of the eye lens is too long, or
 - (ii) the eyeball has become too small.
- This defect can be corrected by using a convex lens of appropriate power.
14. Lens with power -4.0 dioptre is used as a correction for distant vision.
- $$f = \frac{1}{P}, \quad f = \frac{1}{-4} = -0.25\text{m or } -25\text{cm}$$
15. Scattering of light is the phenomenon of light responsible for the Tyndall effect. We observe the Tyndall effect when the sunlight passes through the canopy of dense forest.
16. If the Earth had no atmosphere, the sky would be black because there would be no scattering of sunlight.

Short Answer Type Questions

17. (i) Hypermetropia
(ii) The two causes of this defect are:
- (1) The focal length of the eye lens is too long.
 - (2) The eyeball has become too small.



Correction for Hypermetropic eye



White light splits into its component colours while passing through a glass prism, this phenomenon is called dispersion of light.

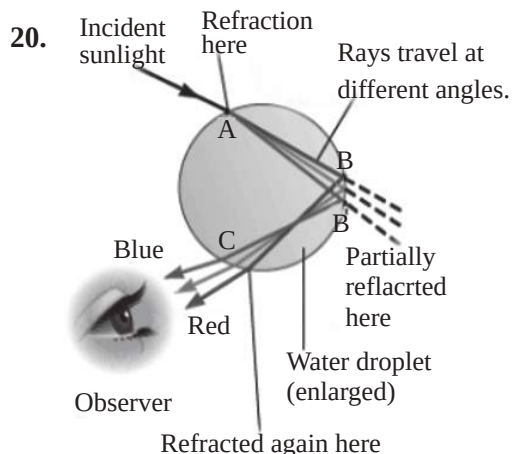
According to Snell's law, the refractive index (n) of a medium is given by $n = \frac{c}{v}$, where c is the speed of light in a vacuum and v is the speed of light in the medium. Since violet light has a shorter wavelength, it travels slowly through the prism compared to other colours. Consequently, it has a higher refractive index and bends the most while passing through a prism.

19. Power of accommodation is the ability of the eye lens to focus near and far objects clearly on the retina by adjusting its focal length of crystalline lens.

This adjustment in focal length of the eye lens is achieved by ciliary muscles.

- (i) When the eye focuses on a near object, ciliary muscles contract. As a result, the lens becomes thick and the focal length of the lens decreases.
- (ii) When the eye focuses on a distant object, the ciliary muscles are relaxed.

As a result, lens become thin and the focal length of the lens increases.



The two necessary conditions to observe the rainbow are:

- (1) The sun comes out after a rainfall.
- (2) The observer stands with the sun towards his/her back

21. The Tyndall effect can be observed when a fine beam of light enters a room filled with smoke through a small hole.

When a beam of light enters a smoke-filled room, it encounters tiny particles of smoke or dust. These particles scatter the light in different directions. The scattered light from the particles makes the path of

the beam visible. This is why you can see the beam of light as it travels through the smoke-filled air.

The colour of the scattered light is dependent on the particle size: smaller particles scatter shorter wavelengths more effectively, often resulting in a blue appearance, while larger particles scatter light more evenly across the spectrum.

22. Power of accommodation is the ability of the eye lens to focus near and far objects clearly on the retina by adjusting its focal length of crystalline lens.

Image distance does not change when we increase the distance of the object from the eye but the focal length of the lens increases.

Ciliary muscles are responsible for the change in focal length of the lens.

23. A rainbow appears in the sky when sunlight interacts with water droplets during or after rain, with the observer facing away from the sun.

The formation of a rainbow involves refraction, dispersion, internal reflection, and a second refraction of light within water droplets.

On the Moon, a rainbow cannot be seen due to the absence of an atmosphere and water droplets necessary for the rainbow's formation.

3

CHAPTER

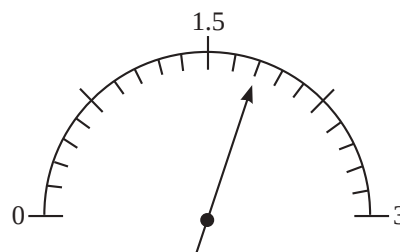
Electricity



Previous Year Questions

Multiple Choice Type Questions

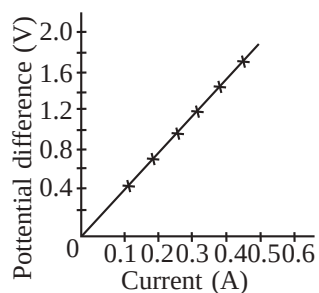
- If the current I through a resistor is increased by 100% (at constant temperature), the increase in power dissipated will be [CBSE 2023]
 (a) 100% (b) 200%
 (c) 300% (d) 400%
- For verifying Ohm's law, we design an electric circuit diagram in which we show the arrangement of different circuit components. We find that with respect to the resistor, the: [CBSE 2023]
 (a) ammeter is connected in parallel and the voltmeter in series.
 (b) ammeter is connected in series and the voltmeter in parallel.
 (c) ammeter and voltmeter are both connected in series.
 (d) ammeter and voltmeter are both connected in parallel.
- If four identical resistors, of resistance 8 ohm, are first connected in series so as to give an effective resistance R_s , and then connected in parallel so as to give an effective resistance R_p , then the ratio $\frac{R_s}{R_p}$ is [CBSE 2023]
 (a) 32 (b) 2
 (c) 0.5 (d) 16
- In the following diagram, the position of the needle is shown on the scale of a voltmeter. The least count of the voltmeter and the reading shown by it respectively are: [CBSE 2023]



- (a) 0.15 V and 1.6 V (b) 0.05 V and 1.6 V
 (c) 0.15 V and 1.8 V (d) 0.05 V and 1.8 V
- At the time of short circuit, the electric current in the circuit: [CBSE 2020]
 (a) Vary continuously (b) does not change
 (c) reduces substantially (d) Increases heavily
- Two bulbs of 100 W and 40 W are connected in series. The current through the 100 W bulb is 1 A. The current through the 40 W bulb will be: [CBSE 2020]
 (a) 0.4 A (b) 0.6 A
 (c) 0.8 A (d) 1 A
- A cylindrical conductor of length ' l ' and uniform area of cross section ' A ' has resistance ' R '. The area of cross section of another conductor of same material and same resistance but of length ' $2l$ ' is [CBSE 2020]
 (a) $\frac{A}{2}$ (b) $\frac{3A}{2}$
 (c) $2A$ (d) $3A$
- The maximum resistance which can be made using four resistors each resistance $\frac{1}{2} \Omega$ is [CBSE 2020]
 (a) 2Ω (b) 1Ω
 (c) 2.5Ω (d) 8Ω

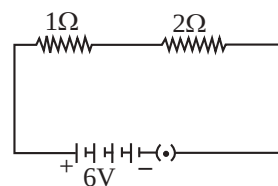
Short Answer Type Questions

9. Show how you would connect three resistors each of resistance 6Ω , so that the combination has a resistance of 9Ω . Also justify your answer. [CBSE 2024]
10. (a) State Joule's law of heating. Express it mathematically when an appliance of resistance R is connected to a source of voltage V and the current I flows through the appliance for a time t . [CBSE 2022]
- (b) A 5Ω resistor is connected across a battery of 6 volts. Calculate the energy that dissipates as heat in 10 s. [CBSE 2022]
11. (a) Calculate the resistance of a metal wire of length 2m and area of cross-section $1.55 \times 10^{-6} \Omega \cdot \text{m}^2$ (Resistivity of the metal is $2.8 \times 10^{-8} \Omega \cdot \text{m}$) [CBSE 2022]
- (b) Why are alloys preferred over pure metals to make the heating elements of electrical heating devices?
12. For a heater, rated 4kW and 220 V, calculate the following [CBSE 2022]
- (a) The current
- (b) Energy consumed in 2 hours
- (c) If 1 kWh is priced at Rs. 4.50, then the cost of energy consumed
13. (a) State Ohm's Law. Represent it mathematically.
- (b) Define 1 ohm.
- (c) What is the resistance of a conductor through which a current of 0.5 A flows when a potential difference of 2 V is applied across its ends? [CBSE 2022]
14. A V-I graph for a nichrome wire is given below. What do you infer from this graph? Draw a labeled circuit diagram to obtain such a graph. [CBSE 2020]

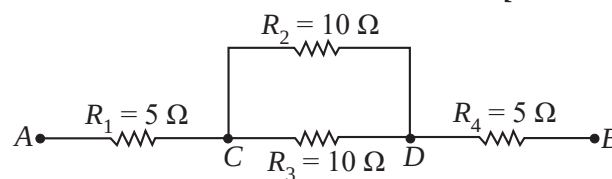


Long Answer Type Questions

15. In the given circuit calculate the power consumed in watts in the resistor of 3Ω [CBSE 2024]



16. An electric oven is designed to work on the mains voltage of 220 V. This oven consumes 11 units of electrical energy in 5 hours. Calculate: [CBSE 2024]
- (a) power rating of the oven
- (b) current drawn by the oven
- (c) resistance of the oven when it is red hot
17. (i) Write the relation between resistance R and electrical resistivity ρ of the material of a conductor in the shape of cylinder of length l and area of cross-section A . Hence derive the SI unit of electrical resistivity. [CBSE 2024]
- (ii) The resistance of a metal wire of length 3 m is 60 Ω . If the area of cross-section of the wire is $4 \times 10^{-7} \text{ m}^2$, calculate the electrical resistivity of the wire. [CBSE 2024]
- (iii) State how would electrical resistivity be affected if the wire (of part 'ii') is stretched so that its length is doubled. Justify your. [CBSE 2024]
18. An electric iron consumes energy at a rate of 880 W when heating is at the maximum rate and 330 W when the heating is at the minimum. If the source voltage is 220 V, calculate the current and resistance in each case. [CBSE 2023]
19. (a) Three resistors R_1 , R_2 and R_3 are connected in parallel and the combination is connected to a battery, an ammeter, a voltmeter and a key. Draw suitable circuit diagram to show the arrangement of these circuit components along with the direction of current flowing. [CBSE 2022]
- (b) Calculate the equivalent resistance of the following network: [CBSE 2022]





Explanations

Previous Year Questions

Multiple Choice Type Questions

1. If the current I through a resistor is increased by 100%, the increase in power dissipated will be 300%. It is given that the current flowing the resistor is I is increased by 100%.

$$P \propto I$$

$$\text{So, } \frac{P_2}{P_1} = \left(\frac{I_1^2}{I_2^2} \right) = \frac{(2I)^2}{I^2} = 4$$

$$\Rightarrow P_2 = 4P_1$$

The percentage increase in power is equal to,

$$\Rightarrow \frac{P_2 - P_1}{P_1} \times 100 \Rightarrow \frac{4P - P}{P} \times 100 = 300\%$$

Hence, the increase in power is 300% if the current is increased by 100%.

2. (b) Ammeter is connected in series and voltmeter is connected in parallel
3. (d) Reason four identical resistors of resistance 8Ω are connected in series:

$$\text{So, } R_s = 8 + 8 + 8 + 8 = 32\Omega$$

When connected in parallel:

$$\frac{1}{R_p} = \frac{1}{8} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8}$$

$$\frac{1}{R_p} = \frac{1+1+1+1}{8} = \frac{4}{8} = \frac{1}{2}$$

$$\text{So, } R_p = 2\Omega$$

$$\frac{R_s}{R_p} = \frac{32}{2} = 16$$

4. (c) 0.15 V and 1.8 V

Reason: Least count of any scale is the value measured by a single division on it.

$$10 \text{ div} = 1.5 \text{ V}$$

$$\text{So, } 1 \text{ div} = \frac{1.5}{10} = 0.15 \text{ V}$$

Hence, least count = 0.15 V

Reading shown = least count $\times$ no. of div.

$$= 0.15 \times 12 = 1.8 \text{ V}$$

Hence, Reading shown = 1.8 V

5. (d) At the time of short circuit, the electric current in the circuit increases heavily.
6. (d) The current remains the same in series circuit.

$$7. \therefore R = \rho \frac{l_1}{A_1} = \rho \frac{l_2}{A_2} \Rightarrow \frac{l_1}{A_1} = \frac{l_2}{A_2}$$

Upon substituting the value we get,

$$\frac{l}{A} = \frac{2l}{A_2} \Rightarrow A_2 = 2A$$

8. (d) The maximum resistance is obtained by connecting all the four resistors in series.

Short Answer Type Questions

9. Two 6Ω resistors are connected in parallel.

$$\text{Resistance} = \frac{1}{\frac{1}{6} + \frac{1}{6}} = \frac{6 \times 6}{6 + 6} = 3\Omega$$

If 3rd resistor of 6Ω and 3Ω are connected in series, it becomes

$$6\Omega + 3\Omega = 9\Omega.$$

10. (a) This law state that heat produced in a resistor is
Directly proportional to the square of current for a given resistance/($H \propto I^2$)
directly proportional to resistance for a given current/($H \propto R$)
directly proportional to resistance for a given current/($H \propto t$)

$$H = VIt$$

- (b) $V = 6\text{V}$; $R = 5\Omega$; $t = 10\text{s}$

Energy dissipated as heat in $t = 10\text{s}$ is

$$H = \frac{V^2}{R} t = \frac{(6\text{V})^2}{5\Omega} \times 10\text{s} = 72\text{J}$$

11. (a) Resistance of a conductor is given by

$$R = \frac{\rho l}{A}$$

Thus, putting all the values, we get

$$R = \frac{2.8 \times 10^{-8} \Omega - \text{m} \times 2\text{m}}{1.55 \times 10^{-6} \text{m}^2}$$

$$R = \frac{5.6 \times 10^{-2}}{1.55} = 3.61 \times 10^{-2} \Omega$$

Thus, the resistance of the metal wire is

$$3.61 \times 10^{-2} \Omega$$

- (b) Alloys are used in electrical heating devices rather than pure metals because the resistivity of an alloy is more than the resistivity of pure metal. Also, the melting point of an alloy is high, so it does not melt or oxidize easily even at a higher temperature.

12. Given power, $P = 4 \text{ kW}$, Voltage, $V = 220 \text{ V}$

- (a) We know that, $P = VI$.

Therefore current

$$I = \frac{P}{V} = \frac{4 \text{ kW}}{220 \text{ V}} = \frac{4000 \text{ W}}{220 \text{ V}} = 18.18 \text{ A}$$

- (b) Energy consumed in 2h = $Pt = 4 \text{ kW} \times 2\text{h} = 8 \text{ kWh}$

- (c) If cost of 1 kWh = Rs. 4.60, then cost of 8 kWh of energy = Rs. $4.60 \times 8 = \text{Rs. } 36.80$.

13. (a) When external conditions like temperature remain constant, the potential difference across the ends of a conductor is directly proportional to the current flowing through the conductor

$$V = i \times R$$

- (b) Resistance is said to be 1 ohm if potential difference of 1V is applied across the ends of the conductor and current flowing through it is 1A.

- (c) Solve using, $V = iR$

14. The nichrome cable's V-I graph exhibits a straight line. It signifies that the resistance of the wire stays unchanged while the current supply has been varied. That is an ohmic conductor because it follows the ohm's law. Mathematically, ohm's law can be represented as:

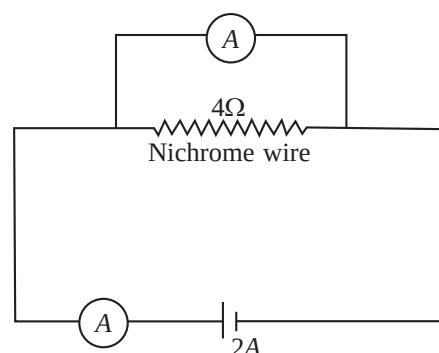
$$V = IR \text{ or } R = \frac{V}{I}$$

Here, R is resistance.

V is the voltage,

I is the current. The resistance of a wire will be:

$$\Rightarrow R = \frac{V}{I} = \frac{0.4}{0.1} = 4 \Omega$$



Long Answer Type Questions

15. 1Ω and 6Ω resistors are connected in series.

The equivalent resistance of the circuit.

$$R = R_1 + R_2$$

$$R = 1 + 2$$

$$R = 3 \Omega$$

According to Ohm's law,

$$V = IR$$

V is the voltage

I is the current

R is the resistance

$$I = \frac{V_1}{R} \Rightarrow I = \frac{6}{3} = 2 \text{ A}$$

By the given expression, power is

$$P_1 = I^2 R$$

$$P_1 = 2^2 \times 2$$

$$P_1 = 8 \text{ W}$$

Hence, the power used in the 2Ω resistor a 6 V battery in series with 1Ω and 2Ω resistors is 8 W .

16. (a) Energy = 11 units = $11 \text{ kWh} = 11000 \text{ Wh}$

Time = 5 hours

$$\text{Power} = \frac{\text{Energy}}{\text{Time}} = 2200 \text{ W}$$

- (b) $P = Vi$

Therefore, $i = 10 \text{ A}$

- (c) $V = i \times R$

$$220 = 10 \times R$$

$$R = 22 \text{ ohm}$$

17. (a)

$$(i) R = \frac{\rho L}{A}$$

$$\rho = R \left(\frac{A}{L} \right) = \text{ohm} \times \text{m}^2 / \text{m}$$

$$= \text{ohm} \times \text{m}$$

Therefore, unit of rho is rho meter.

- (ii) Calculate using the formula of resistivity:

$$R = \frac{\rho L}{A}$$

(iii) Electrical resistivity is independent of dimensions of the wire. Therefore, it will not change

18. Power = $VI = \frac{V^2}{R}$,

Where V is voltage, I is current, R is resistance.

First case,

$$\text{Power} = 880\text{W} \Rightarrow 880 = 220 \times I \Rightarrow I = 4\text{A}$$

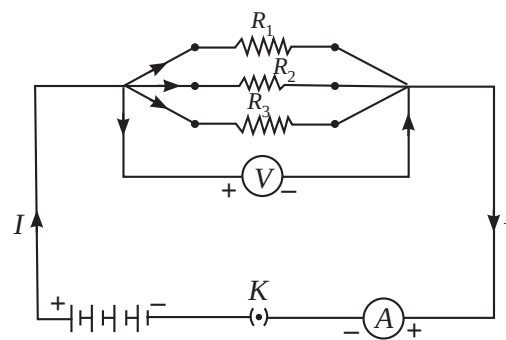
$$\Rightarrow 880 = \frac{220^2}{R} \Rightarrow R = 55\Omega$$

Second case,

$$\text{Power} = 440\text{ W} = 220 \times I \Rightarrow I = 2\text{A}$$

$$\Rightarrow 440 = \frac{220^2}{R} \Rightarrow R = 110\Omega$$

19. (a)



Circuit diagram with given component direction

(b) Resistance between C and D is given by

$$\frac{1}{R_{CD}} = \frac{1}{10} + \frac{1}{10} = \frac{2}{10} = \frac{1}{5}$$

$$R_{CD} = 5\Omega$$

$$D \text{ and } B = R_4 = 5\Omega$$

$\therefore$ Total resistance is

$$R_S = R_{CD} + R_1 + R_4$$

$$R_{\text{Total}} = 5\Omega + 5\Omega + 5\Omega = 15\Omega$$

4 CHAPTER

Magnetic Effects of Electric Current



Previous Year Questions

Multiple Choice Questions

1. These questions consist of two statements Assertion (A) and Reason (R). Answer these questions selecting the appropriate option given below:

[CBSE 2024]

- (a) Both (A) and (R) are true and (R) is the correct explanation of (A).
(b) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
(c) (A) is true, but (R) is false.
(d) (A) is false, but (R) is true.

Assertion (A): The deflection of a compass needle placed near a current carrying wire decreases when the magnitude of an electric current in the wire is increased.

Reason (R): Strength of the magnetic field at a point due to a current carrying conductor increases on increasing the current in the conductor.

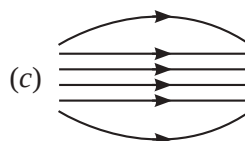
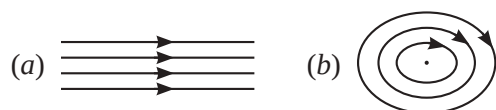
2. Strength of magnetic field produced by a current carrying solenoid DOES NOT depend upon:

[CBSE 2024]

- (a) number of turns in the solenoid
(b) direction of the current flowing through it
(c) radius of solenoid
(d) material of core of the solenoid

3. The pattern of the magnetic field produced inside a current carrying solenoid is:

[CBSE 2024]

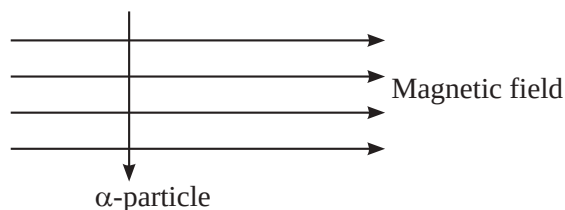


(c)

(d) None of these

4. An alpha particle enters a uniform magnetic field as shown. The direction of force experienced by the alpha particle is:

[CBSE 2023]



- (a) Towards right (b) Towards left
(c) Into the page (d) Out of the page

5. **Assertion (A):** A current carrying straight conductor experiences a force when placed perpendicular to the direction of magnetic field.

Reason (R): The net charge on a current carrying conductor is always zero.

[CBSE 2023]

- (a) Both (A) and (R) are true and (R) is the correct explanation of (A)
(b) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(c) (A) is true but (R) is false
(d) (A) is false but (R) is true

Short Answer Type Questions

6. (a) Distinguish between the terms 'overloading' and 'short-circuiting' as used in domestic circuits.
(b) Why are the coils of electric toasters made of any alloy rather than a pure metal?

[CBSE 2024]

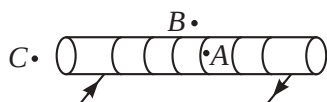
7. (a) (i) Why is an alternating current (A.C.) considered to be advantageous over direct current (D.C.) for the long distance transmission of electric power?
- (ii) How is the type of current used in household supply different from the one given by a battery of dry cells?
- (iii) How does an electric fuse prevent the electric circuit and the appliances from a possible damage due to short circuiting or overloading?

[CBSE 2023]

OR

- (b) For the current carrying solenoid as shown, draw magnetic field lines and give reason to explain that out of the three points A, B and C, at which point the field strength is maximum and at which point it is minimum?

[CBSE 2023]



8. (a) Name the poles P, Q, R and S of the magnets in the following figures 'a' and 'b':

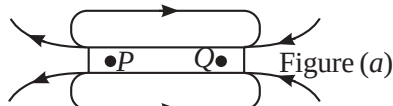


Figure (a)

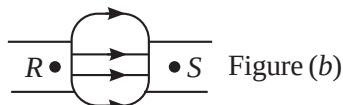


Figure (b)

- (b) State the inference drawn about the direction of the magnetic field lines on the basis of these diagrams.

[CBSE 2022]

OR

When is the force experienced by a current - carrying straight conductor placed in a uniform magnetic field

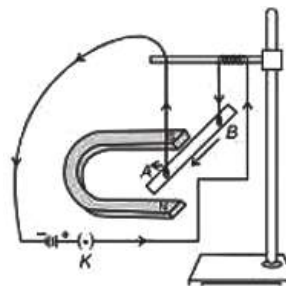
[CBSE 2022]

- (i) Maximum:
(ii) Minimum?

9. Case Study Based Questions:

A student was asked to perform an experiment to study the force on a current carrying conductor in a magnetic field. He took a small aluminium rod AB, a strong horse shoe magnet, some connecting wires, a battery and a switch and connected them as shown. He observed that on passing current, the rod gets displaced. On reversing the direction of current,

the direction of displacement also gets reversed. On the basis of your understanding of this phenomenon, answer the following questions:



- (a) Why does the rod get displaced on passing current through it?
- (b) State the rule that determines the direction of the force on the conductor AB.
- (c) (i) If the U shaped magnet is held vertically and the aluminium rod is suspended horizontally with its end B towards due north, then on passing current through the rod from B to A as shown, in which direction will the rod be displaced?
- (ii) Name any two devices that use current carrying conductors and magnetic field. [CBSE 2022]

OR

Draw the pattern of magnetic field lines produced around a current carrying straight conductor held vertically on a horizontal cardboard. Indicate the direction of the field lines as well as the direction of current flowing through the conductor.

[CBSE 2022]

10. (a) What is a solenoid? Draw a sketch of the pattern of field lines of the magnetic field through and around a current carrying solenoid.
- (b) Consider a circular loop of wire lying in the plane of the table. Let the current pass through the loop clockwise. Apply the right hand rule to find out the direction of the magnetic field inside and outside the loop.

11. What is the function of a galvanometer in a circuit?

[CBSE 2019]

12. State Fleming's left hand rule.

[CBSE 2018]

Long Answer Type Questions

13. (i) Two magnetic field lines do not intersect each other. Why?

- (ii) How is a uniform magnetic field in a given region represented? Draw a diagram in support of your answer. **[CBSE 2024]**

14. Name and state the rule to determine the direction of a: **[CBSE 2024]**

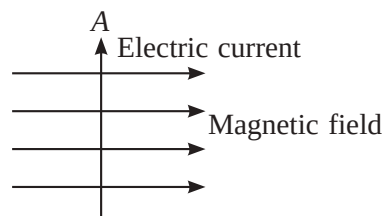
- (i) magnetic field produced around a current carrying straight conductor.
(ii) force experienced by a current carrying straight conductor placed in a magnetic field which is perpendicular to it.

15. Draw the pattern of the magnetic field lines due to a straight current carrying conductor indicating the direction of current in the conductor and the direction of the corresponding magnetic field lines. **[CBSE 2024]**

16. Draw a diagram to show the pattern of magnetic field lines on a horizontal sheet of paper due to a straight conductor passing through its centre and carrying current vertically upwards. Mark on it
(i) the direction of current in the conductor and
(ii) the corresponding magnetic field lines. State right hand thumb rule and check whether the directions marked by you are in accordance with this rule or not. **[CBSE 2024]**

17. State Fleming's left hand rule. Apply this rule to determine the direction of force experienced by a

straight current carrying conductor AB placed in a uniform magnetic field as shown. **[CBSE 2024]**



18. Define the term solenoid. Draw the pattern of magnetic field lines around a current carrying solenoid. State how this magnetic field can be used to magnetise a piece of magnetic material, like soft iron. **[CBSE 2024]**

- 19.** (a) What is an electromagnet? List any two uses.
(b) Draw a labelled diagram to show how an electromagnet is made.
(c) State the purpose of soft iron core used in making an electromagnet. **[CBSE 2020]**
(d) List two ways of increasing the strength of an electromagnet if the material of the electromagnet is fixed.

20. What is a solenoid? Draw the pattern of magnetic field lines of

- (i) A current carrying solenoid and
(ii) A bar magnet.

List two distinguishing features between the two fields. **[CBSE 2019]**



Explanations

Previous Year Questions

Multiple Choice Type Questions

1. Option (d) is correct.

Explanation: In reality, when the current in a wire increases, the magnetic field around the wire becomes stronger and the deflection of a compass needle placed near the wire increases, not decreases. Therefore, the relationship between the deflection of the compass needle and the current in the wire is opposite to what is stated in Assertion (A).

2. Option (c) is correct.

Explanation: The strength of the magnetic field in a solenoid is primarily influenced by factors such as the number of turns in the coil, the current flowing through the coil and the permeability of the material inside the coil. The length of the solenoid is not a factor that directly affects the strength of the magnetic field.

3. Option (a) is correct.

4. (d) Using Fleming's left hand rule.

5. (b)

Short Answer Type Questions

6. (a) Short circuiting - When neutral and live wire come in direct contact.

Overloading - When too many appliances are connected to a single socket drawing much more current or power than permissible.

- (b) Resistivity of an alloy is higher than its constituent metal and alloys do not oxidize as easily as constituent metal at high temperature. That is why the coils of electric toasters are made of an alloy rather than a pure metal.

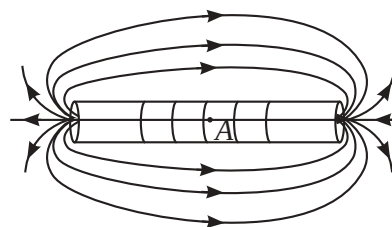
7. (a) (i) Alternating current (A.C.) can be transmitted over long distance with a very low loss of energy in comparison to Direct current (D.C.).
(ii) The type of current used in household supply is alternating current which continuously vary in direction and magnitude whereas the current given by the battery of dry cells is direct current which is of constant magnitude

and direction

- (iii) A fuse is a safety device having a short length of a thin wire made of tin-lead alloy which has a lower melting point. This wire melts and breaks the circuit if the current exceeds the safe value. This saves costly electric appliances and buildings from damage.

OR

- (b) Field strength is maximum at point A as it lies inside the solenoid where field lines are most dense. Field strength is minimum at point B as it lies outside the solenoid.



8. In figure 'a'

- (a) P-North pole Q - South pole

In figure 'b'

R - North pole

S - South pole

- (b) On the basis of diagrams we conclude that outside the magnet, magnetic field lines emerge from north pole and merge at south pole.

OR

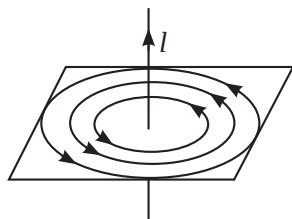
- (i) The force experienced by a current-carrying straight conductor is maximum when the conductor is placed perpendicular to the direction of magnetic field.
(ii) The force experienced by a current-carrying straight conductor is minimum when the conductor is placed along the direction of magnetic field whether parallel or antiparallel.
9. (a) Because a current carrying conductor placed in a magnetic field experiences force.
(b) **Fleming's left hand rule:** If the forefinger, the second finger and the thumb of the left hand are stretched at right angles to each other, with the forefinger pointing in the direction of the field and

the second finger in the direction of the current then the thumb indicates the direction of the force.

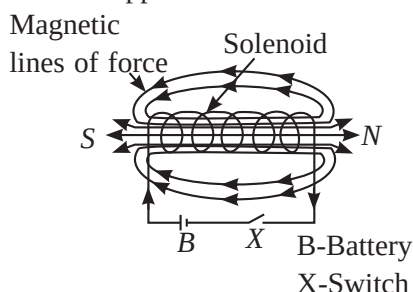
- (c) (i) Downwards
(ii) Electric motor and electric generator

OR

The magnetic field lines around a straight current carrying conductor are concentric circles on planes perpendicular to the direction of current.

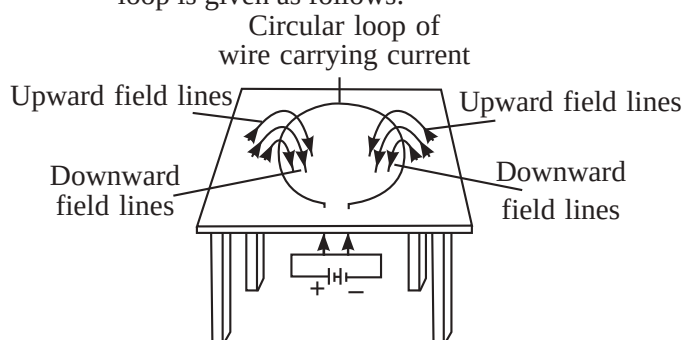


10. (a) A solenoid is a long coil (shaped like a cylinder) containing a large number of close turns of insulated copper wire.



Field lines of the magnetic field through and around a current carrying solenoid.

- (b) Direction of magnetic field inside and outside the loop is given as follows:



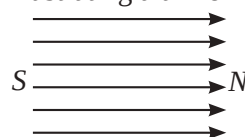
11. To detect the presence of electric current in a circuit.
12. Hold the forefinger the centre finger and the thumb of your left hand at right angles to one another. If the forefinger points in the direction of magnetic field, and centre finger points in the direction of current, the thumb gives direction of motion conductor.

Long Answer Type Questions

13. (i) Magnetic field lines do not intersect each other because if they did, it would imply the existence

of two different directions of the magnetic field at the same point, which is not physically meaningful. The basic principle is that a magnetic field line represents the direction of the magnetic force experienced by a hypothetical north pole placed at that point. If two field lines were to intersect, it would suggest conflicting directions for the magnetic force at the intersection point, violating the rules of a well-defined magnetic field. The non-intersection of magnetic field lines is a fundamental property of magnetic fields and is consistent with the nature of magnetic interactions.

- (ii) A uniform magnetic field in a given region is represented by straight and parallel magnetic field lines. These field lines indicate that the magnetic field has the same strength and direction throughout the entire region. Here is a simple diagram illustrating a uniform magnetic field:



In this representation, the arrows indicate the direction of the magnetic field and the fact that they are straight and parallel suggests uniformity. The spacing between the lines represents the strength of the magnetic field, with closer lines indicating a stronger field. In a truly uniform magnetic field, the spacing between the lines would be consistent throughout the region.

14. (i) **Name of the Rule:** The rule that determines the direction of the magnetic field produced around a current-carrying straight conductor is known as the Right-Hand Thumb Rule.

Statement of the Rule: When the right-hand thumb rule is applied to a straight current-carrying conductor:

If the thumb points in the direction of the current (the flow of positive charge), the curled fingers around the conductor represent the direction of the magnetic field. In summary, when the thumb points along the current, the fingers curl in the direction of the magnetic field lines around the conductor. This rule helps establish the relationship between the current direction and the magnetic field direction in a straight conductor.

- (ii) The rule to determine the direction of the force experienced by a current-carrying straight conductor placed in a magnetic field, which is

perpendicular to it, is known as the Fleming's Left-Hand Rule.

Statement of the Rule: When using Fleming's Left-Hand Rule:

The thumb represents the direction of the force (F).
The index finger represents the direction of the magnetic field (B).

The middle finger represents the direction of the current (I).

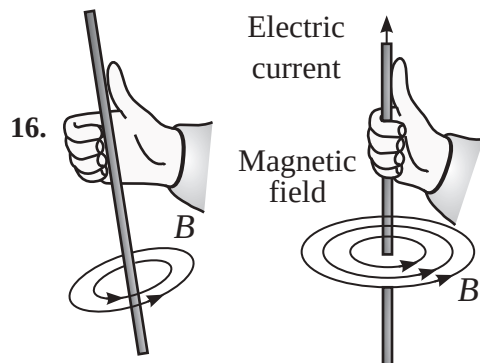
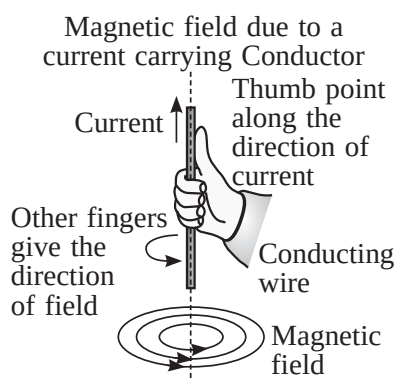
By aligning these three directions according to the rule, the force direction on the conductor can be determined. The rule is applicable when the magnetic field, current and force are mutually perpendicular to each other.

15. The magnetic field around a straight current-carrying conductor forms concentric circles centered around the conductor. To indicate the direction of the magnetic field and current, we can use the right-hand thumb rule:

1. Right-Hand Thumb Rule for Conventional Current:

If you point your thumb in the direction of the current (the flow of positive charge), the direction in which your fingers encircle the conductor represents the direction of the magnetic field.

2. Direction of Magnetic Field Lines: The magnetic field lines form circles around the conductor and are perpendicular to the direction of the current.



Right hand thumb rule: According to this rule, "If you imagine holding a current carrying wire in your right-hand with your thumb pointing towards the direction of electric current flow then the direction in which your fingers curl gives the direction of lines of force of the magnetic field".

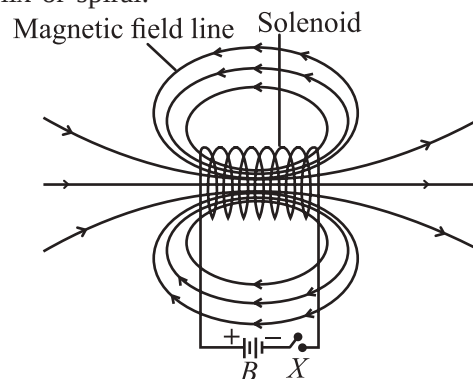
Yes it is accordance to the right hand rule.

17. (a) Extend your forefinger, middle finger and thumb of your left hand perpendicular to each other. If the forefinger signifies the magnetic field's direction and the middle finger denotes the current direction, then the thumb points out the motion of the conductor.

As current flows through the conductor, it experiences displacement, confirming the presence of a force acting on the conductor. This phenomenon is illustrated by the Left-Hand Rule, a principle commonly employed in physics to understand the relationship between the magnetic field, current direction and the resulting motion of a conductor.

- (b) The direction of force is into the plane.

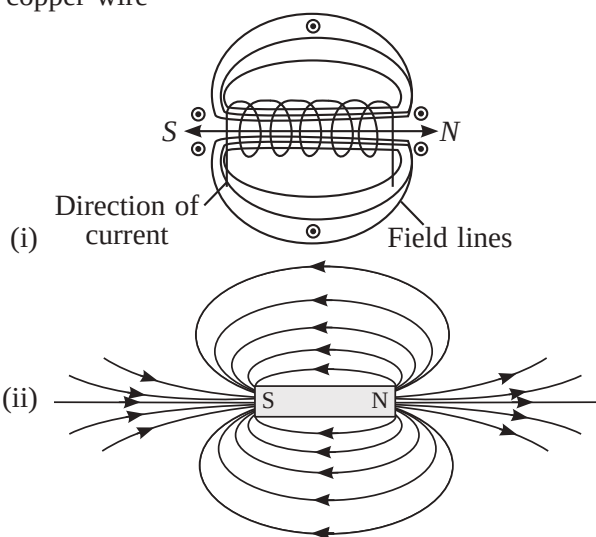
18. **Solenoid:** A coil of wire wound in the form of a helix or spiral.



Magnetic field inside the solenoid is very strong and make the soft iron core an electromagnet.

19. (a) An electromagnet is a temporary magnet made from a piece of soft iron when current flows in the coil wound around it.
- (b) Uses of electromagnet:
- (i) For removing pieces of iron from wounds
 - (ii) For lifting and transporting the large masses of iron scrap, grinders, plates etc.
- (c) If a core of soft iron is placed inside a solenoid, the magnetic field strength inside the solenoid is greatly increased.
- (d) (i) By increasing the number of turns of winding
- (ii) By increasing the current

20. A solenoid is a long cylindrical coil containing a large number of closely spaced turns of insulated copper wire



Distinguish between the two fields are

- (a) The strength of magnetic field due to solenoid can be changed while the magnetic field strength due to bar magnet cannot be changed.
- (b) Solenoid produces magnetic field so long as current flows in its coils while bar magnet produces a permanent magnetic field.